



SACS

Concrete

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The reinforced concrete program modules, Concrete I and Concrete II, allow for the modeling of reinforced concrete beam, column and slab elements in the SACS model. These modules give the user analysis, design and code check capabilities for concrete structures or structures containing concrete components. The concrete modules may also be used with nonlinear foundation analysis when coupled with PSI module or dynamic response analysis when coupled with the Dynpac and Dynamic Response modules.

Both concrete modules require the SACS IV program module containing the pre-processor module Pre, the solver module Solve and the post processor module Post. Concrete analysis and code check results are accessed through the post processor module, Post, and may be executed as part of SACS IV or as an individual analysis step. This manual addresses the reinforced concrete modeling, analysis and code check capabilities of the SACS system

1.2 PROGRAM FEATURES

The SACS system contains two concrete modules, Concrete I and Concrete II. The Concrete I module allows for the modeling, analysis and code check of reinforced concrete elements using a first-order analysis. The Concrete II module allows for nonlinear second-order or $P\Delta$ analysis which includes the effects of sway deflections.

Note: Second-order or PD analysis requires both concrete program modules .

1.2.1 First-Order Analysis (Concrete I)

The Concrete I module requires that concrete model data and analysis options be specified in a SACS input model file.

Some of the main features and capabilities of this module are:

1. Beam elements and bi-axial beam-column elements.
2. Rectangular, circular, tee, I shape, right and left L cross sections available.
3. Supports asymmetric column cross sections with symmetric or asymmetric reinforcement.
4. One way, two way and general slab elements supported.
5. Row, box, circular and prismatic reinforcement patterns.
6. Elements may have multiple reinforcement patterns.
7. Results and code check per Building Code Requirements for Reinforced Concrete ACI 318R-89 (Revised 1992).
8. Column axial/moment interaction capacity curve plots.

-
9. Automatic determination of moment magnification to account for slenderness of compression members.
 10. Allows user specified moment magnifier.
 11. Shear and torsion reinforcement check.
 12. Checks punching shear capacity of two way slabs located near interior, exterior and/or corner columns.
 13. Minimum and maximum reinforcement check.
 14. Capacity and unity check reports pertaining to concrete cross sections.
 15. May be coupled with nonlinear foundation analysis.

1.2.2 Second-Order Nonlinear Analysis (Concrete II)

The Concrete II module allows for a nonlinear second-order or PÄ analysis to more exactly account for the influence of axial loads and variable moment of inertia on member end moments and stiffness. The effects of deflections on stiffness and forces and moments are also taken into account. The Concrete II module requires no special input considerations.

Some of the capabilities of the Concrete II module are:

1. Nonlinear second-order analysis including effects of sway deflections.
2. Determines effective stiffness based on cracked cross-section moment of inertia.
3. Iterates on element stiffness, joint deflection and member rotation.

1.3 CONCRETE MODEL COMPONENTS

Reinforced concrete model components and analysis data are specified in the standard SACS IV model file. Concrete model data may be generated by various SACS program modules including Precede, Datagen or a text editor.

Note: The model may contain concrete or a mixture of concrete and steel components.

In order to execute the concrete analysis, the following additional data pertaining to the concrete components of the structure must be specified:

1. Concrete Analysis Options
2. Concrete Post Processor Options
3. Concrete Material and Section Property Data
4. Concrete Element Data

1.4 CONCRETE ANALYSIS OPTIONS

Analysis options specific to concrete elements are specified on the 'CNCOPT' input line and include the following:

1. Analysis Type (columns 13-14)
 - a. First-order frame analysis braced against side sway (default). For this analysis type, moment magnification is used to calculate the factored moment for compression elements. The dead load moment magnifier is calculated based on ACI 318R recommendations while sway load moments are not magnified.
 - b. First-order frame analysis unbraced against side sway is denoted by 'UN' in columns 13-14. Moment magnification is used to calculate the factored moment for compression elements. The dead load moment magnifier is calculated based on ACI 318R recommendations and moments due to other loading are magnified by the sway load magnifier input in columns 53-57.
 - c. Nonlinear second-order frame analysis is designated by 'NL'. This analysis option requires the Concrete II program module.
 - d. Nonlinear second order frame analysis including PSI is designated by 'NP' in columns 13-14. This analysis option requires both the PSI and Concrete II program modules.
2. Sidesway Moment Magnifier (columns 53-57)

The magnifier applied to moments due to loading which induces appreciable sidesway must be specified by the user. This value is used for analysis of unbraced frames only (option 'UN'). See Section for designating load cases as load cases that cause no sidesway and load cases that cause appreciable sidesway.

1.5 CONCRETE REPORT OPTIONS

Report options specific to concrete elements may be specified in the SACS model file on the 'CNCOPT' input line but are not required. Report options specified in the model are used as defaults by the Post program and may be modified in the Post input file.

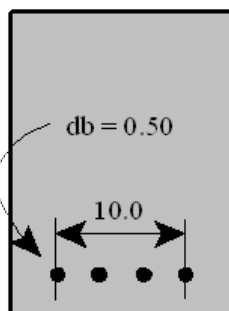
Note: A Post input file is not necessary if the report options specified in the model file are to be used.

1. The desired concrete reports are designated on the 'CNCOPT' line in columns 33-51. The following concrete reports are available:
 - a. Column member detail report designated by 'CD'.
 - b. Beam member detail report designated by 'BD'.

-
- c. Concrete element unity check report designated by 'EL'.
 - d. Unity check range and group summary report designated by 'UR'.
 - e. Concrete slab detail report designated by 'SD'.

1.5.1 Excluding Non-concrete Elements from Reports

Both steel and concrete elements are reported when internal load and member end force reports are selected on the 'OPTIONS' line. Non-concrete elements may be excluded from these reports by specifying 'CO' as one of the report options on the 'CNCOPT' line.

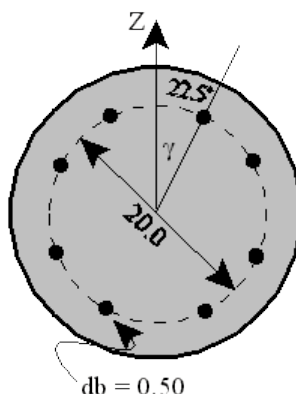


1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
12	34	56	78	90	12	34	56	78	90	12	34	56	78	90	12
SECT ROW1				RRP				0.50				4			
												10.0			

2.1.1.2 Circular Reinforcement Pattern

Circular reinforcement patterns are used to define equally spaced steel rebars of the same diameter arranged in a circle. The pattern name must be specified in columns 6-12 along with the pattern type identifier 'CRP' in columns 16-18.

The default rebar diameter, the total number of rebars in the pattern, the diameter of the pattern row measured from center to center of the outermost rebars and the angle between the local Z axis and the first clock-wise rebar must be specified in columns 50-55, 56-60, 67-71 and 72-76, respectively. For example, the pattern shown on the right can be defined using the 'SECT' line as follows:

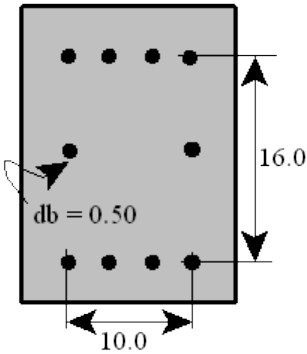


1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
12	34	56	78	90	12	34	56	78	90	12	34	56	78	90	12
SECT CIRCLE1				CRP				0.50				8			
												20.0 22.5			

2.1.1.3 Box Reinforcement Pattern

Box reinforcement patterns are used to define reinforcement arranged in a rectangular pattern. The pattern name must be specified in columns 6-12 along with the pattern type identifier 'BRP' in columns 16-18.

The default rebar diameter, the number of rebars in the upper horizontal row, the number of rebars in the left vertical column, the width of the box and the height of the box measured from center to center of the outermost rebars must be specified in columns 50-55, 56-60, 67-71 and 72-76, respectively. For example, the pattern shown can be defined using the following ‘SECT’ input line:

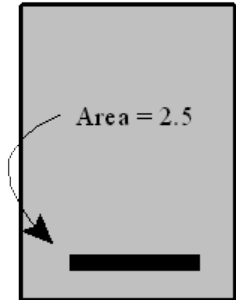


1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
SECT		BOX1		BRP		0.50		4		3		10.0		16.0	

2.1.1.4 Prismatic Reinforcement Section

Prismatic reinforcement sections are used to define the reinforcement as a single block of steel. The prismatic section name must be specified in columns 6-12 along with the identifier ‘PRP’ in columns 16-18.

The total area of reinforcement steel must be specified in columns 50-55. For example, the pattern shown above may be defined as follows:



1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
SECT		PRISM1		PRP		2.50									

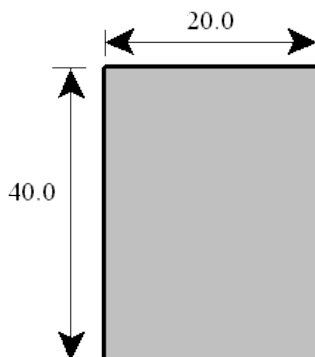
Section properties for beam elements are defined by the section referenced on the 'GRUPC' input line of the group the element is assigned to. All sections referenced by a concrete property group must be defined in the model file using a Concrete Cross Section line with the 'SECT' header.

1. Circular
2. Rectangular
3. Tee
4. Right L
5. Left L
6. I-Section

Circular cross sections may be used to define beam and column elements that have a circular cross section. The section name must be specified in columns 6-12 along with the cross section type identifier 'CCS' in columns 16-18. An 'R' must be input in column 15 if the section is reinforced with steel rebar.

A diagram of a circle with a horizontal line passing through its center. The line has arrows at both ends pointing to the circle's edge. The number "24.0" is written above the line in the middle.

Rectangular cross sections may be used to define beam and column elements that are rectangular or square shaped. The section name must be specified in columns 6-12 along with the cross section type identifier 'CRS' in columns 16-18. An 'R' must be input in column 15 if the section is reinforced with steel rebar. See Section 2.1.13 for defining section reinforcement.

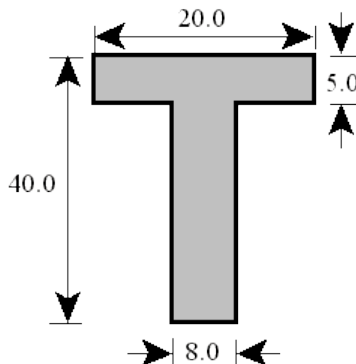


1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890							
SECT	RECT1	CRS		40.0	20.0		

2.1.2.3 Tee Cross Section

For tee beam and column cross sections, the section name must be specified in columns 6-12 along with the cross section type identifier 'CTS' in columns 16-18. An 'R' must be input in column 15 if the section is reinforced with steel rebar. See Section 2.1.13 for section reinforcement.

For tee cross sections, the height, web width, effective flange width and flange thickness are input in columns 50-55, 56-60, 61-66 and 67-71, respectively. Stiffness properties may be optionally input in columns 19-48. The cross section shown can be defined using the Concrete Cross Section line labeled 'SECT' as follows:

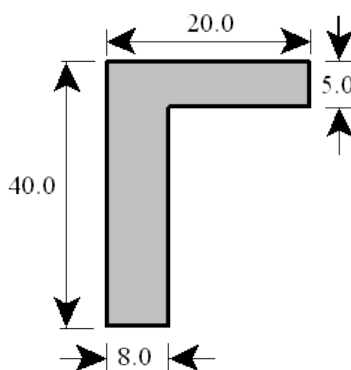


	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
	SECT	TEEL	CTS		40.0	8.0	20.0	5.0

2.1.2.4 Right L Cross Section

The right L section may be used to define L shaped columns or beams. The section name must be specified in columns 6-12 along with the cross section type identifier ‘CRL’ in columns 16-18. An ‘R’ must be input in column 15 if the section is reinforced with steel rebar. See Section 2.1.13 for defining section reinforcement.

For right L cross sections, the height, web width, effective flange width and flange thickness are input in columns 50-55, 56-60, 61-66 and 67-71, respectively. Stiffness properties may be optionally input in columns 19-48. The cross section shown can be defined using the Concrete Cross Section line labeled ‘SECT’ as follows:

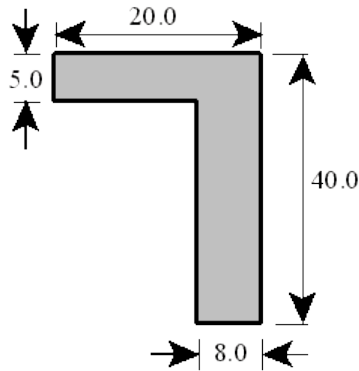


1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1234567890123456789012345678901234567890123456789012345678901234567890															
SECT RIGHT1								CRL		40.0		8.0		20.0	

2.1.2.5 Left L Cross Section

Like the right L section, the left L section may be used to define L shaped columns or beams. The section name must be specified in columns 6-12 along with the cross section type identifier ‘CLL’ in columns 16-18. An ‘R’ must be input in column 15 if the section is reinforced with steel rebar. See Section 2.1.13 for defining section reinforcement.

For left L cross sections, the height, web width, effective flange width and flange thickness are input in columns 50-55, 56-60, 61-66 and 67-71, respectively. Stiffness properties may be optionally input in columns 19-48. The left L cross section shown on the right can be defined using the following Cross Section line labeled ‘SECT’:

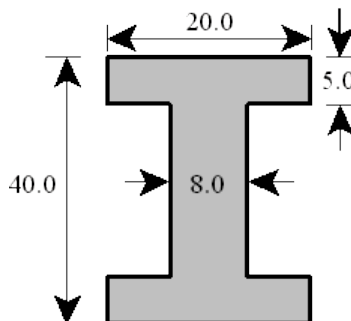


1		2		3		4	
1234567890123456789012345678901234567890123456789012345678901234567890							
SECT LEFT1		CLL				40.0 8.0 20.0 5.0	

2.1.2.6 I Cross Section

The I section may be used to define I shaped columns or beams. The section name must be specified in columns 6-12 along with the cross section type identifier 'CIS' in columns 16-18. An 'R' must be input in column 15 if the section is reinforced with steel rebar. See Section 2.1.13 for defining section reinforcement.

For I cross sections, the height, web width, effective flange width and flange thickness are input in columns 50-55, 56-60, 61-66 and 67-71, respectively. Stiffness properties may be optionally input in columns 19-48. The cross section shown can be defined using the Concrete Cross Section line labeled 'SECT' as follows:



1		2		3		4	
1234567890123456789012345678901234567890123456789012345678901234567890							
SECT ISECT1		CIS				40.0 8.0 20.0 5.0	

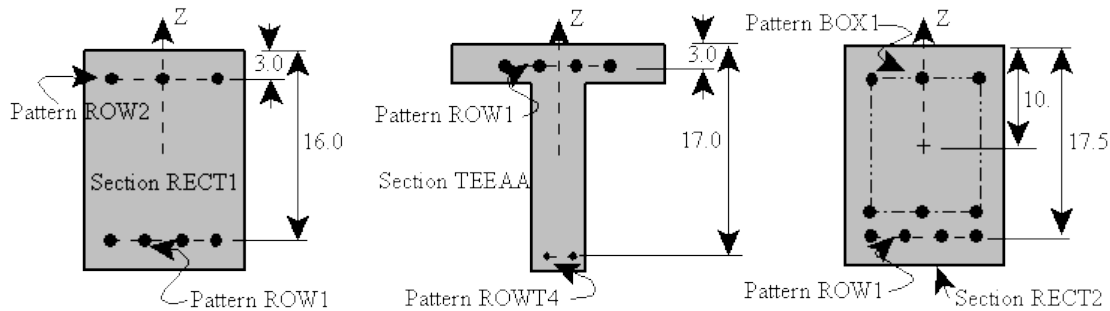
2.1.3 Defining Section Reinforcement

The steel reinforcement patterns used in the concrete cross section are defined on the 'SECT2' line immediately following the 'SECT' line on which the cross section properties are defined.

Up to three reinforcement patterns may be used to define the reinforcement of the concrete section. The first pattern data is specified in columns 9-32 on the 'SECT2' line, the second in columns 33-56 and the third in columns 57-80. For each pattern used, the pattern section ID, the dimension from the top of the concrete section to the pattern CG and the yield strength of the rebars must be specified. The rebar diameter for the pattern may be overridden by specifying a rebar diameter override.

1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890							
SECT TUBE1	RCCS			24.0			
SECT2	CIRCLE1	10.0	50.0				

16



1		2		3		4	
12345678901		2345678901		2345678901		2345678901	
SECT RECT1		RCRS				20.0 15.0	
SECT2 ROW1		16.0 50.0		ROW2		3.0 50.0	
SECT RECT2		RCRS				20.0 17.0	
SECT2 BOX1		10.0 50.0		ROW1		17.5 50.0	
SECT TEEAA		RCTS				20.0 8.0 20.0 6.0	
SECT2 ROW1		3.0 50.0		ROWT4		17.0 50.0	

Note: The top of the section is defined by the positive local Z axis. Also, all rebar pattern sections referenced by 'SECT2' lines must be defined in the model using Reinforcement Pattern Section 'SECT' input lines.

2.1.4 Overriding Pattern Rebar Diameter

By default, the rebar diameter designated on the 'SECT' line defining the pattern is used when a rebar pattern is specified on the 'SECT2' input line. The rebar diameter may be overridden for a particular concrete section by specifying a rebar diameter override on the 'SECT2' line. For example, for section RECT1, the diameter of rebars defined by pattern ROW1 are re-designated as 0.625 in columns 28-32 on the 'SECT2' input line.

1		2		3		4	
12345678901		2345678901		2345678901		2345678901	
SECT RECT1		RCRS				20.0 15.0	
SECT2 ROW1		16.0 50.0 0.625		ROW2		3.0 50.0	
SECT RECT2		RCRS				20.0 17.0	
SECT2 BOX1		10.0 50.0		ROW1		17.5 50.0	

Note: The diameter was overridden only for section RECT1, the default rebar diameter is used for section RECT2.

2.1.5 Material Properties

For concrete column and beam elements, the steel rebar material properties such as modulus of elasticity and density are specified on the 'GRUPC' line in columns 18-23 and 24-29, respectively. Concrete properties such as modulus of elasticity, density and compressive strength are also specified on the 'GRUPC' line in columns 31-35, 36-40 and 41-45, respectively. The concrete group to which the member is assigned is designated on the 'MEMBER' input line.

*Note: If the concrete modulus of elasticity is left blank on the 'GRUPC' input line, the value will be calculated automatically as $E_c = 33rc^{1.5} * \text{Sqrt}(f'_c)$, where rc is the density of concrete and f'_c is the compressive strength.*

2.1.6 Shear Properties

The shear properties of a concrete element are derived from the shear reinforcement information specified on the concrete 'GRUPC' line defining the group to which the element is assigned. The yield stress, spacing, shear reinforcement type ('T' for ties or stirrups or 'S' for spiral) and shear reinforcement bar diameter are designated in columns 60-64, 65-69, 70 and 71-76, respectively.

2.1.7 Designating Element Type

Beam elements may be classified as either a column or beam element. In either case, the element type or member classification is designated in column 47 on the concrete group line defining the property group to which the member is assigned.

2.1.7.1 Column Members

Members that act as beam-columns, i.e. carry both axial force and moment, are assigned to a property group classified as a column group. Columns braced against sidesway are assigned member class '1' while columns unbraced against sidesway are assigned member class '2'. The effective stiffness for first-order analysis, and the initial effective stiffness for second-order analysis is taken as:

$$EI_{eff} = E_c I_g \left(0.2 + 1.2 \rho_t \frac{E_s}{E_c} \right)$$

Column elements are code checked for axial load and resultant moment interaction, column buckling and axial load, shear and torsion interaction. For first order analysis, the resultant moment is magnified to account for column buckling and general stability.

Braced columns that are to be omitted from the post processing reports are classified as type '4' while unbraced columns to be omitted are assigned type '5'.

2.1.7.2 Beam Members

Concrete beams are elements that are subject to flexure without axial load. Beam elements are assigned to groups designated as member class '3' in column 47. The effective stiffness for first-order analysis, and the initial effective stiffness for second-order analysis is taken as:

$$EI_{eff} = 0.5 E_c I_g$$

Beam elements are code checked for moment about the local Y axis, shear along the local Z axis and torsion. Beam elements that are to be omitted from post processing may be assigned member class '6'.

Note: Although axial load and bending about the local Z axis are ignored for beam elements, a warning message is issued if the axial load or local Z bending exceeds 3% of the element's capacity.

2.1.8 Segmented Member Groups

Like segmented steel cross sections, segmented concrete cross sections are defined by a series of 'GRUP' input lines with the same group label. The member classification is determined from the value input on the first 'GRUP' line of the series.

2.2 SLAB ELEMENT PROPERTY DATA

Each slab or concrete plate element in the SACS model is assigned to a group which contains the concrete material and cross-section property data and steel reinforcement properties for all elements assigned to that group. Slabs with identical structural, material and code check properties may be assigned to the same concrete plate group.

2.2.1 Defining Slab Reinforcement Patterns

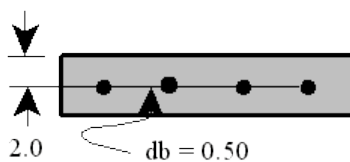
Concrete plate or slab steel reinforcement patterns are defined using Slab Reinforcement Bar Description input lines labeled 'PSTIF'. A slab reinforcement or stiffener pattern section may be referenced by any concrete plate group and a concrete plate group may reference up to two slab reinforcement patterns.

Note: All slab reinforcement or stiffener patterns referenced on concrete plate group input lines must be defined within the model input file.

Reinforcement arranged in a horizontal row of equally spaced steel rebars of the same diameter may be defined using a 'PSTIF' line. The pattern name must be specified in columns 11-17 along with the pattern type identifier 'RBR' in columns 7-9.

The rebar diameter, distance from the rebar centroid to the slab edge (top, bottom or both as designated in the slab group input) are specified in columns 21-27, and 28-34, respectively. The rebar yield stress, elastic modulus and density are input in columns 35-41, 42-48 and 49-55, respectively.

For example, the pattern shown on the right can be defined using the following 'PSTIF' input line:



1								2								3								4								5								6								7								8							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55									
PSTIF											RBR			MESH1			0.50			2.0			60.0			29.0			490.0																																		

Note: All rebars within a defined pattern must be the same diameter. Also, the rebar spacing is defined as part of the concrete plate group input.

2.2.2 Slab Cross-Section and Material Properties

Section and material properties for slab elements are defined on the ‘PGRUPC’ input line of the group the element is assigned to.

When defining concrete cross section properties using a ‘PGRUPC’ input line, the group name and cross section and material properties must be stipulated. The slab group name and slab thickness are input in columns 7-9 and 11-16, respectively. Material properties including modulus of elasticity, Poisson’s ratio, concrete compressive strength and concrete density are input in columns 18-23, 24-29, 30-35 and 73-80, respectively.



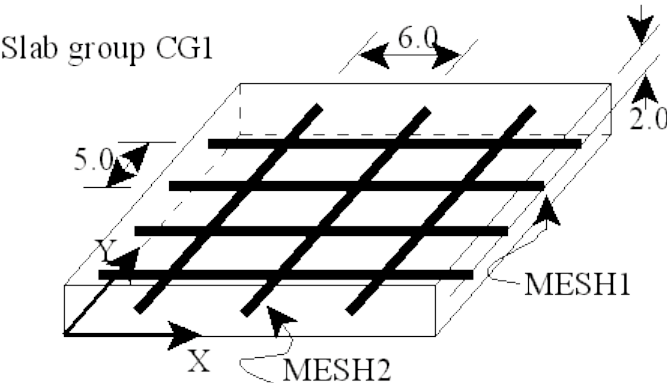
The slab cross section shown may be defined using the Concrete Plate Group line labeled ‘PGRUPC’ as follows:

1								2								3								4								5								6								7								8																							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
PGRUPC								CG1								4.0								C 29.0								0.25 4.0																145.0																															

2.2.3 Slab Reinforcement

The slab reinforcement patterns used in the concrete slab are defined on the ‘PGRUPC’ line defining the concrete plate group properties.

Up to two slab reinforcement patterns defined as ‘PSTIF’ input data may be used to define the reinforcement of the concrete plate section. The first pattern data is specified in columns 42-56 on the ‘PGRUPC’ line, and the second pattern in columns 58-72. For each stiffener pattern used, the pattern section ID, the average spacing of the rebars, the direction of the rebars (i.e. along local X or local Y axis) and the surface to which the location of the centroid of the rebar is measured must be specified.



For example, the concrete plate group CG1 shown above is reinforced with reinforcement stiffener ‘MESH1’ in the local X direction spaced at 5.0 center to center. The rebars are located 2.0 from the top

of the slab. Reinforcement section 'MESH2' is used at 6.0 centers in the local Y direction. The following input line defines concrete slab group 'CG1'.

1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
PGRUPCCG1	4.0	C	29.0	0.25	4.0	MESH1	5.0	XT	MESH2	6.0	YT	145.0			

Note: Slab reinforcement pattern sections may be referenced by more than one concrete plate group.

2.2.4 Slab Types

The type of slab is designated in column 17 on the 'PGRUPC' input line. Concrete slab types supported are:

1. One-way Local X
2. One-way Local Y
3. General
4. Two-way
5. Two-way with punching shear check

2.2.4.1 One-way Slab

One-way slabs are concrete plate elements that bend in one direction only. The bending axis is assumed to be normal to the slab span direction.

Slabs spanning in the local X direction are assumed to have bending stiffness about the local Y axis only and are designated as slab type 'A' in column 17. Slabs spanning in the local Y direction are assumed to have bending stiffness about the local X axis only and are designated as slab type 'B' in column 17.

1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
PGRUPCCP1	4.0	A	29.0	0.25	4.0									145.0	
PGRUPCCP2	4.0	B	29.0	0.25	4.0									145.0	

For code check purposes, one-way slabs are assumed to behave like a series of beam elements. The element is checked for beam shear and bending. The effect of axial load on bending and shear capacity is ignored.

2.2.4.2 Two-way Slabs

Two-way slab elements are concrete plate elements that bend in double curvature and have bending stiffness about both the local X and Y axes. Two-way slabs are designated by 'C' in column 17.

Two-way slabs are checked for beam shear in the local XZ and YZ planes along with bending about the local X and Y axes. No interaction between moment about X and Y axes is considered. The effect of axial load on bending and shear capacity is ignored.

2.2.4.3 Two-way Slab with Punching Shear Considered

Two-way slab elements located near columns or concentrated loads must be checked for punching shear in addition to beam shear and designated as slab type 'E' in column 17.

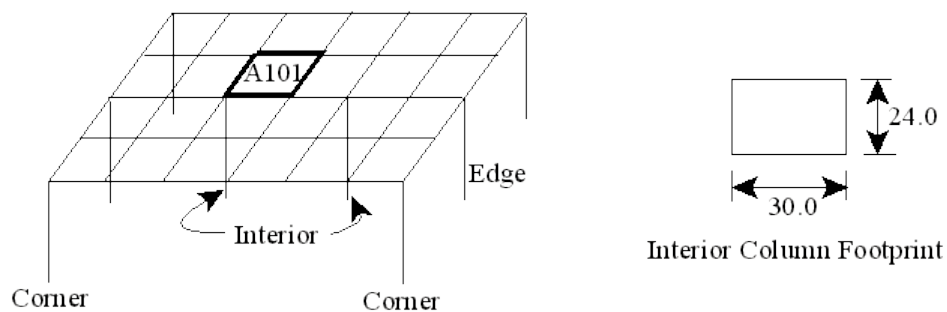
2.2.4.3.1 Punching Shear Critical Section

For slab element type 'E', the punching shear critical section data must be designated on the 'PGRUP2' line immediately following the 'PGRUPC' line.

The punching shear critical section is usually in the vicinity of a column or concentrated load. The critical section location refers to the location of the column or concentrated load with respect to the overall structure and is input in columns 8-10. Enter 'INT' if the critical section is interior, 'EDX' if it is located at an edge of the structure parallel to the plate local X axis, 'EDY' if it is located at an edge of the structure parallel to the plate local Y axis or 'COR' if it is located at one of the corners of the structure.

Note: The critical section location refers to the classification of the column or concentrated load based on its location within the structure not its location with respect to a slab element to which it may be attached.

The figure below illustrates the classification of critical sections defined by columns.



The critical section shape must be designated in column 11. Input 'R' for rectangular section and 'C' for circular sections. For rectangular critical section shapes, the width of the critical section along the plate local X axis is stipulated in columns 12-17 while the width along the plate local Y axis is designated in columns 18-23. For circular sections, the diameter is entered in columns 12-17.

For plate A101 shown in the above figure, the following 'PGRUPC' and 'PGRUP2' input lines are used:

1		2		3		4		5		6		7		8	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
PGRUPCCP1 4.0 E 29.0 0.25 4.0 MESH1 6.0 XT MESH1 6.0 XT 145.0															
PGRUP2 INTR 30.0 24.0															

Note: The critical section width at a column is simply the width or diameter of the column. For concentrated loads, the critical section dimensions are the footprint dimensions.

2.2.4.4 General Concrete Plate Elements

General concrete plate elements are elements that bend in double curvature and have bending stiffness about both the local X and Y axes. General concrete plate elements are designated by 'D' in column 17.

The axial load and bending interaction capacity is determined for axial load along the local X axis and bending about the local Y axis and for axial load along the local Y with bending about the local X axis. General slabs are checked for beam shear in the local XZ and YZ planes. The effect of axial load on bending and shear capacity is considered.

2.3 LOAD CASE TYPES FOR FIRST-ORDER ANALYSIS

When executing a first order analysis, column elements are checked using the magnified factored moment M_c . The magnified factored moment is determined from the following:

$$M_c = \delta_b M_b + \delta_s M_s$$

where δ_b , used to account for column buckling, is applied to the component of the moment due to loading that does not induce sway in the structure (M_b), and δ_s , used to account for basic stability, is applied to the component of the moment due to loading that causes appreciable sidesway (M_s).

In general, the moment M_b results from gravity loads while the moment M_s results from lateral loading. For the purpose of determining the components of the moment, the program determines M_b as the moment resulting from load cases designated as gravity or dead loading while M_s is the moment resulting from all other load cases.

2.3.1 Designating Gravity Load Cases

Load cases that contain gravity loading or loading that causes no sideways should be designated as a gravity load case by specifying them on the LCSEL line using the 'PD' function in columns 7-8. The following designates that load cases 'DED1' and 'WLK1' are gravity load cases:

1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890							
LCSEL PD	DED1 WLK1						

For first-order analysis, the portion of a column element's moment due to a load case designated as a gravity load will be magnified by db.

Note: The application of gravity loads to an un-braced frame in an unsymmetrical pattern or to an unsymmetrical frame may result in some minor sidesway. In general, the minor effects of this sway component may be neglected. For any load case using first-order analysis, the user must determine whether the load case will cause any “appreciable” sidesway component.

By default a load case is considered to cause “appreciable” sidesway unless it is designated as a gravity load case on the ‘LCSEL’ input line. For first-order analysis, the portion of a column element’s moment due to load cases causing sidesway will be magnified by δ_s .

Load combinations consisting of factored basic load conditions or previously defined load combinations must be defined by the user using the 'LCOMB' input line. Load factors for dead load, live load, earthquake load, etc. should be specified along with the load condition number.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
LDCOMB	3	140.0	1	170.0	2			

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3 COMMENTARY

3.1 TERMS AND DEFINITIONS

The following terms and definitions pertain to the variables used throughout the commentary:

A_b	Area of reinforcement bar
A_{con}	Area of the equivalent concrete compression zone
A_s	Area of tension reinforcement
A_t	Area of one leg of a closed stirrup resisting torsion
A_v	Area of shear reinforcement within spacing s
A_{web}	Equivalent concrete compression zone area used for shear
c	distance from the extreme compression fiber to the neutral axis measured perpendicular to the neutral axis
c_b	distance from the extreme compression fiber to the neutral axis for the balanced strain condition
CG	Distance from the fiber of extreme compression to the centroid of the concrete section neglecting reinforcement measured normal to the neutral axis
CG_{con}	Distance from the fiber of extreme compression to the CG of the equivalent concrete compression zone measured normal to the neutral axis
C_t	Shear and torsion stress property factor
d	Distance from extreme compression fiber to centroid of longitudinal tension reinforcement
d_b	Distance from the fiber of extreme compression to the rebar measured normal to the neutral axis
d_{max}	Distance from the fiber of extreme compression to the farthest rebar measured normal to the neutral axis
E_c	Concrete modulus of elasticity
E_s	Modulus of elasticity of steel
f_c	Compressive strength of concrete

f_s	Stress in steel reinforcement
f_y	Steel yield strength
I_b	Moment of inertia of reinforcement bar
I_{con}	Moment of inertia of equivalent concrete compression zone
I_g	Moment of inertia of the gross concrete section ignoring reinforcement
M_n	Nominal moment capacity
M_u	Magnified factored moment resultant from first order analysis, factored moment resultant from P-delta analysis
M_y	Factored moment about the local Y axis
M_z	Factored moment about the local Z axis
P_n	Nominal axial load capacity at a given eccentricity
P_u	Factored axial force
s	spacing of shear or torsion reinforcements
T_c	Nominal torsion strength provided by concrete
T_n	Nominal torsion capacity
T_s	Nominal torsion strength provided by torsion reinforcement
T_u	Factored torsional moment
V_c	Nominal shear strength provided by concrete
V_n	Nominal shear capacity
V_s	Nominal shear strength provided by shear reinforcement
V_u	Factored shear force
x	shorter dimension of rectangular portion of cross section
x_1	shorter center to center dimension of closed stirrup
y	longer dimension of rectangular portion of cross section

y_1	longer center to center dimension of closed stirrup
β_1	Factor used to define height of concrete compression zone
β_d	Ratio of factored dead axial load to total axial load
ϵ_c	Maximum usable strain of concrete in compression
ρ	ratio of tension reinforcement steel area to concrete equivalent compression zone area
$\rho_{\max}$	maximum allowable ratio of tension reinforcement steel to concrete equivalent compression zone area
ρ_t	ratio of reinforcement steel area to concrete area

3.2 CONCRETE ELEMENT EFFECTIVE STIFFNESS

The effective stiffness used for first-order analysis and the initial effective stiffness used for second-order analysis is based on the classification of the concrete beam or column element as follows:

$$EI_{eff\text{ Beam}} = 0.5 E_c I_g$$

$$EI_{eff\text{ Column}} = E_c I_g \left(0.2 + 1.2 \rho_t \frac{E_s}{E_c} \right)$$

For one-way and two-way slabs the effective bending stiffness is considered to be the same as that of a beam element of the same dimensions. The effective stiffness of the general concrete plated element is considered to be the same as that of a column element of the same dimensions.

For second-order analysis, the above effective stiffness is used only for the first stiffness iteration of each load case. For subsequent iterations, the element stiffness is based on the cracked cross section as follows:

$$EI_{eff} = E_c (I_{\text{con}} + A_{\text{con}}(c - CG_{\text{con}})^2) + E_s \sum_{k=1}^n (I_{b_k} + A_{b_k}(d_{b_k} - c)^2)$$

3.3 ANALYSIS AND DESIGN ASSUMPTIONS

The analysis and design of concrete elements is based on the ultimate strength design method recommended by ACI.

3.3.1 Stress and Strain in Concrete

The maximum usable strain in concrete is 0.003. The compressive stress distribution is assumed to be rectangular with a constant stress of $0.85f'_c$ uniformly distributed over an equivalent compression zone. The height of the compression zone is assumed to be $\beta_1 c$, where c is the distance from the fiber of maximum concrete strain to the neutral axis measured perpendicular to the neutral axis.

The concrete tensile strength is neglected for axial and flexural considerations.

3.3.2 Stress and Strain in Steel Reinforcement

The strain in steel reinforcement is assumed to be directly proportional to the distance from the neutral axis. When the neutral axis is located within the concrete section, the stress in steel reinforcement is taken as:

$$f_s = E_s \epsilon_c \frac{d_b - c}{c} \leq f_y$$

When the neutral axis is located outside the concrete section, the stress in steel reinforcement is taken as:

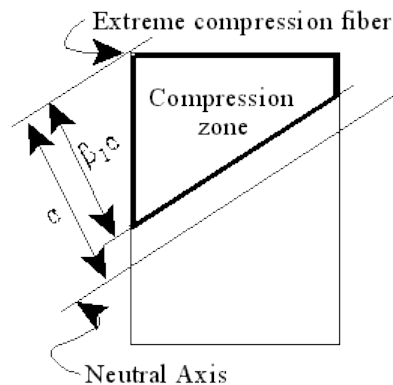
$$f_s = f_y \frac{(d_b - c)}{(d_{\max} - c)}$$

where $d_{\max}$ is the distance from the fiber of maximum compressive strain to the farthest rebar.

3.3.3 Equivalent Concrete Compression Zone

The equivalent concrete compression zone used to calculate axial and flexural capacity, shear capacity of rectangular and circular sections, reinforcement ratio and the maximum area of tension steel, is taken as the area of concrete between the extreme compression fiber and a line parallel to the neutral axis a distance b_1c from the fiber of extreme compression measured perpendicular to the neutral axis.

For flanged sections, the equivalent concrete compression zone used to calculate the shear capacity of concrete, A_{web} , is calculated using the area of the web between the extreme compression fiber and a line parallel to the neutral axis a distance b_1c from the fiber of extreme compression measured perpendicular to the neutral axis.



3.3.4 Balanced Steel Condition

To ensure ductile failure, the balanced steel condition is used to determine the maximum area of tension steel allowed. For the balanced condition, the neutral axis is located such that the strain at the top of the compression zone is 0.003 and the strain in the tension steel farthest from the fiber of extreme compression corresponds to the yield strength. The location of the balanced condition neutral axis, c_b , is determined from:

$$c_b = \frac{E_s \epsilon_c}{f_y + E_s \epsilon_c} d_{\max}$$

where $d_{\max}$ is the distance from the fiber of extreme compression to the farthest reinforcement bar.

3.3.5 Slenderness Effects on Columns

The effects of slenderness on compression members may be considered using a second-order PΔ analysis or by magnifying the resultant moments from a first-order linear analysis.

The Concrete II module performs a second-order nonlinear analysis in which the moments and axial loads include the effects of lateral deflection and sway. The resultant moments also include any secondary moments due to column buckling and/or member end rotation that may occur in slender columns.

3.3.5.1 First-Order Analysis Moment Magnification

Compression columns braced against sidesway, i.e. member class '1', are considered to be slender for a particular axis if $Kl/r > 34 - 12(M_1/M_2)$ where M_1 is the smaller factored end moment due to non-sway loading and M_2 is the larger factored end moment due to non-sway loading. Compression columns not braced against sidesway, i.e. member class '2', are considered to be slender for a particular axis if $Kl/r > 22$.

The effects of slenderness on compression columns are accounted for in a linear analysis by multiplying the factored moment due to non-sway loading, M_b , by the moment magnifier β_b and the moment due to sway loading, M_s , by the moment magnifier δ_s as follows:

$$M_u = \sqrt{(\delta_{by} M_{by} + \delta_s M_{sy})^2 + (\delta_{bz} M_{bz} + \delta_s M_{sz})^2}$$

where δ_{by} and δ_{bz} are moment magnification factors used to account for column buckling about the local Y and Z axes, respectively and δ_s is a magnification factor used to account for lateral drift or sway.

For braced frames, analysis option 'BR', δ_s is taken as 1.0 while the sidesway moment magnifier specified on the 'CNCOPT' line is used for unbraced frames (analysis option 'UN').

The column buckling magnifiers δ_{by} and δ_{bz} are calculated using the following:

$$\delta_{by} = \frac{C_{my}}{1 - \frac{P_n}{\phi P_{cy}}} \quad \delta_{bz} = \frac{C_{mz}}{1 - \frac{P_n}{\phi P_{cz}}}$$

where C_m is taken as 1.0 for columns unbraced against sidesway (member class '2') and taken from the following for columns braced against sidesway (member class '1'):

$$C_m = 0.6 + 0.4 \frac{M_1}{M_2}$$

The critical buckling load P_{cy} and P_{cz} are determined from:

$$P_{cy} = \frac{\pi}{(k_y \ell)^2} \times \frac{\frac{E_c I_{gy}}{5} + E_s I_{sey}}{1 + \beta_d}$$

$$P_{cz} = \frac{\pi}{(k_z \ell)^2} \times \frac{\frac{E_c I_{gz}}{5} + E_s I_{sez}}{1 + \beta_d}$$

where I_{sey} and I_{sez} are the moments of inertia of the steel rebars taken about the section local Y and Z centroidal axes, respectively.

3.4 CONCRETE MEMBER CAPACITY

3.4.1 Moment Capacity of Beam Elements

Beam elements are checked for uniaxial bending about the local Y axis. The unity check ratio for bending is computed as:

$$UC = \frac{M_y}{\phi M_n}$$

where the capacity reduction factor ϕ is 0.90. The moment capacity for members classified as beam elements is calculated as:

$$M_n = 0.85 f'_c A_{con} (CG - CG_{con}) + \sum_{k=1}^n (f_{sk} A_{bk} (d_{bk} - CG))$$

3.4.2 Axial and Moment Capacity of Column Elements

Column elements are checked for biaxial bending and axial load interaction as follows:

$$UC = \frac{P_u}{\phi P_n} = \frac{M_u}{\phi M_n}$$

where M_u is the magnified factored moment resultant from a first-order analysis or the factored moment resultant from a second-order analysis. The capacity reduction factor ϕ is taken as 0.9 for columns in tension, 0.75 for members in compression with spiral shear reinforcement and 0.70 for members in compression with ties or stirrups. For columns in compression with axial load P_u less than $0.10f'_c A_g$, the value of ϕ is increased linearly as follows:

$$\phi = \phi_{min} + 0.2 \frac{0.1 f'_c A_g - \phi_{min} P_n}{0.1 f'_c A_g} \leq 0.90$$

where ϕ_{min} is 0.75 for members with spiral shear reinforcement and 0.70 for members with ties or stirrups.

The axial and moment capacities are determined based the eccentricity e , defined as M_u/P_u , from the following equations:

$$P_n = 0.85 f'_c A_{\text{con}} + \sum_{k=1}^n (f_{s_k} A_{b_k})$$

$$M_n = -P_n(c - CG) + 0.85 f'_c A_{\text{con}}(c - CG_{\text{con}}) + \sum_{k=1}^n (f_{s_k} A_{b_k}(c - d_{b_k}))$$

such that:

$$\frac{M_n}{P_n} = e$$

3.4.3 Moment Capacity of Slab Elements

One-way slab elements are checked as beam elements for uniaxial bending about the axis perpendicular to the span direction. Two-way slabs are considered as beam elements for bending about the local X and Y axes separately. The unity check ratio for bending is computed from one or both of the following:

$$UC = \frac{M_x}{\phi M_n} \quad UC = \frac{M_y}{\phi M_n}$$

where the capacity reduction factor ϕ is 0.90.

3.4.4 Axial & Moment Capacity of General Concrete Plate Elements

General concrete plate elements are checked for biaxial bending and axial load interaction as follows:

$$UC = \frac{P_x}{\phi P_{nx}} = \frac{M_y}{\phi M_{ny}} \quad UC = \frac{P_y}{\phi P_{ny}} = \frac{M_x}{\phi M_{nx}}$$

where the capacity reduction factor ϕ is taken as 0.9 for plates in tension and 0.70 for plates in compression. For plates in compression with axial load P_u less than $0.10f'_c A_g$, the value of ϕ is increased linearly as follows:

$$\phi = \phi_{\min} + 0.2 \frac{0.1 f'_c A_g - \phi_{\min} P_n}{0.1 f'_c A_g} \leq 0.90$$

where $\phi_{\min}$ is 0.70.

3.4.5 Column Buckling

Column elements under axial compression are checked for column buckling as follows:

$$UC = \frac{P_n}{\phi P_{cy}} \quad UC = \frac{P_n}{\phi P_{cz}}$$

where ϕ is 0.70 for columns with spiral shear reinforcement and 0.75 for columns with ties or stirrups. The critical buckling loads P_{cy} and P_{cz} are determined from:

$$P_{cy} = \frac{\pi}{(k_y \ell)^2} \times \frac{\frac{E_c I_{gy}}{5} + E_s I_{sey}}{1 + \beta_d}$$

$$P_{cz} = \frac{\pi}{(k_z \ell)^2} \times \frac{\frac{E_c I_{gz}}{5} + E_s I_{sez}}{1 + \beta_d}$$

where I_{sey} and I_{sez} are the moment of inertias of the steel rebars taken about the section local Y and Z centroidal axes, respectively.

3.4.6 Shear Capacity

The nominal shear capacity V_n is computed by:

$$V_n = V_c + V_s$$

where V_s and V_c are the nominal shear strength of the shear reinforcement and concrete, respectively, and are taken as:

$$V_s = \frac{A_v f_y d}{s} \quad V_c = 2 \sqrt{f'_c} A_{web} \frac{N}{R}$$

For beam, one-way slab and two way slab elements, because the effects of axial load and torsion on shear capacity are neglected, the value of N/R is taken as unity. Beam type members are checked for shear along the local Z axis, using 0.85 for ϕ , as follows:

$$UC = \frac{V_z}{\phi V_{nz}}$$

N/R is also taken as unity for column elements subject to neither axial load nor torsion and for general concrete plate elements subject to no axial load. For column and general concrete plate elements subject to axial compression or axial tension, N is determined from:

$$N_{\text{compression}} = 1 + \frac{P_u}{2000 A_g} \quad N_{\text{tension}} = 1 - \frac{P_u}{500 A_g} \geq 0$$

For column elements subject to factored torsional moment, T_u , less than or equal to $\phi(0.5 \sqrt{f'_c}) S_x 2y$, R is taken as unity. Otherwise, R is taken from the following:

$$R = \sqrt{1 + \left(2.5 C_t \frac{T_u}{V_u} \right)^2}$$

where C_t is $A_{web}/\Sigma x^2 y$ and V_u is the factored shear along the axis in question. Column members are checked for shear along the local Y and Z axes as follows:

$$UC = \frac{V_z}{\phi V_{nz}} \quad UC = \frac{V_y}{\phi V_{ny}}$$

where ϕ is 0.85.

3.4.6.1 Punching Shear Capacity of Two-Way Slabs

Two-way slabs located near columns or concentrated loads may need to be checked for punching shear at a critical section around the column or concentrated load area. The punching shear capacity of the slab is taken as the following:

$$V_s = 0 \quad V_c = Y \sqrt{f'_c} b_o d$$

where Y is the smallest of $(2 + 4/\beta_c)$ and $\alpha_s d/b_o$ but not greater than 4.0, where b_o is the perimeter of the critical section and β_c is the ratio of the longer and shorter sides of the critical section. The value of α_s is taken as 40 for interior, 30 for edge and 20 for corner location of the critical section.

3.4.7 Torsional Capacity

Beam and column type members are checked for torsional moment using ϕ as 0.85 in the following:

$$UC = \frac{T_u}{\phi T_n}$$

The nominal torsional moment capacity T_n is computed by:

$$T_n = T_c + T_s$$

where T_s and T_c are the nominal shear strength of the shear reinforcement and concrete, respectively, and are taken as:

$$T_s = \frac{A_t \alpha_t x_1 y_1 f_y}{s} \quad T_c = \frac{0.8 N \sqrt{f'_c} \Sigma x^2 y}{U}$$

where $\alpha_t = [0.66 + 0.33 (y_1/x_1)] \leq 1.50$, C_t is $A_{web}/\Sigma x^2 y$ and N is taken as unity for elements subject to axial compression. For elements subject to axial tension, N is determined from:

$$N = 1 - \frac{P_u}{500 A_g} \geq 0$$

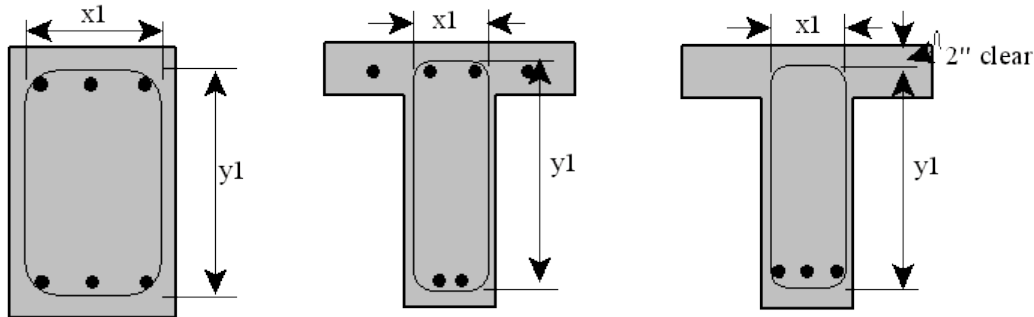
Note: For circular cross sections, the value of $\Sigma x^2 y$ is taken as $3/8(\pi r^3)$.

U is unity when T_u is zero and is defined by the following equation when the magnitude of T_u is greater than zero:

$$U = \sqrt{1 + \left(\frac{0.4 V_u}{C_t T_u} \right)^2}$$

Note: T_c is calculated using shear along the local Z axis for beam type elements. For members classified as column elements, the concrete torsional moment capacity is calculated using shear along the local Y axis and shear along the Z axis separately. T_c is taken as the smaller of the two values.

To calculate x_1 and y_1 , the stirrup is assumed to wrap around the outermost longitudinal rebars within the width of the web. When only one row of reinforcement is defined, the height of the stirrup is determined by assuming 2.0 inch (5.0 cm) clearance on the top or bottom opposite of the reinforcement row.



For circular cross-sections the values of x_1 and y_1 are equal and are determined from the following:

$$x_1, y_1 = \sqrt{\frac{\pi d_s^2}{4}}$$

where d_s is the center to center diameter of the closed stirrup.

3.5 MAXIMUM REINFORCEMENT

Concrete members are checked to ensure that the element meets the minimum and maximum reinforcement requirements as specified by ACI code using the following:

$$\rho_{\min} \leq \rho \leq \rho_{\max}$$

where $\rho_{\min}$ is $200/f_y$, ρ is A_s/A_{con} and $\rho_{\max}$ is $A_{s\max}/A_{\text{con}}$ for beam elements and for columns with axial load less than the smaller of $0.1f_c'A_g$ and ϕP_b , where $A_{s\max}$ is given by the following equation:

$$A_{s\max} = \frac{0.75C_c + C_s}{f_y}$$

where C_c is the total force of the equivalent concrete compression zone taken as $0.85f_c'A_{\text{con}}$ and C_s is the total force in the compression steel.

For column elements with axial load greater than the smaller of $0.1f_c'A_g$ and ϕP_b , $\rho_{\min}$ is 0.01, r is A_{tot}/A_g and $\rho_{\max}$ is 0.08.

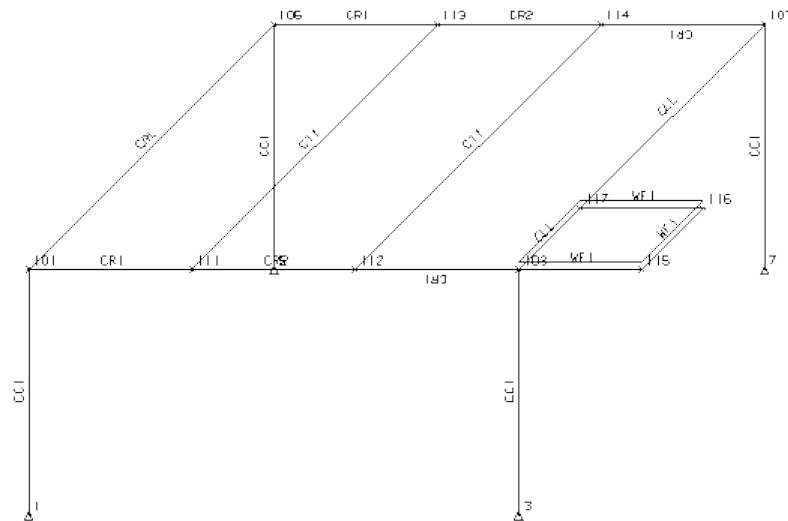
4 SAMPLE PROBLEMS

The sample problems illustrate various capabilities for concrete structures. Two first order analyses are detailed.

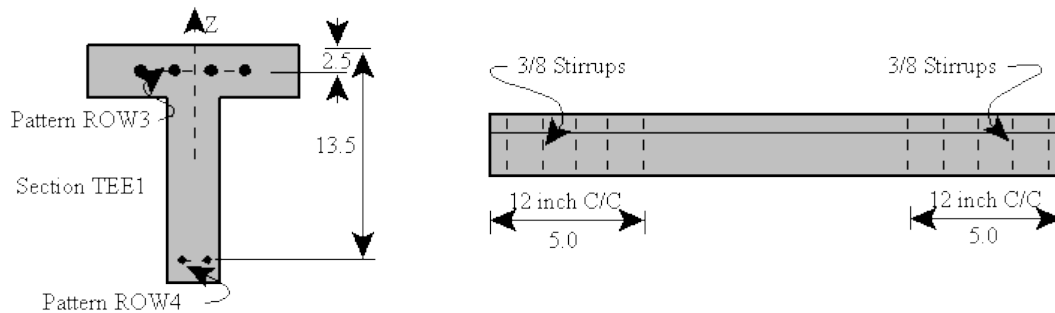
1. The first sample problem is a braced (against sidesway) frame structure consisting of circular concrete columns and rectangular, tee and L shaped concrete beams. The model also includes steel plate and steel beam wide flange elements. This sample contains member and plate offsets along with member end releases. Three basic load conditions, of which one is designated as causing no appreciable sidesway, and three load combinations were specified.
2. Sample Problem 2 illustrates the same model as sample problem executed as an unbraced structure (i.e. Includes moment magnification).
3. Sample Problem 3 illustrates the analysis of concrete one-way and two-way slab elements.

4.1 SAMPLE PROBLEM 1

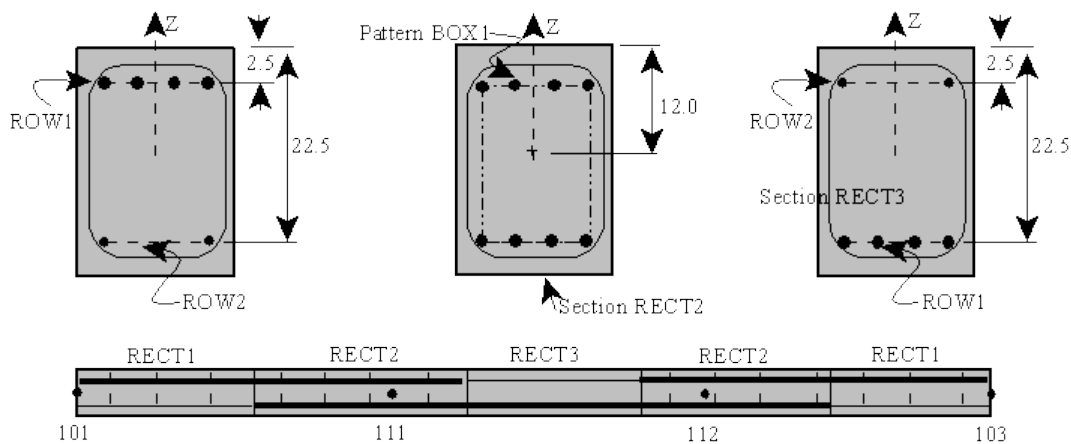
Sample Problem 1 is a first-order analysis of the frame structure shown below. The structure is assumed to be braced against sidesway. Circular concrete columns ('CC1'), rectangular ('CR1', 'CR2'), tee ('CT1') and L-shaped ('CRL', 'CLL') concrete beams along with steel plate and beam elements are modeled.



The tee beams (member group 'CT1') are segmented so that shear stirrups are spaced differently at the ends than in the middle of the member as follows:



Groups 'CR1' and 'CR2' are rectangular cross sections with varying reinforcement along the members. The stirrup spacing is also variable along the member length.



Three basic load conditions and three load combinations are specified. Load case 1 consist of the self weight of the structure and load case 2 contains live loading. Wind load is applied along the global X-axis in load case 3. The load combinations are defined according the ACI-318R guidelines as follows:

Combination	LCase	Factor	LCase	Factor	LCase	Factor
4	1	1.4	2	1.7		
5	1	1.05	2	1.275	3	1.275
6	1	0.9	3	1.3		

The SACS model file is shown below followed by a description of selected portions:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
CONCRETE SAMPLE PROBLEM 1								
A	OPTIONS	EN	LR	2 1 0 0	PT	PTPT		
B	CNCOPT	BR		CD BD EL UR			1.000	
	LCSEL	4	5	6				
C	LCSEL PD	1						
	SECT							
D	SECT BOX1	BRP			0.50	4	210.00	19.00
E	SECT CIRCLE	CRP			0.75	8	11.25	
F	SECT ROW1	RRP			0.50	4	10.00	
G	SECT ROW2	RRP			0.375	2	10.00	
	SECT ROW3	RRP			0.375	4	18.00	
	SECT ROW4	RRP			0.375	2	3.50	
H	SECT CYLIND	RCCS			15.00			
I	SECT2 CIRCLE	7.5060.00						
	SECT LEFTL	RCLL			16.008.000	24.005.000		
	SECT2 ROW3	2.5060.00	ROW4	13.5060.00				
J	SECT RECT1	RCRS			24.0015.00			
K	SECT2 ROW1	2.5060.00	ROW2	22.5060.00				
L	SECT RECT2	RCRS			24.0015.00			
M	SECT2 BOX1	12.0060.00						
	SECT RECT3	RCRS			24.0015.00			
	SECT2 ROW1	22.5060.00	ROW2	2.5060.00				
	SECT RIGHTL	RCRL			16.008.000	24.005.000		
	SECT2 ROW3	2.5060.00	ROW4	13.5060.00				
	SECT TEE1	RCTS			16.008.000	24.005.000		
	SECT2 ROW3	2.5060.00	ROW4	13.5060.00				
	GRUP							
N	GRUPCCC1 CYLIND	29.000490.00	145.0 4.00 1		1.001.0060.00	24.00S 0.375		
	GRUPCCLL LEFTL	29.000490.00	145.0 4.00 3		1.001.0060.00	24.00T 0.375		
O	GRUPCCR1 RECT1	29.000490.00	145.0 4.00 3		1.001.0060.00	12.00T 0.3753.00		
O	GRUPCCR1 RECT2	29.000490.00	145.0 4.00 3		1.001.0060.00	18.00T 0.375		
P	GRUPCCR2 RECT2	29.000490.00	145.0 4.00 3		1.001.0060.00	18.00T 0.3751.67		
P	GRUPCCR2 RECT3	29.00 490.00	145.0 4.00 3		1.001.00			
P	GRUPCCR2 RECT2	29.000490.00	145.0 4.00 3		1.001.0060.00	18.00T 0.3751.67		
	GRUPCCRL RIGHTL	29.000490.00	145.0 4.00 3		1.001.0060.00	24.00T 0.375		
	GRUPCCT1 TEE1	29.000490.00	145.0 4.00 3		1.001.0060.00	12.00T 0.3755.00		
	GRUPCCT1 TEE1	29.00 490.00	145.0 4.00 3		1.001.00			
	GRUPCCT1 TEE1	29.000490.00	145.0 4.00 3		1.001.0060.00	12.00T 0.3755.00		
	GRUP WF1 W6X15		29.0011.6036.00 1		1.001.00	0.50 490.00		
	MEMBER							
	MEMBER1 1 101 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 3 103 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 7 107 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 5 105 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 101 105 CRL							
	MEMBER OFFSETS				4.00		4.00	
	MEMBER1 103 117 CLL							
	MEMBER OFFSETS				4.00		4.00	
	MEMBER0 101 111 CR1							
	MEMBER0 103 112 CR1							
	MEMBER0 105 113 CR1							
	MEMBER0 107 114 CR1							
	MEMBER0 111 112 CR2							
	MEMBER0 113 114 CR2							
	MEMBER1 111 113 CT1							
	MEMBER OFFSETS				4.00		4.00	
	MEMBER1 112 114 CT1							
	MEMBER OFFSETS				4.00		4.00	
	MEMBER1 103 115 WF1							

1		2		3		4		5		6		7		8		
1234567890123456789012345678901234567890123456789012345678901234567890																
MEMBER OFFSETS								4.00		4.00						
MEMBER1 117 107 CLL																
MEMBER OFFSETS								4.00		4.00						
MEMBER1 117 116 WF1																
MEMBER OFFSETS								4.00		4.00						
PLATE																
PLATE AAAA 103 115 117 116 0.25 1																
PLATE OFFSETS								4.00		4.00						
PLATE OFFSETS								4.00		4.00						
JOINT																
JOINT 1 -10.000-10.000 0.000										111						
JOINT 3 10.000-10.000 0.000										111						
JOINT 5 -10.000 10.000 0.000										111						
JOINT 7 10.000 10.000 0.000										111						
JOINT 101 -10.000-10.000 10.000																
JOINT 103 10.000-10.000 10.000																
JOINT 105 -10.000 10.000 10.000																
JOINT 107 10.000 10.000 10.000																
JOINT 111 -3.333-10.000 10.000																
JOINT 112 3.333-10.000 10.000																
JOINT 113 -3.333 10.000 10.000																
JOINT 114 3.333 10.000 10.000																
JOINT 115 15.000-10.000 10.000																
JOINT 116 15.000 -5.000 10.000																
JOINT 117 10.000 -5.000 10.000																
LOAD																
LOADCN 1																
LOAD Z 1 101 -0.208 -0.208										GLOB UNIF		SELFWT				
LOAD Z 3 103 -0.238 -0.238										GLOB UNIF		SELFWT				
LOAD Z 7 107 -0.268 -0.268										GLOB UNIF		SELFWT				
LOAD Z 5 105 -0.298 -0.298										GLOB UNIF		SELFWT				
LOAD Z 103 115 -0.015 -0.015										GLOB UNIF		SELFWT				
LOAD Z 115 116 -0.015 -0.015										GLOB UNIF		SELFWT				
LOAD Z 117 116 -0.015 -0.015										GLOB UNIF		SELFWT				
LOAD Z 101 105 -0.220 -0.220										GLOB UNIF		SELFWT				
LOAD Z 111 113 -0.220 -0.220										GLOB UNIF		SELFWT				
LOAD Z 112 114 -0.220 -0.220										GLOB UNIF		SELFWT				
LOAD Z 117 107 -0.220 -0.220										GLOB UNIF		SELFWT				
LOAD Z 103 117 -0.220 -0.220										GLOB UNIF		SELFWT				
LOAD Z 101 111 -0.375 -0.375										GLOB UNIF		SELFWT				
LOAD Z 111 112 -0.375 -0.375										GLOB UNIF		SELFWT				
LOAD Z 103 112 -0.375 -0.375										GLOB UNIF		SELFWT				
LOAD Z 105 113 -0.375 -0.375										GLOB UNIF		SELFWT				
LOAD Z 113 114 -0.375 -0.375										GLOB UNIF		SELFWT				
LOAD Z 107 114 -0.375 -0.375										GLOB UNIF		SELFWT				
LOADCN 2																
LOAD Z 101 105 -0.250 -0.250										GLOB UNIF		LIVE1				
LOAD Z 103 117 -0.250 -0.250										GLOB UNIF		LIVE1				
LOAD Z 111 113 -0.500 -0.500										GLOB UNIF		LIVE1				
LOAD Z 112 114 -0.500 -0.500										GLOB UNIF		LIVE1				
LOAD Z 117 107 -0.250 -0.250										GLOB UNIF		LIVE1				
LOAD 103 0.62500										GLOB JOIN		LIVE2				
LOAD 115 0.62500										GLOB JOIN		LIVE2				
LOAD 116 0.62500										GLOB JOIN		LIVE2				
LOAD 117 0.62500										GLOB JOIN		LIVE2				
LOADCN 3																
LOAD X 1 101 0.278 0.278										GLOB UNIF		WIND-X				
LOAD X 5 105 0.278 0.278										GLOB UNIF		WIND-X				
LCOMB																
Q	LCOMB 4 1 1.40 2 1.70															
	LCOMB 5 1 1.05 2 1.2753 1.275															
	LCOMB 6 1 0.90 3 1.30															
END																

Note: The above model file was created using the Precede program. Precede is a highly efficient 3D graphics interactive modeler that can create SACS concrete and/or steel models including geometry,

properties, loading, analysis options and code check parameters and incorporates extensive data checking features.

The following is a description of selected input lines used the SACS model file for Sample Problem 1. The lines are referenced by the letter in the left margin of the input listing.

- A. The OPTIONS line specifies the analysis options, namely:
 - a. English units are designated by 'EN' in columns 14-15.
 - b. Code check for steel elements will be based on AISC/API LRFD code ('LR' in columns 25-26).
 - c. Prismatic beam elements will be divided into two post processing segments and each segment of segmented elements will be considered as a post processing segment by '2' and '1' in columns 30 and 32.
 - d. For steel elements, the stress corresponding to the maximum unity check ratio will be reported ('PT' in columns 49-50).
 - e. Member end forces and joint reactions will also be reported.
- B. The concrete analysis options are designated on the CNCOPT input line as follows:
 - a. 'BR' in columns 13-14 designates that a first-order linear analysis is to be performed and the structure is to be considered as braced against sidesway.
 - b. Member detailed reports for concrete columns and beams along with element unity check and unity check range reports are requested.
- C. Load condition one is classified as a 'DEAD' load case in columns 47-50. Loading in this load case will be assumed to cause no appreciable sidesway in the structure.
- D. The reinforcement pattern named BOX1 is defined using the SECT input line with 'BOX1' in columns 6-12.
 - a. The pattern is designated as a box pattern type by 'BRP' in columns 16-18.
 - b. The rebar diameter, 0.50, and the number of rebars in a horizontal row and vertical column are specified as 4 and 2 respectively.
 - c. The width and height of the box, measured from the centers of the outer most rebars, are specified as 10.0 and 19.0 in columns 67-71 and 72-76, respectively.

E. The circular reinforcement pattern named CIRCLE is defined using the SECT input line with 'CIRCLE' in columns 6-12.

- a. The pattern is designated as a circular pattern type by 'CRP' in columns 16-18.
- b. The rebar diameter, 0.75, and the number of rebars, 8, are specified in columns 50-55 and 56-60, respectively.
- c. The pattern diameter and the angle from the local Z axis to the first clock-wise rebar are designated as 11.25 and 0.0 in columns 67-71 and 72-76, respectively.

F. The reinforcement pattern named ROW1 is defined using the SECT input line with 'ROW1' in columns 6-12.

- a. The pattern is designated as a row pattern type by 'RRP' in columns 16-18.
- b. The rebar diameter, the number of rebars in a horizontal row, and the width measured from the centers of the outer most rebars, are specified as 0.50, 4 and 10.0 in columns 50-55, 56-60 and 67-71, respectively.

G. The reinforcement pattern named ROW2 is defined using the SECT input line with 'ROW2' in columns 6-12.

- a. The pattern is designated as a row pattern type by 'RRP' in columns 16-18.
- b. The rebar diameter, the number of rebars in the row, and the width measured from the centers of the outer most rebars, are specified as 0.375, 2 and 10.0 in columns 50-55, 56-60 and 67-71, respectively.

H. The circular cross section named CYLIND is defined by the SECT input line with 'CYLIND' in columns 6-12.

- a. The cross section shape is stipulated as circular by 'CCS' in columns 16-18. The diameter is entered in columns 50-55.
- b. The cross section is reinforced, 'R' in column 15, with reinforcement patterns designated on the 'SECT2' line immediately following.

I. The 'SECT2' line defines the reinforcement applied to the cross section named CYLIND which is defined on the previous 'SECT' line.

-
- a. The CYLIND cross section is reinforced using rebar pattern named CIRCLE as designated in columns 9-15.
 - b. The distance from the top of the cross section to the center of rebar pattern CIRCLE and the yield strength of the rebars are designated as 7.50 and 60.0 in columns 16-20 and 21-25, respectively.

J. The rectangular cross section named RECT1 is defined by the SECT line with 'RECT1' in columns 6-12.

- a. The cross section shape is stipulated as rectangular by 'CRS' in columns 16-18. The height and width, 24 and 15, are input in columns 50-55 and 56-60, respectively.
- b. The cross section is reinforced, 'R' in column 15, with reinforcement patterns designated on the 'SECT2' line immediately following.

K. The 'SECT2' input line defines the reinforcement applied to the cross section named RECT1 which is defined on the previous 'SECT' line.

- a. The RECT1 cross section is reinforced using rebar patterns ROW1 and ROW2 as designated in columns 9-15 and 33-39, respectively.
- b. The distance from the top of the cross section to the center of rebar pattern ROW1 and the yield strength of the rebars are designated as 2.50 and 60.0 in columns 16-20 and 21-25, respectively.
- c. The distance from the top of the cross section to the center of rebar pattern ROW2 and the yield strength of the rebars are designated as 22.50 and 60.0 in columns 40-44 and 45-49, respectively.

L. The rectangular cross section named RECT2 is defined by the SECT input line with 'RECT2' in columns 6-12.

- a. The cross section shape is stipulated as rectangular by 'CRS' in columns 16-18. The height and width, 24 and 15, are input in columns 50-55 and 56-60, respectively.
- b. The cross section is reinforced, 'R' in column 15, with reinforcement patterns designated on the 'SECT2' line immediately following.

M. The 'SECT2' line defines the reinforcement applied to the cross section named RECT2 which is defined on the previous 'SECT' line.

- a. The RECT2 cross section is reinforced using rebar pattern BOX1 as designated in columns 9-15.

b. The distance from the top of the cross section to the center of rebar pattern BOX1 and the yield strength of the rebars are designated as 12.0 and 60.0 in columns 16-20 and 21-25, respectively.

N. Concrete properties are defined on GRUP input lines with 'C' specified in column 5. The properties for all members assigned to concrete group named CC1 are defined on the GRUP line with 'CC1' specified in cols. 6-8.

- a. The cross section is defined by section CYLIND.
- b. The reinforcement steel elastic modulus, and density are specified in columns 18-23 and 24-29, respectively.
- c. The concrete elastic modulus will be calculated by the program and the density and compressive strength are specified as 145.0 and 4.0.
- d. Members assigned to group CC1 will be considered braced column elements as designated by '1' in column 47.
- e. $\frac{3}{8}$ diameter spiral shear reinforcement at 24.0 spacing is specified by 24.0, S and 0.375 in columns 65-69, 70 and 71-76, respectively. The yield stress of the shear reinforcement is defined as 60.0 in columns 60-64.

O. The properties for all members assigned to the segmented concrete group named CR1 are defined on the GRUP input lines with 'CR1' specified in cols. 6-8

- a. The cross section for the first segment is defined by section RECT1.
- b. Members assigned to group CR1 will be considered beam elements as designated by '3' in column 47.
- c. For the first segment, $\frac{3}{8}$ diameter stirrup or tied shear reinforcement at 12.0 spacing is specified by 12.0, T and 0.375 in columns 65-69, 70 and 71-76, respectively. The yield stress of the shear reinforcement is defined as 60.0 in columns 60-64.
- d. The length of the first segment is 3.0 (columns 77-80).
- e. The cross section for the second segment is defined by section RECT2 as specified in columns 10-16 on the second group line with 'CR1' entered in columns 6-8.
- f. For the second segment, $\frac{3}{8}$ diameter stirrup or tied shear reinforcement at 18.0 spacing is specified by 18.0, T and 0.375 in

columns 65-69, 70 and 71-76, respectively. The yield stress of the shear reinforcement is defined as 60.0 in columns 60-64.

g. The length of the second segment is left blank.

P. The properties for all members assigned to the segmented concrete group named CR2 are defined on the GRUP lines with 'CR2' specified in cols. 6-8.

a. The cross section for the first and third segment is defined by section RECT2.

b. Members assigned to group CR2 will be considered beam elements as designated by '3' in column 47.

c. For the first and third segments, $\frac{3}{8}$ diameter stirrup or tied shear reinforcement at 18.0 spacing is specified by 18.0, T and 0.375 in columns 65-69, 70 and 71-76, respectively. The yield stress of the shear reinforcement is defined as 60.0 in columns 60-64.

d. The length of the first and third segments is 1.67 as specified in columns 77-80 on the first and third group input lines defining group CR2.

e. The cross section for the second segment is defined by section RECT3 as specified in columns 10-16 on the second group line with 'CR2' entered in columns 6-8.

f. No shear reinforcement is defined for the second segment.

g. The length of the second segment is left blank.

Q. Load combinations made up of one or more load cases and/or combinations are defined after the LCOMB header line. Load case 4 is a combination consisting of load case 1 multiplied by 1.4 and 1.7 times load case 2.

The ensuing is a portion of the listing file for the analysis and code check of this sample problem.

***** CONCRETE CIRCULAR SECTION PROPERTIES *****

											***** EFFECTIVE CROSS SECTION PROPERTIES *****				
GRUP	SECTION	OD	*** STEEL ***	*****	CONCRETE	*****	MEMB	K-FACTOR		SECT	AXIAL	***** MOMENTS OF INERTIA *****			
			ES	DENS	EC	DENS	FC	CLASS	Y	Z	LEN	AREA	X-X	Y-Y	Z-Z
			1000		1000										
		IN	KSI	LB/FT3	KSI	LB/FT3	KSI			FT	IN**2	IN**4	IN**4	IN**4	
CC1	CYLIND	15.0	29.0	490.00	3.6	145.00	4.00	CL	1.0	1.0	0.0	176.71	4970.1	971.6	971.6

**** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE CIRCULAR ****

* SHEAR REINFORCEMENT *						***** REINFORCEMENT BARS *****									
GRUP	SECTION	FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR	AXIAL	***** MOMENTS OF INERTIA *****			
										DIA	AREA	X-X	Y-Y	Z-Z	
		KSI	IN		IN		IN	KSI		IN	IN**2	IN**4	IN**4	IN**4	
CC1	CYLIND	60.00	24.0	S	0.375	CIRCLE	7.50	60.00	S	0.75	3.53	0.2	56.0	56.0	

***** CONCRETE RECTANGULAR SECTION PROPERTIES *****

***** EFFECTIVE CROSS SECTION PROPERTIES *****																
GRUP	SECTION	HT	WT	*** STEEL ***		***** CONCRETE *****		MEMB	K-FACTOR SECT			AXIAL	***** MOMENTS OF INERTIA *****			
				ES	DENS	EC	DENS		CLASS	Y	Z		LEN	X-X	Y-Y	Z-Z
				1000		1000										
		IN	IN	KSI	LB/FT3	KSI	LB/FT3	KSI				FT	IN**2	IN**4	IN**4	IN**4
CR1	RECT1	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	3.0	360.00	16503.8	8640.0	3375.0
CR1	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0
CR2	RECT3	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0

**** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE RECTANGULAR ****

* SHEAR REINFORCEMENT *						***** REINFORCEMENT BARS *****									
GRUP	SECTION	FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR DIA	AXIAL AREA	***** X-X	MOMENTS OF INERTIA Y-Y	***** Z-Z	
		KSI	IN		IN		IN	KSI		IN	IN**2	IN**4	IN**4	IN**4	
CR1	RECT1	60.00	12.0	T	0.375	ROW1	2.50	60.00	S	0.50	0.79	0.0	0.0	10.9	
CR1	RECT1	60.00	12.0	T	0.375	ROW2	22.50	60.00	S	0.38	0.23	0.0	0.0	5.7	

***** CONCRETE CIRCULAR SECTION PROPERTIES *****																
***** EFFECTIVE CROSS SECTION PROPERTIES *****																
GRUP	SECTION	OD	*** ES 1000	STEEL *** DENS	***** EC 1000	CONCRETE ***** DENS	MEMB FC	K-FACTOR CLASS	SECT Y	Z	LEN	AXIAL AREA	***** X-X	MOMENTS OF INERTIA Y-Y	***** Z-Z	
		IN	KSI	LB/FT3	KSI	LB/FT3	KSI				FT	IN**2	IN**4	IN**4	IN**4	
CC1	CYLIND	15.0	29.0	490.00	3.6	145.00	4.00	CL	1.0	1.0	0.0	176.71	4970.1	971.6	971.6	
***** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE CIRCULAR *****																
* SHEAR REINFORCEMENT * ***** REINFORCEMENT BARS *****																
GRUP	SECTION	FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR DIA	AXIAL AREA	***** X-X	MOMENTS OF INERTIA Y-Y	***** Z-Z	*****	
		KSI	IN		IN		IN	KSI		IN	IN**2	IN**4	IN**4	IN**4		
CC1	CYLIND	60.00	24.0	S	0.375	CIRCLE	7.50	60.00	S	0.75	3.53	0.2	56.0	56.0		
***** CONCRETE RECTANGULAR SECTION PROPERTIES *****																
***** EFFECTIVE CROSS SECTION PROPERTIES *****																
GRUP	SECTION	HT	WT	*** ES 1000	STEEL *** DENS	***** EC 1000	CONCRETE ***** DENS	MEMB FC	K-FACTOR CLASS	SECT Y	Z	LEN	AXIAL AREA	***** X-X	MOMENTS OF INERTIA Y-Y	***** Z-Z
		IN	IN	KSI	LB/FT3	KSI	LB/FT3	KSI				FT	IN**2	IN**4	IN**4	IN**4
CR1	RECT1	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	3.0	360.00	16503.8	8640.0	3375.0
CR1	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0
CR2	RECT3	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0
***** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE RECTANGULAR *****																
* SHEAR REINFORCEMENT * ***** REINFORCEMENT BARS *****																
GRUP	SECTION	FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR DIA	AXIAL AREA	***** X-X	MOMENTS OF INERTIA Y-Y	***** Z-Z	*****	
		KSI	IN		IN		IN	KSI		IN	IN**2	IN**4	IN**4	IN**4		
CR1	RECT1	60.00	12.0	T	0.375	ROW1	2.50	60.00	S	0.50	0.79	0.0	0.0	10.9		
CR1	RECT1	60.00	12.0	T	0.375	ROW2	22.50	60.00	S	0.38	0.23	0.0	0.0	5.7		
CR1	RECT2	60.00	18.0	T	0.375	BOX1	12.00	60.00	S	0.50	1.57	0.0	141.8	21.8		
CR2	RECT2	60.00	18.0	T	0.375	BOX1	12.00	60.00	S	0.50	1.57	0.0	141.8	21.8		
CR2	RECT3	50.00	0.0	T	0.000	ROW1	22.50	60.00	S	0.50	0.79	0.0	0.0	10.9		
CR2	RECT3	50.00	0.0	T	0.000	ROW2	2.50	60.00	S	0.38	0.23	0.0	0.0	5.7		
CR2	RECT2	60.00	18.0	T	0.375	BOX1	12.00	60.00	S	0.50	1.57	0.0	141.8	21.8		

*** SACS POST PROCESSOR COMMENTS ***

** THE USER SHOULD TAKE NOTE OF THE FOLLOWING COMMENTS REGARDING THE SACS POST PROCESSOR OUTPUT **

BEAMS

- (1) INTERNAL LOADS FOR MEMBERS ARE PRESENTED IN THE CLASSICAL ENGINEERING SIGN CONVENTION AS DESCRIBED BY TIMOSHENKO
- (2) IF THE AXIAL LOAD ON A MEMBER EXCEEDS THE AISC ALLOWABLE BUCKLING LOAD, THEN THE AXIAL UNITY CHECK VALUE FOR THE MEMBER IS SET EQUAL TO 100 TO INDICATE THAT THE MEMBER HAS BUCKLED
- (3) THE MAXIMUM COMBINED UNITY CHECK CAN BE THE MAXIMUM SHEAR UNITY CHECK IF IT IS GREATER THAN THE MAXIMUM UNITY CHECK DUE TO BENDING AND AXIAL LOAD
- (4) THE FOLLOWING ABBREVIATIONS ARE USED TO DESCRIBE THE CRITICAL UNITY CHECK CONDITIONS:
 - SHEAR - SHEAR
 - CM+BN - COMPRESSION WITH BENDING
 - TN+BN - TENSION WITH BENDING
 - WEB-SH - WEB SHEAR
 - FLG-SH - FLANGE SHEAR
 - EULER - EULER BUCKLING
- (5) THE FOLLOWING ABBREVIATIONS ARE USED TO DESCRIBE THE CRITICAL UNITY CHECK CONDITIONS FOR CONCRETE:
 - COLBUC - COLUMN BUCKLING
 - CM+BN - COMPRESSION WITH BENDING IN COLUMN ELEMENT
 - TN+BN - TENSION WITH BENDING IN COLUMN ELEMENT
 - SHEAR - SHEAR
 - TORS - TORSION
 - BEND-Y - PURE BENDING IN BEAM ELEMENT ABOUT LOCAL Y AXIS
 - REINF - REINFORCEMENT RATIO

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE COLUMN ELEMENTS

MEMBER	GRP	LOAD CASE	DIST FROM		AXIAL LOAD KIPS	RESULT MOMENT IN-KIP	MOMENT ANGLE DEG	MOM MAGN-IFIER	SHEAR FY KIPS	SHEAR FZ KIPS	TORSION MX IN-KIP	***** ALLOWABLES *****			***** UNITS *****		
			END FT	LOAD KIPS								AXIAL KIPS	BEND IN-KIP	SHEAR-Y KIPS	SHEAR-Z KIPS	TORS. IN-KIP	CHECK RATIO
1- 101 CC1	4	0.0	-27.1	0.0	0.000	1.000	2.3	2.9	0.0	510.55	0.00	22.54	22.54	38.6	0.13		
	5	0.0	-19.6	0.0	0.000	1.000	-0.6	2.2	0.0	510.55	0.00	22.19	22.19	38.6	0.10		
	6	0.0	-8.3	0.0	0.000	1.000	-1.7	0.6	0.0	510.55	0.00	21.66	21.66	38.6	0.08		
	4	4.5	-25.8	199.2	37.778	1.000	2.3	2.9	0.0	147.32	1136.62	22.48	22.48	38.6	0.18		
	5	4.5	-18.7	118.6	5.497	1.000	1.0	2.2	0.0	179.85	1144.26	22.14	22.14	38.6	0.10		
	6	4.5	-7.4	58.8	-54.389	1.000	-0.1	0.6	0.0	142.93	1130.57	21.62	21.62	38.6	0.05		
	4	9.0	-24.5	398.4	37.778	1.000	2.3	2.9	0.0	58.67	953.27	22.41	22.41	38.6	0.42		
	5	9.0	-17.7	260.0	24.746	1.000	2.6	2.2	0.0	64.83	953.18	22.09	22.09	38.6	0.27		
6	9.0	-6.6	68.9	-6.475	1.000	1.6	0.6	0.0	103.00	1076.07	21.58	21.58	38.6	0.07			
3- 103 CC1	4	0.0	-23.4	0.0	0.000	1.000	-2.3	2.9	0.0	510.55	0.00	22.36	22.36	38.6	0.13		
	5	0.0	-18.2	0.0	0.000	1.000	-2.6	2.2	0.0	510.55	0.00	22.12	22.12	38.6	0.12		
	6	0.0	-10.2	0.0	0.000	1.000	-1.6	0.6	0.0	510.55	0.00	21.75	21.75	38.6	0.07		
	4	4.5	-21.9	199.4	-37.808	1.000	-2.3	2.9	0.0	122.19	1113.30	22.29	22.29	38.6	0.18		
	5	4.5	-17.1	183.7	-49.981	1.000	-2.6	2.2	0.0	99.51	1068.00	22.07	22.07	38.6	0.17		
	6	4.5	-9.2	90.6	-67.812	1.000	-1.6	0.6	0.0	110.53	1087.35	21.70	21.70	38.6	0.08		
	4	9.0	-20.4	398.8	-37.808	1.000	-2.3	2.9	0.0	49.38	967.79	22.22	22.22	38.6	0.41		
	5	9.0	-16.0	367.5	-49.981	1.000	-2.6	2.2	0.0	42.38	973.76	22.02	22.02	38.6	0.38		
6	9.0	-8.3	181.3	-67.812	1.000	-1.6	0.6	0.0	44.34	970.53	21.66	21.66	38.6	0.19			
7- 107 CC1	4	0.0	-26.9	0.0	0.000	1.000	-2.2	-2.9	0.0	510.55	0.00	22.53	22.53	38.6	0.13		
	5	0.0	-20.9	0.0	0.000	1.000	-2.6	-2.2	0.0	510.55	0.00	22.25	22.25	38.6	0.12		
	6	0.0	-10.3	0.0	0.000	1.000	-1.6	-0.6	0.0	510.55	0.00	21.75	21.75	38.6	0.07		
	4	4.5	-25.2	198.8	217.671	1.000	-2.2	-2.9	0.0	143.65	1133.50	22.45	22.45	38.6	0.18		
	5	4.5	-19.6	183.2	229.903	1.000	-2.6	-2.2	0.0	118.52	1105.52	22.19	22.19	38.6	0.17		
	6	4.5	-9.2	90.7	247.816	1.000	-1.6	-0.6	0.0	110.01	1086.66	21.70	21.70	38.6	0.08		
	4	9.0	-23.5	397.5	217.671	1.000	-2.2	-2.9	0.0	56.75	956.79	22.37	22.37	38.6	0.41		
	5	9.0	-18.4	366.4	229.903	1.000	-2.6	-2.2	0.0	48.57	966.78	22.13	22.13	38.6	0.38		
6	9.0	-8.1	181.4	247.816	1.000	-1.6	-0.6	0.0	43.37	970.63	21.65	21.65	38.6	0.19			
5- 105 CC1	4	0.0	-28.0	0.0	0.000	1.000	2.3	-2.9	0.0	510.55	0.00	22.58	22.58	38.6	0.13		
	5	0.0	-20.3	0.0	0.000	1.000	-0.6	-2.2	0.0	510.55	0.00	22.22	22.22	38.6	0.10		
	6	0.0	-9.0	0.0	0.000	1.000	-1.7	-0.6	0.0	510.55	0.00	21.69	21.69	38.6	0.08		
	4	4.5	-26.2	199.1	142.339	1.000	2.3	-2.9	0.0	149.62	1138.30	22.49	22.49	38.6	0.17		
	5	4.5	-18.9	118.8	174.639	1.000	1.0	-2.2	0.0	182.37	1144.41	22.15	22.15	38.6	0.10		
	6	4.5	-7.8	58.8	234.387	1.000	-0.1	-0.6	0.0	151.26	1137.71	21.63	21.63	38.6	0.05		
	4	9.0	-24.3	398.3	142.339	1.000	2.3	-2.9	0.0	58.28	954.09	22.40	22.40	38.6	0.42		
	5	9.0	-17.5	260.1	155.389	1.000	2.6	-2.2	0.0	64.17	954.25	22.09	22.09	38.6	0.27		
6	9.0	-6.6	68.9	186.439	1.000	1.6	-0.6	0.0	103.00	1076.06	21.58	21.58	38.6	0.07			

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE BEAM ELEMENTS

MEMBER	GRP	LOAD CASE	DIST FROM END FT	AXIAL LOAD KIPS	MOMENT MY IN-KIP	SHEAR FZ KIPS	TORSION MX IN-KIP	REINF. RATIO	MOMENT MY IN-KIP	SHEAR FZ KIPS	TORSION MX IN-KIP	REINF. RATIO	CRIT. COND	UNITY CHECK RATIO
101- 105 CRL	4	0.0	0.0	-2.9	-112.7	7.4	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56
	5	0.0	0.0	-2.2	-84.5	5.5	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42
	6	0.0	0.0	-0.6	-38.5	2.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15
	4	10.0	-2.9	332.7	0.0	0.0	-2.5	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.53
	5	10.0	-2.2	249.5	0.0	0.0	-1.8	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.15
	6	10.0	-0.6	80.0	0.0	0.0	0.1	0.005	217.5	8.2	74.6	0.033	BEND-Y	0.37
	4	20.0	-2.9	-101.6	-7.3	0.0	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56
	5	20.0	-2.2	-76.2	-5.5	0.0	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42
	6	20.0	-0.6	-39.0	-2.0	0.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15
101- 111 CR1	4	0.0	0.0	-2.3	-273.6	17.1	-248.8	0.004	894.9	29.8	340.3	0.022	TORS	0.73
	5	0.0	0.0	-2.6	-141.9	12.1	-186.6	0.004	894.9	29.2	343.2	0.022	TORS	0.54
	6	0.0	0.0	-1.6	-10.8	4.6	-40.1	0.004	894.9	43.4	306.2	0.022	TORS	0.13
	4	3.0	-2.3	315.0	15.6	0.0	-248.8	0.010	348.8	15.5	282.0	0.024	SHEAR	1.01
	5	3.0	-2.6	273.7	11.0	0.0	-186.6	0.010	348.8	15.2	287.6	0.024	BEND-Y	0.78
	6	3.0	-1.6	136.7	3.6	0.0	-40.1	0.010	348.8	17.8	250.2	0.024	BEND-Y	0.39
	4	3.0	-2.3	315.0	15.6	0.0	-248.8	0.009	925.4	19.1	271.2	0.024	TORS	0.92
	5	3.0	-2.6	273.7	11.0	0.0	-186.6	0.009	925.4	18.7	275.3	0.024	TORS	0.68
	6	3.0	-1.6	136.7	3.6	0.0	-40.1	0.009	925.4	26.9	243.5	0.024	TORS	0.16
	4	6.7	-2.3	957.4	13.6	0.0	-248.8	0.009	925.4	18.1	279.5	0.024	BEND-Y	1.03
	5	6.7	-2.6	723.9	9.5	0.0	-186.6	0.009	925.4	17.6	283.5	0.024	BEND-Y	0.78
	6	6.7	-1.6	267.5	2.4	0.0	-40.1	0.009	925.4	26.9	275.2	0.024	BEND-Y	0.29
103- 112 CR1	4	0.0	0.0	-2.3	-210.6	16.6	314.7	0.004	894.9	27.2	351.7	0.022	TORS	0.89
	5	0.0	0.0	-2.6	-266.9	13.2	236.0	0.004	894.9	27.7	349.7	0.022	TORS	0.67
	6	0.0	0.0	-1.6	-190.7	6.1	36.7	0.004	894.9	43.4	274.3	0.022	BEND-Y	0.21
	4	3.0	-2.3	359.0	15.0	0.0	314.7	0.010	348.8	14.5	305.1	0.024	SHEAR	1.04
	5	3.0	-2.6	186.1	12.0	0.0	236.0	0.010	348.8	14.7	300.0	0.024	SHEAR	0.82
	6	3.0	-1.6	10.7	5.1	0.0	36.7	0.010	348.8	17.8	216.4	0.024	SHEAR	0.29
	4	3.0	-2.3	359.0	15.0	0.0	314.7	0.009	925.4	17.1	286.9	0.024	TORS	1.10
	5	3.0	-2.6	186.1	12.0	0.0	236.0	0.009	925.4	17.6	283.7	0.024	TORS	0.83
	6	3.0	-1.6	10.7	5.1	0.0	36.7	0.009	925.4	26.9	205.1	0.024	SHEAR	0.19
	4	6.7	-2.3	978.2	13.1	0.0	314.7	0.009	925.4	16.2	293.2	0.024	TORS	1.07
	5	6.7	-2.6	682.1	10.6	0.0	236.0	0.009	925.4	16.7	290.1	0.024	TORS	0.81
	6	6.7	-1.6	207.4	3.9	0.0	36.7	0.009	925.4	26.9	229.9	0.024	BEND-Y	0.22

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE BEAM ELEMENTS

MEMBER	GRP	LOAD CASE	DIST	***** APPLIED *****			***** ALLOWABLES *****				CRIT. COND	UNITY CHECK RATIO		
			FROM	AXIAL	MOMENT	SHEAR	TORSION	REINF.	MOMENT	SHEAR			TORSION	REINF.
			END FT	LOAD KIPS	MY IN-KIP	FZ KIPS	MX IN-KIP	RATIO	MY IN-KIP	FZ KIPS			MX IN-KIP	RATIO
105- 113 CR1	4	0.0	-2.3	-267.9	17.0	260.3	0.004	894.9	29.3	343.0	0.022	TORS	0.76	
	5	0.0	-2.6	-137.7	12.0	195.2	0.004	894.9	28.7	345.6	0.022	TORS	0.56	
	6	0.0	-1.6	-11.0	4.6	39.5	0.004	894.9	43.4	305.0	0.022	TORS	0.13	
	4	3.0	-2.3	315.8	15.4	260.3	0.010	348.8	15.3	286.7	0.024	SHEAR	1.01	
	5	3.0	-2.6	274.3	10.9	195.2	0.010	348.8	15.0	292.3	0.024	BEND-Y	0.79	
	6	3.0	-1.6	136.7	3.6	39.5	0.010	348.8	17.8	248.9	0.024	BEND-Y	0.39	
	4	3.0	-2.3	315.8	15.4	260.3	0.009	925.4	18.7	274.7	0.024	TORS	0.95	
	5	3.0	-2.6	274.3	10.9	195.2	0.009	925.4	18.2	278.6	0.024	TORS	0.70	
	6	3.0	-1.6	136.7	3.6	39.5	0.009	925.4	26.9	242.3	0.024	TORS	0.16	
	4	6.7	-2.3	952.2	13.5	260.3	0.009	925.4	17.7	282.6	0.024	BEND-Y	1.03	
	5	6.7	-2.6	720.0	9.4	195.2	0.009	925.4	17.2	286.5	0.024	BEND-Y	0.78	
	6	6.7	-1.6	267.7	2.4	39.5	0.009	925.4	26.9	274.3	0.024	BEND-Y	0.29	
111- 112 CR2	4	0.0	-2.3	954.5	2.0	32.9	0.009	925.4	26.9	272.7	0.024	BEND-Y	1.03	
	5	0.0	-2.6	721.8	0.8	24.7	0.009	925.4	26.9	302.2	0.024	BEND-Y	0.78	
	6	0.0	-1.6	267.6	0.4	-1.7	0.009	925.4	26.9	165.7	0.024	BEND-Y	0.29	
	4	1.7	-2.3	986.1	1.1	32.9	0.009	925.4	26.9	300.0	0.024	BEND-Y	1.07	
	5	1.7	-2.6	731.0	0.1	24.7	0.009	925.4	26.9	317.2	0.024	BEND-Y	0.79	
	6	1.7	-1.6	269.4	-0.2	-1.7	0.009	925.4	26.9	223.5	0.024	BEND-Y	0.29	
	4	1.7	-2.3	986.1	1.1	32.9	0.004	952.1	29.1	223.9	0.022	BEND-Y	1.04	
	5	1.7	-2.6	731.0	0.1	24.7	0.004	952.1	29.1	232.0	0.022	BEND-Y	0.77	
	6	1.7	-1.6	269.4	-0.2	-1.7	0.004	952.1	29.1	172.6	0.022	BEND-Y	0.28	
	4	5.0	-2.3	996.6	-0.6	32.9	0.004	952.1	29.1	229.7	0.022	BEND-Y	1.05	
	5	5.0	-2.6	710.3	-1.2	24.7	0.004	952.1	29.1	217.1	0.022	BEND-Y	0.75	
	6	5.0	-1.6	239.6	-1.3	-1.7	0.004	952.1	29.1	36.4	0.022	BEND-Y	0.25	
	4	5.0	-2.3	996.6	-0.6	32.9	0.009	925.4	26.9	312.1	0.024	BEND-Y	1.08	
	5	5.0	-2.6	710.3	-1.2	24.7	0.009	925.4	26.9	287.1	0.024	BEND-Y	0.77	
	6	5.0	-1.6	239.6	-1.3	-1.7	0.009	925.4	26.9	109.9	0.024	BEND-Y	0.26	
	4	6.7	-2.3	975.6	-1.5	32.9	0.009	925.4	26.9	289.6	0.024	BEND-Y	1.05	
	5	6.7	-2.6	680.2	-1.8	24.7	0.009	925.4	26.9	258.9	0.024	BEND-Y	0.74	
	6	6.7	-1.6	207.7	-1.9	-1.7	0.009	925.4	26.9	102.6	0.024	BEND-Y	0.22	

SACS-IV SYSTEM CONCRETE ELEMENT UNITY CHECK REPORT
ACI 318R-89 (REVISED 92)

MEMBER	GRP	MAX. CRIT.		LOAD COND NO.	DIST FROM END FT	* * * * *				U N I T Y		C H E C K C O M P O N E N T S										* * * * *				
		UNITY CHECK	COND			*SHEAR* FZ	*TORSION* MX	*REINF* RATIO	*BEAMS* MY	AXIAL FX	EULER Y-Y	BUCKLING Z-Z	SHEAR FY	BEND FY	MOM MAGN	CMY	CMZ	C O L U M N S								
1-	101	CC1	0.42	CM+BN	4	9.00	0.130	0.000	0.559			0.418	0.010	0.010	0.101	0.418	1.00	0.60	0.60							
3-	103	CC1	0.41	CM+BN	4	9.00	0.131	0.000	0.559			0.412	0.008	0.008	0.102	0.412	1.00	0.60	0.60							
7-	107	CC1	0.41	CM+BN	4	9.00	0.130	0.000	0.559			0.415	0.010	0.010	0.101	0.415	1.00	0.60	0.60							
5-	105	CC1	0.42	CM+BN	4	9.00	0.130	0.000	0.559			0.417	0.010	0.010	0.101	0.417	1.00	0.60	0.60							
101-	105	CRL	1.53	BEND-Y	4	10.00	0.006	0.031	0.161	1.529																
103-	117	CLL	0.86	BEND-Y	4	5.00	0.251	0.126	0.161	0.856																
101-	111	CR1	1.03	BEND-Y	4	6.67	0.752	0.890	0.368	1.035																
103-	112	CR1	1.10	TORS	4	3.00	0.877	1.097	0.368	0.388																
105-	113	CR1	1.03	BEND-Y	4	6.67	0.761	0.921	0.368	1.029																
107-	114	CR1	1.05	BEND-Y	4	6.67	0.773	0.978	0.368	1.048																
111-	112	CR2	1.08	BEND-Y	4	5.00	0.023	0.105	0.368	1.077																
113-	114	CR2	1.07	BEND-Y	4	5.00	0.024	0.040	0.368	1.074																
111-	113	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136																
112-	114	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136																
117-	107	CLL	1.38	BEND-Y	4	7.50	0.330	0.704	0.161	1.384																

SACS-IV CONCRETE MEMBER UNITY CHECK RANGE SUMMARY

GROUP I - UNITY CHECKS GREATER THAN 1.33

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****								SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST	
		UNITY CHECK	LOAD CASE		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS	KLX/RY	KLZ/RZ	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE			
101-	105	CRL	1.53	4	10.0	-2.9	332.7	0.8	0.0	52.26	35.55	1.15	5	0.56	4			
117-	107	CLL	1.38	4	7.5	-2.9	301.1	-0.8	0.0	39.20	26.66	1.04	5	0.86	4			

GROUP II - UNITY CHECKS GREATER THAN 1.00 AND LESS THAN 1.33

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****						SLENDERNES RATIO		SECOND-HIGHEST		THIRD-HIGHEST	
		UNITY	LOAD		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS		KLY/R	KLZ/R	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE
101-	111	CR1	1.03	4	6.7	-2.3	957.4	1.5	0.0	13.6	11.55	18.48	1.01	4	0.92	4
103-	112	CR1	1.10	4	3.0	-2.3	359.0	-0.1	0.0	15.0	11.55	18.48	1.07	4	1.04	4
105-	113	CR1	1.03	4	6.7	-2.3	952.2	-1.6	0.0	13.5	11.55	18.48	1.01	4	0.95	4
107-	114	CR1	1.05	4	6.7	-2.3	969.7	1.5	0.0	13.1	11.55	18.48	1.01	4	1.00	4
111-	112	CR2	1.08	4	5.0	-2.3	996.6	1.1	0.0	-0.6	11.55	18.47	1.07	4	1.05	4
113-	114	CR2	1.07	4	5.0	-2.3	994.3	-1.2	0.0	-0.7	11.55	18.47	1.07	4	1.05	4
111-	113	CT1	1.14	4	15.0	0.0	247.2	0.4	0.0	-5.7	52.26	43.86	1.14	4	1.11	4
112-	114	CT1	1.14	4	15.0	0.0	247.1	-0.3	0.0	-5.7	52.26	43.86	1.14	4	1.11	4

GROUP III - UNITY CHECKS GREATER THAN 0.00 AND LESS THAN 0.50

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****						SLENDERNES RATIO		SECOND-HIGHEST		THIRD-HIGHEST	
		UNITY	LOAD		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS		KLY/R	KLZ/R	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE
1-	101	CC1	0.42	4	9.0	-24.5	314.9	244.1	2.3	2.9	28.80	28.80	0.27	5	0.18	4
3-	103	CC1	0.41	4	9.0	-20.4	315.1	-244.5	-2.3	2.9	28.80	28.80	0.38	5	0.19	6
7-	107	CC1	0.41	4	9.0	-23.5	-314.7	-242.9	-2.2	-2.9	28.80	28.80	0.38	5	0.19	6
5-	105	CC1	0.42	4	9.0	-24.3	-315.3	243.3	2.3	-2.9	28.80	28.80	0.27	5	0.17	4

SACS-IV SYSTEM CONCRETE MEMBER GROUP SUMMARY REPORT
ACI 318R-89 (REVISED 92)

GRP ID	CRITICAL MEMBER	LOAD COND	MAX.		DIST FROM END FT	***** APPLIED LOADS *****				***** ALLOWABLE LOADS *****					CRIT COND	EFFECTIVE KL/R	
			UNITY CHECK	LOAD		AXIAL KIPS	BEND IN-KIP	TORSION IN-KIP	REINF RATIO	AXIAL KIPS	EULER KIPS	BEND IN-KIP	TORSION IN-KIP	REINF RATIO		Y	Z
CC1	1-	101	4	0.42	9.0	-24.51	398.4	0.0	0.017	58.7	2412.7	953.3	38.6	0.031	CM+BN	28.8	28.8
CRL	101-	105	4	1.53	10.0	-2.89	332.7	-2.5	0.005			217.5	78.0	0.033	BEND-Y	52.3	35.6
CLL	117-	107	4	1.38	7.5	-2.89	301.1	53.9	0.005			217.5	76.6	0.033	BEND-Y	39.2	26.7
CR1	103-	112	4	1.10	3.0	-2.26	359.0	314.7	0.009			925.4	286.9	0.024	TORS	11.5	18.5
CR2	111-	112	4	1.08	5.0	-2.26	996.6	32.9	0.009			925.4	312.1	0.024	BEND-Y	11.5	18.5
CT1	111-	113	4	1.14	15.0	-0.03	247.2	-2.9	0.005			217.5	2.7	0.033	BEND-Y	52.3	43.9

4.2 SAMPLE PROBLEM 2

Sample Problem 2 is a first-order analysis of the same frame structure used in Sample Problem 1. For this sample problem, however, the structure is assumed to be unbraced against sidesway.

The SACS model file is shown. See Sample Problem 1 for a detailed description of selected portions of the input file:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
	CONCRETE SAMPLE PROBLEM 1							
A	OPTIONS	EN	LR	2 1 0 0	PT	PTPT		
B	CNCOPT	UN		CD BD EL UR CO			1.500	
	LCSEL	4	5	6				
	SECT							
C	SECT BOX1	BRP			0.50	4	210.00	19.00
D	SECT CIRCLE	CRP			0.75	8	11.25	
E	SECT ROW1	RRP			0.50	4	10.00	
F	SECT ROW2	RRP			0.375	2	10.00	
	SECT ROW3	RRP			0.375	4	18.00	
	SECT ROW4	RRP			0.375	2	3.50	
G	SECT CYLIND	RCCS			15.00			
H	SECT2 CIRCLE	7.50	60.00					
	SECT LEFTL	RCLL			16.00	8.000	24.00	5.000
	SECT2 ROW3	2.50	60.00	ROW4	13.50	60.00		
I	SECT RECT1	RCRS			24.00	15.00		
J	SECT2 ROW1	2.50	60.00	ROW2	22.50	60.00		
K	SECT RECT2	RCRS			24.00	15.00		
L	SECT2 BOX1	12.00	60.00					
	SECT RECT3	RCRS			24.00	15.00		
	SECT2 ROW1	22.50	60.00	ROW2	2.50	60.00		
	SECT RIGHTL	RCRL			16.00	8.000	24.00	5.000
	SECT2 ROW3	2.50	60.00	ROW4	13.50	60.00		
	SECT TEE1	RCTS			16.00	8.000	24.00	5.000
	SECT2 ROW3	2.50	60.00	ROW4	13.50	60.00		
	GRUP							
M	GRUPCCC1 CYLIND	29.000	490.00	145.0	4.00	1	1.001.00	0.060.0024.00s 0.375
	GRUPCCLL LEFTL	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0024.00T 0.375
N	GRUPCCR1 RECT1	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0012.00T 0.3753.00
N	GRUPCCR1 RECT2	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0018.00T 0.375
O	GRUPCCR2 RECT2	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0018.00T 0.3751.67
O	GRUPCCR2 RECT3	29.00	490.00	145.0	4.00	3	1.001.00	
O	GRUPCCR2 RECT2	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0018.00T 0.3751.67
	GRUPCCRL RIGHTL	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0024.00T 0.375
	GRUPCCT1 TEE1	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0012.00T 0.3755.00
	GRUPCCT1 TEE1	29.00	490.00	145.0	4.00	3	1.001.00	
	GRUPCCT1 TEE1	29.000	490.00	145.0	4.00	3	1.001.00	0.060.0012.00T 0.3755.00
	GRUP WF1 W6X15			29.00	11.60	36.00	1	1.001.00
								0.50 490.00

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
MEMBER								
MEMBER1 1 101 CC1								
MEMBER OFFSETS							-12.00	
MEMBER1 3 103 CC1								
MEMBER OFFSETS							-12.00	
MEMBER1 7 107 CC1								
MEMBER OFFSETS							-12.00	
MEMBER1 5 105 CC1								
MEMBER OFFSETS							-12.00	
MEMBER1 101 105 CRL								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 103 117 CLL								
MEMBER OFFSETS					4.00		4.00	
MEMBER0 101 111 CR1								
MEMBER0 103 112 CR1								
MEMBER0 105 113 CR1								
MEMBER0 107 114 CR1								
MEMBER0 111 112 CR2								
MEMBER0 113 114 CR2								
MEMBER1 111 113 CT1								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 112 114 CT1								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 103 115 WF1								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 115 116 WF1								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 117 107 CLL								
MEMBER OFFSETS					4.00		4.00	
MEMBER1 117 116 WF1								
MEMBER OFFSETS					4.00		4.00	
PLATE								
PLATE AAAA 103 115 117 116 0.25 1								
PLATE OFFSETS					4.00		4.00	
PLATE OFFSETS					4.00		4.00	
JOINT								
JOINT 1 -10.000-10.000 0.000						111		
JOINT 3 10.000-10.000 0.000						111		
JOINT 5 -10.000 10.000 0.000						111		
JOINT 7 10.000 10.000 0.000						111		
JOINT 101 -10.000-10.000 10.000								
JOINT 103 10.000-10.000 10.000								
JOINT 105 -10.000 10.000 10.000								
JOINT 107 10.000 10.000 10.000								
JOINT 111 -3.333-10.000 10.000								
JOINT 112 3.333-10.000 10.000								
JOINT 113 -3.333 10.000 10.000								
JOINT 114 3.333 10.000 10.000								
JOINT 115 15.000-10.000 10.000								
JOINT 116 15.000 -5.000 10.000								
JOINT 117 10.000 -5.000 10.000								
LOAD								
P LOADCN 1					DEAD			
LOAD Z 1 101		-0.208		-0.208		GLOB UNIF	SELFWT	
LOAD Z 3 103		-0.238		-0.238		GLOB UNIF	SELFWT	
LOAD Z 7 107		-0.268		-0.268		GLOB UNIF	SELFWT	
LOAD Z 5 105		-0.298		-0.298		GLOB UNIF	SELFWT	
LOAD Z 103 115		-0.015		-0.015		GLOB UNIF	SELFWT	
LOAD Z 115 116		-0.015		-0.015		GLOB UNIF	SELFWT	
LOAD Z 117 116		-0.015		-0.015		GLOB UNIF	SELFWT	
LOAD Z 101 105		-0.220		-0.220		GLOB UNIF	SELFWT	

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
LOAD Z	117	107	-0.220	-0.220		GLOB UNIF	SELFWT	
LOAD Z	103	117	-0.220	-0.220		GLOB UNIF	SELFWT	
LOAD Z	101	111	-0.375	-0.375		GLOB UNIF	SELFWT	
LOAD Z	111	112	-0.375	-0.375		GLOB UNIF	SELFWT	
LOAD Z	103	112	-0.375	-0.375		GLOB UNIF	SELFWT	
LOAD Z	105	113	-0.375	-0.375		GLOB UNIF	SELFWT	
LOADCN	2							
LOAD Z	101	105	-0.250	-0.250		GLOB UNIF	LIVE1	
LOAD Z	103	117	-0.250	-0.250		GLOB UNIF	LIVE1	
LOAD Z	111	113	-0.500	-0.500		GLOB UNIF	LIVE1	
LOAD Z	112	114	-0.500	-0.500		GLOB UNIF	LIVE1	
LOAD Z	117	107	-0.250	-0.250		GLOB UNIF	LIVE1	
LOAD	103		0.62500			GLOB JOIN	LIVE2	
LOAD	115		0.62500			GLOB JOIN	LIVE2	
LOAD	116		0.62500			GLOB JOIN	LIVE2	
LOAD	117		0.62500			GLOB JOIN	LIVE2	
LOADCN	3							
LOAD X	1	101	0.278	0.278		GLOB UNIF	WIND-X	
LOAD X	5	105	0.278	0.278		GLOB UNIF	WIND-X	
LDCOMB								
Q	LDCOMB	4 140.00	1 170.00	2				
	LDCOMB	5 105.00	1 127.50	2 127.50	3			
	LDCOMB	6 90.00	1 130.00	3				
	END							

The concrete options on the CNCOPT line are as follows:

- 'UN' in columns 13-14 designates that a first-order linear analysis is to be performed and the structure is to be considered as un-braced against sidesway.
- The sidesway moment magnifier applied to moments due to loading designated as causing appreciable sidesway is specified in columns 71-75.
- Member detailed reports for concrete columns and beams along with element unity check and unity check range reports are requested.
- The internal load report as specified on the OPTIONS line shall contain only concrete elements as designated by 'CO' in columns 48-51.

A portion of the output file for the analysis is listed on the following pages.

CONCRETE SAMPLE PROBLEM 2

DATE 16-MAY-1996 TIME 09:59:26 PRE PAGE 1

PRE VERSION IV.F.002

** PROGRAM OPTIONS **

UNITS

...ENGLISH

EXECUTION

...UNITY CHECK

API LOAD & RESISTANCE FACTOR DESIGN

ACI 318R - 1989 (REVISED 1992)

...NO SEGMENTS FOR PRISMATIC MEMBERS 2

...NO SEGMENTS/SECTION FOR NON-PRISMATIC MEMBERS 1

REPORTS SELECTED

...ELEMENT STRESS AT MAXIMUM UNITY CHECK...PRINT

...ELEMENT INTERNAL LOADS.....PRINT

...JOINT REACTIONS.....PRINT

LOAD

...NO. BASIC LOAD COND. 3

...NO. COMB. LOAD COND. 3

CONCRETE OPTIONS

... ANALYSIS TYPEUNBRACED

... SIDESWAY MOMENT MAGNIFICATION . 1.50

... COLUMN DETAILED PRINTYES

... BEAM DETAILED PRINTYES

... ELEMENT UNITY CHECK OPTIONYES

... UNITY CHECK RANGE REPORTYES

... CONCRETE ONLY PRINTNO

CONCRETE SAMPLE PROBLEM 2

DATE 16-MAY-1996 TIME 10:06:55 PST PAGE 7

				SACS-IV SYSTEM MEMBER FORCES AND MOMENTS					
				KIPS *****			IN-KIPS *****		
MEMBER NUMBER	MEMBER END	GROUP ID	LOAD CASE	FORCE (X)	FORCE (Y)	FORCE (Z)	MOMENT (X)	MOMENT (Y)	MOMENT (Z)
1- 101	1	CC1	4	-27.1344	2.2598	2.9157	0.0000	0.0000	0.0000
			5	-19.6330	-0.5871	2.1867	0.0000	0.0000	0.0000
			6	-8.2669	-1.6982	0.6339	0.0000	0.0000	0.0000
	101		4	-24.5136	2.2598	2.9157	0.0000	314.8914	244.0608
			5	-17.6675	2.6030	2.1867	0.0000	236.1685	108.8576
			6	-6.5821	1.5543	0.6339	0.0000	68.4645	-7.7705
3- 103	3	CC1	4	-23.3619	-2.2635	2.9172	0.0000	0.0000	0.0000
			5	-18.2391	-2.6057	2.1879	0.0000	0.0000	0.0000
			6	-10.1928	-1.5541	0.6338	0.0000	0.0000	0.0000
	103		4	-20.3631	-2.2635	2.9172	0.0000	315.0622	-244.4608
			5	-15.9901	-2.6057	2.1879	0.0000	236.2966	-281.4196
			6	-8.2650	-1.5541	0.6338	0.0000	68.4516	-167.8395
7- 107	7	CC1	4	-26.9198	-2.2495	-2.9135	0.0000	0.0000	0.0000
			5	-20.9076	-2.5952	-2.1852	0.0000	0.0000	0.0000
			6	-10.2756	-1.5551	-0.6341	0.0000	0.0000	0.0000
	107		4	-23.5430	-2.2495	-2.9135	0.0000	-314.6622	-242.9438
			5	-18.3750	-2.5952	-2.1852	0.0000	-235.9967	-280.2818
			6	-8.1048	-1.5551	-0.6341	0.0000	-68.4813	-167.9463
5- 105	5	CC1	4	-28.0397	2.2532	-2.9194	0.0000	0.0000	0.0000
			5	-20.3120	-0.5921	-2.1895	0.0000	0.0000	0.0000
			6	-9.0042	-1.6978	-0.6337	0.0000	0.0000	0.0000
	105		4	-24.2849	2.2532	-2.9194	0.0000	-315.2914	243.3437
			5	-17.4959	2.5980	-2.1895	0.0000	-236.4685	108.3198
			6	-6.5904	1.5548	-0.6337	0.0000	-68.4348	-7.7231
101- 105	101	CRL	4	-2.8879	0.0008	7.3762	-2.4527	-112.6679	0.7503
			5	-2.1659	0.0006	5.5321	-1.8395	-84.5009	0.5627
			6	-0.6275	-0.0001	1.9779	0.0566	-38.5264	0.1882
	105		4	-2.8879	0.0008	-7.2838	-2.4527	-101.5786	0.9454
			5	-2.1659	0.0006	-5.4628	-1.8395	-76.1840	0.7090
			6	-0.6275	-0.0001	-1.9821	0.0566	-39.0294	0.1727
103- 117	103	CLL	4	-2.3590	0.0013	5.7201	-3.5570	-46.9771	-0.7911
			5	-1.7693	0.0009	4.2901	-2.6677	-35.2329	-0.5933
			6	-0.5132	0.0000	2.0624	0.1911	-41.8420	-0.1309
	117		4	-2.3590	0.0013	2.0552	-3.5570	186.2817	-0.7161
			5	-1.7693	0.0009	1.5414	-2.6677	139.7113	-0.5371
			6	-0.5132	0.0000	1.0724	0.1911	52.2019	-0.1329

CONCRETE SAMPLE PROBLEM 2

DATE 16-MAY-1996 TIME 10:06:55 PST PAGE 10

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE COLUMN ELEMENTS

MEMBER	GRP	LOAD CASE	DIST FROM END FT	AXIAL LOAD KIPS	RESULT MOMENT IN-KIP	MOMENT ANGLE DEG	MOM MAGN- IPIER	SHEAR FY KIPS	SHEAR FZ KIPS	TORSION MX IN-KIP	***** ALLOWABLES *****			UNITY CHECK RATIO		
											AXIAL KIPS	*****				
												BEND IN-KIP	SHEAR-Y KIPS		SHEAR-Z KIPS	TORS. IN-KIP
1- 101	CC1	4	0.0	-27.1	0.0	0.000	1.500	2.3	2.9	0.0	510.55	0.00	22.54	22.54	38.6	0.13
		5	0.0	-19.6	0.0	0.000	1.500	-0.6	2.2	0.0	510.55	0.00	22.19	22.19	38.6	0.10
		6	0.0	-8.3	0.0	0.000	1.500	-1.7	0.6	0.0	510.55	0.00	21.66	21.66	38.6	0.08
		4	4.5	-25.8	298.8	37.778	1.500	2.3	2.9	0.0	90.89	1050.57	22.48	22.48	38.6	0.28
		5	4.5	-18.7	177.9	5.497	1.500	1.0	2.2	0.0	115.47	1100.15	22.14	22.14	38.6	0.16
		6	4.5	-7.4	88.2	-54.389	1.500	-0.1	0.6	0.0	88.17	1046.30	21.62	21.62	38.6	0.08
		4	9.0	-24.5	597.6	37.778	1.500	2.3	2.9	0.0	40.16	977.23	22.41	22.41	38.6	0.61
		5	9.0	-17.7	390.1	24.746	1.500	2.6	2.2	0.0	44.00	970.57	22.09	22.09	38.6	0.40
		6	9.0	-6.6	103.4	-6.475	1.500	1.6	0.6	0.0	60.42	949.32	21.58	21.58	38.6	0.11
3- 103	CC1	4	0.0	-23.4	0.0	0.000	1.500	-2.3	2.9	0.0	510.55	0.00	22.36	22.36	38.6	0.13
		5	0.0	-18.2	0.0	0.000	1.500	-2.6	2.2	0.0	510.55	0.00	22.12	22.12	38.6	0.12
		6	0.0	-10.2	0.0	0.000	1.500	-1.6	0.6	0.0	510.55	0.00	21.75	21.75	38.6	0.07
		4	4.5	-21.9	299.1	-37.808	1.500	-2.3	2.9	0.0	73.96	1012.40	22.29	22.29	38.6	0.30
		5	4.5	-17.1	275.6	-49.981	1.500	-2.6	2.2	0.0	59.09	950.78	22.07	22.07	38.6	0.29
		6	4.5	-9.2	135.9	-67.812	1.500	-1.6	0.6	0.0	64.60	953.48	21.70	21.70	38.6	0.14
		4	9.0	-20.4	598.2	-37.808	1.500	-2.3	2.9	0.0	33.28	977.50	22.22	22.22	38.6	0.61
		5	9.0	-16.0	551.2	-49.981	1.500	-2.6	2.2	0.0	28.47	979.98	22.02	22.02	38.6	0.56
		6	9.0	-8.3	271.9	-67.812	1.500	-1.6	0.6	0.0	29.43	967.87	21.66	21.66	38.6	0.28
7- 107	CC1	4	0.0	-26.9	0.0	0.000	1.500	-2.2	-2.9	0.0	510.55	0.00	22.53	22.53	38.6	0.13
		5	0.0	-20.9	0.0	0.000	1.500	-2.6	-2.2	0.0	510.55	0.00	22.25	22.25	38.6	0.12
		6	0.0	-10.3	0.0	0.000	1.500	-1.6	-0.6	0.0	510.55	0.00	21.75	21.75	38.6	0.07
		4	4.5	-25.2	298.2	217.671	1.500	-2.2	-2.9	0.0	88.52	1045.45	22.45	22.45	38.6	0.29
		5	4.5	-19.6	274.8	229.903	1.500	-2.6	-2.2	0.0	72.06	1006.41	22.19	22.19	38.6	0.27
		6	4.5	-9.2	136.0	247.816	1.500	-1.6	-0.6	0.0	64.34	953.91	21.70	21.70	38.6	0.14
		4	9.0	-23.5	596.3	217.671	1.500	-2.2	-2.9	0.0	38.56	977.33	22.37	22.37	38.6	0.61
		5	9.0	-18.4	549.6	229.903	1.500	-2.6	-2.2	0.0	32.84	980.56	22.13	22.13	38.6	0.56
		6	9.0	-8.1	272.1	247.816	1.500	-1.6	-0.6	0.0	28.86	967.61	21.65	21.65	38.6	0.28
5- 105	CC1	4	0.0	-28.0	0.0	0.000	1.500	2.3	-2.9	0.0	510.55	0.00	22.58	22.58	38.6	0.13
		5	0.0	-20.3	0.0	0.000	1.500	-0.6	-2.2	0.0	510.55	0.00	22.22	22.22	38.6	0.10
		6	0.0	-9.0	0.0	0.000	1.500	-1.7	-0.6	0.0	510.55	0.00	21.69	21.69	38.6	0.08
		4	4.5	-26.2	298.7	142.339	1.500	2.3	-2.9	0.0	92.27	1053.67	22.49	22.49	38.6	0.28
		5	4.5	-18.9	178.1	174.639	1.500	1.0	-2.2	0.0	117.20	1103.30	22.15	22.15	38.6	0.16
		6	4.5	-7.8	88.1	234.387	1.500	-0.1	-0.6	0.0	93.79	1058.47	21.63	21.63	38.6	0.08
		4	9.0	-24.3	597.4	142.339	1.500	2.3	-2.9	0.0	39.79	977.17	22.40	22.40	38.6	0.61
		5	9.0	-17.5	390.1	155.389	1.500	2.6	-2.2	0.0	43.46	970.68	22.09	22.09	38.6	0.40
		6	9.0	-6.6	103.3	186.439	1.500	1.6	-0.6	0.0	60.61	948.93	21.58	21.58	38.6	0.11

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE BEAM ELEMENTS

MEMBER	GRP	LOAD CASE	DIST	***** APPLIED *****			***** ALLOWABLES *****			CRIT. COND	UNITY CHECK RATIO		
			FROM	AXIAL	MOMENT	SHEAR	TORSION	REINF.	MOMENT			SHEAR	TORSION
			END FT	LOAD KIPS	MY IN-KIP	FZ KIPS	MX IN-KIP	RATIO	MY IN-KIP			FZ KIPS	MX IN-KIP
101- 105 CRL	4	0.0	-2.9	-112.7	7.4	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56
	5	0.0	-2.2	-84.5	5.5	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42
	6	0.0	-0.6	-38.5	2.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15
	4	10.0	-2.9	332.7	0.0	-2.5	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.53
	5	10.0	-2.2	249.5	0.0	-1.8	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.15
	6	10.0	-0.6	80.0	0.0	0.1	0.005	217.5	8.2	74.6	0.033	BEND-Y	0.37
	4	20.0	-2.9	-101.6	-7.3	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56
	5	20.0	-2.2	-76.2	-5.5	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42
6	20.0	-0.6	-39.0	-2.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15	
103- 117 CLL	4	0.0	-2.4	-47.0	5.7	-3.6	0.009	331.0	13.1	24.4	0.017	SHEAR	0.44
	5	0.0	-1.8	-35.2	4.3	-2.7	0.009	331.0	13.1	24.4	0.017	SHEAR	0.33
	6	0.0	-0.5	-41.8	2.1	0.2	0.009	331.0	13.1	19.9	0.017	SHEAR	0.16
	4	2.5	-2.4	97.1	3.9	-3.6	0.005	217.5	8.2	24.0	0.033	SHEAR	0.47
	5	2.5	-1.8	72.9	2.9	-2.7	0.005	217.5	8.2	24.0	0.033	SHEAR	0.36
	6	2.5	-0.5	12.6	1.6	0.2	0.005	217.5	8.2	19.8	0.033	SHEAR	0.19
	4	5.0	-2.4	186.3	2.1	-3.6	0.005	217.5	8.2	28.2	0.033	BEND-Y	0.86
	5	5.0	-1.8	139.7	1.5	-2.7	0.005	217.5	8.2	28.2	0.033	BEND-Y	0.64
6	5.0	-0.5	52.2	1.1	0.2	0.005	217.5	8.2	20.1	0.033	BEND-Y	0.24	
101- 111 CRL	4	0.0	-2.3	-273.6	17.1	-248.8	0.004	894.9	29.8	340.3	0.022	TORS	0.73
	5	0.0	-2.6	-141.9	12.1	-186.6	0.004	894.9	29.2	343.2	0.022	TORS	0.54
	6	0.0	-1.6	-10.8	4.6	-40.1	0.004	894.9	43.4	306.2	0.022	TORS	0.13
	4	3.0	-2.3	315.0	15.6	-248.8	0.010	348.8	15.5	282.0	0.024	SHEAR	1.01
	5	3.0	-2.6	273.7	11.0	-186.6	0.010	348.8	15.2	287.6	0.024	BEND-Y	0.78
	6	3.0	-1.6	136.7	3.6	-40.1	0.010	348.8	17.8	250.2	0.024	BEND-Y	0.39
	4	3.0	-2.3	315.0	15.6	-248.8	0.009	925.4	19.1	271.2	0.024	TORS	0.92
	5	3.0	-2.6	273.7	11.0	-186.6	0.009	925.4	18.7	275.3	0.024	TORS	0.68
	6	3.0	-1.6	136.7	3.6	-40.1	0.009	925.4	26.9	243.5	0.024	TORS	0.16
	4	6.7	-2.3	957.4	13.6	-248.8	0.009	925.4	18.1	279.5	0.024	BEND-Y	1.03
	5	6.7	-2.6	723.9	9.5	-186.6	0.009	925.4	17.6	283.5	0.024	BEND-Y	0.78
	6	6.7	-1.6	267.5	2.4	-40.1	0.009	925.4	26.9	275.2	0.024	BEND-Y	0.29

SACS-IV SYSTEM CONCRETE ELEMENT UNITY CHECK REPORT
ACI 318R-89 (REVISED 92)

MEMBER	GRP	MAX. CRIT.		LOAD COND NO.	DIST FROM END FT	*****				U N I T Y		C H E C K C O M P O N E N T S										*****			
		UNITY CHECK	COND			*SHEAR* FZ	*TORSION* MX	*REINF* RATIO	*BEAMS* MY	AXIAL FX	EULER Y-Y	BUCKLING Z-Z	SHEAR FY	BEND	MOM MAGN	CMY	CMZ								
1-	101	CC1	0.61	CM+BN	4	9.00	0.130	0.000	0.559			0.610	0.010	0.010	0.101	0.612	1.50	0.60	0.60						
3-	103	CC1	0.61	CM+BN	4	9.00	0.131	0.000	0.559			0.612	0.008	0.008	0.102	0.612	1.50	0.60	0.60						
7-	107	CC1	0.61	CM+BN	4	9.00	0.130	0.000	0.559			0.611	0.010	0.010	0.101	0.610	1.50	0.60	0.60						
5-	105	CC1	0.61	CM+BN	4	9.00	0.130	0.000	0.559			0.610	0.010	0.010	0.101	0.611	1.50	0.60	0.60						
101-	105	CRL	1.53	BEND-Y	4	10.00	0.006	0.031	0.161	1.529															
103-	117	CLL	0.86	BEND-Y	4	5.00	0.251	0.126	0.161	0.856															
101-	111	CR1	1.03	BEND-Y	4	6.67	0.752	0.890	0.368	1.035															
103-	112	CR1	1.10	TORS	4	3.00	0.877	1.097	0.368	0.388															
105-	113	CR1	1.03	BEND-Y	4	6.67	0.761	0.921	0.368	1.029															
107-	114	CR1	1.05	BEND-Y	4	6.67	0.773	0.978	0.368	1.048															
111-	112	CR2	1.08	BEND-Y	4	5.00	0.023	0.105	0.368	1.077															
113-	114	CR2	1.07	BEND-Y	4	5.00	0.024	0.040	0.368	1.074															
111-	113	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136															
112-	114	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136															
117-	107	CLL	1.38	BEND-Y	4	7.50	0.330	0.704	0.161	1.384															

SACS-IV CONCRETE MEMBER UNITY CHECK RANGE SUMMARY

GROUP I - UNITY CHECKS GREATER THAN 1.33

MEMBER	GRP	MAX.		DIST	***** CRITICAL LOADING *****					SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST		
		UNITY	LOAD		AXIAL	MOMENT	MOMENT	SHEAR	SHEAR	KLY/RX	KLZ/RZ	UNITY	LOAD	UNITY	LOAD	
		CHECK	CASE		END	LOAD	MY	MZ	FY			FZ	CHECK	CASE	CHECK	CASE
101-	105	CRL	1.53	4	10.0	-2.9	332.7	0.8	0.0	52.26	35.55	1.15	5	0.56	4	
117-	107	CLL	1.38	4	7.5	-2.9	301.1	-0.8	0.0	39.20	26.66	1.04	5	0.86	4	

SACS-IV CONCRETE MEMBER UNITY CHECK RANGE SUMMARY

GROUP II - UNITY CHECKS GREATER THAN 1.00 AND LESS THAN 1.33

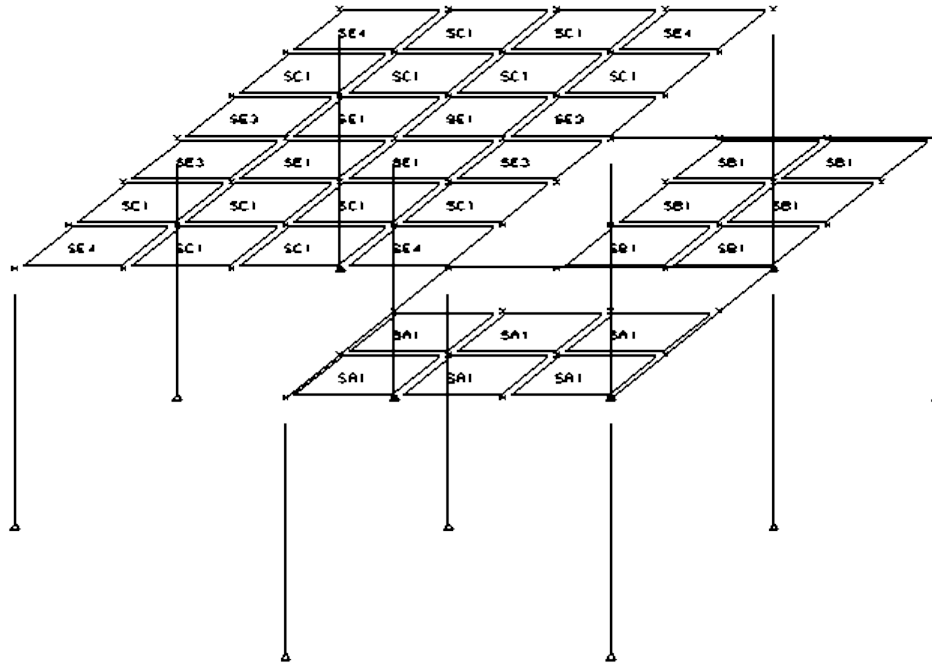
MEMBER	GRP		MAX.	LOAD	DIST	***** CRITICAL LOADING *****				SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST		
			UNITY		FROM	AXIAL	MOMENT	MOMENT	SHEAR	SHEAR	KLY/RV	KLZ/RZ	UNITY	LOAD	UNITY	LOAD
			CHECK	CASE	END	LOAD	MY	MZ	FY	FZ			CHECK	CASE	CHECK	CASE
					FT	KIPS	IN-KIP	IN-KIP	KIPS	KIPS						
101-103	111-112	CR1	1.03	4	6.7	-2.3	957.4	1.5	0.0	13.6	11.55	18.48	1.01	4	0.92	4
			1.10	4	3.0	-2.3	359.0	-0.1	0.0	15.0	11.55	18.48	1.07	4	1.04	4
105-107	113-114	CR1	1.03	4	6.7	-2.3	952.2	-1.6	0.0	13.5	11.55	18.48	1.01	4	0.95	4
			1.05	4	6.7	-2.3	969.7	1.5	0.0	13.1	11.55	18.48	1.01	4	1.00	4
111-113	112-114	CR2	1.08	4	5.0	-2.3	996.6	1.1	0.0	-0.6	11.55	18.47	1.07	4	1.05	4
			1.07	4	5.0	-2.3	994.3	-1.2	0.0	-0.7	11.55	18.47	1.07	4	1.05	4
111-112	113-114	CT1	1.14	4	15.0	0.0	247.2	0.4	0.0	-5.7	52.26	43.86	1.14	4	1.11	4
			1.14	4	15.0	0.0	247.1	-0.3	0.0	-5.7	52.26	43.86	1.14	4	1.11	4

SACS-IV SYSTEM CONCRETE MEMBER GROUP SUMMARY REPORT
ACI 318R-89 (REVISED 92)

GRP ID	CRITICAL MEMBER	LOAD COND	MAX. UNITY CHECK	DIST FROM END FT	***** AXIAL KIPS	***** APPLIED BEND IN-KIP	***** LOADS TORSION IN-KIP	***** REINF RATIO	***** AXIAL KIPS	***** EULER KIPS	***** ALLOWABLE BEND IN-KIP	***** LOADS TORSION IN-KIP	***** REINF RATIO	CRIT COND	EFFECTIVE KL/R Y	Z
CC1	3-103	4	0.61	9.0	-20.36	598.2	0.0	0.017	33.3	2446.8	977.5	38.6	0.031	CM+BN	28.8	28.8
CRL	101-105	4	1.53	10.0	-2.89	332.7	-2.5	0.005			217.5	78.0	0.033	BEND-Y	52.3	35.6
CLL	117-107	4	1.38	7.5	-2.89	301.1	53.9	0.005			217.5	76.6	0.033	BEND-Y	39.2	26.7
CR1	103-112	4	1.10	3.0	-2.26	359.0	314.7	0.009			925.4	286.9	0.024	TORS	11.5	18.5
CR2	111-112	4	1.08	5.0	-2.26	996.6	32.9	0.009			925.4	312.1	0.024	BEND-Y	11.5	18.5
CT1	111-113	4	1.14	15.0	-0.03	247.2	-2.9	0.005			217.5	2.7	0.033	BEND-Y	52.3	43.9

4.3 SAMPLE PROBLEM 3

Sample Problem 3 illustrates the use of concrete one-way and two-way slab elements. The model represents a main slab area that acts as a two-way slab with corner, edge and interior columns. Some two way slabs, (i.e. groups SE1, SE3 and SE4), are checked for punching shear around the columns along with beam shear and bending. The side wings of the model contain one-way slabs spanning in the local X ('SA1') and the local Y ('SB1') axis directions.



Three basic load cases representing dead, live and wind loading, respectively, along with three load combinations were analyzed. The SACS model file is shown followed by a detailed description of selected portions of the input:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
	CONCRETE SAMPLE PROBLEM 3							
	OPTIONS	EN	LR	2 1 0 0	PT	PTPT		
A	CNCOPT	BR		PD BD			1.000	
	LCSEL	4	5	6				
	SECT							
	SECTRBOX1	BRP			0.75	4	411.25	11.25
	SECT SQUARE	RCRS			15.00	15.0		
	SECT2 BOX1	7.50	60.00S					
	SECT LEFTL	RCLL			16.00	8.000	24.00	5.000
	SECT2 ROW3	2.50	60.00S	ROW4	13.50	60.00S		
	SECT RIGHTL	RCRL			16.00	8.000	24.00	5.000
	SECT2 ROW3	2.50	60.00S	ROW4	13.50	60.00S		
	SECTROW3	RRP			0.38	4	18.00	
	SECTROW4	RRP			0.38	2	3.50	
	GRUP							
	GRUPCCC1 SQUARE	29.000	490.00	145.0 4.00 1	1.001.00	60.00	24.00S	0.375
	GRUPCCL1 LEFTL	29.000	490.00	145.0 4.00 3	1.001.00	60.00	24.00T	0.375
	GRUPCCR1 RIGHTL	29.000	490.00	145.0 4.00 3	1.001.00	60.00	24.00T	0.375
	GRUP WF1 W6X15			29.0011.60	36.00 1	1.001.00	0.50	490.00
	MEMBER							
	MEMBER1 2 6 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 3 112 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 4 8 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 7 114 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 13 11 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 14 16 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 15 17 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 22 23 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 24 28 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 25 27 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER1 30 31 CC1							
	MEMBER OFFSETS						-12.00	
	MEMBER0 6 18 CL1							
	MEMBER0 16 19 CL1							
	MEMBER0 18 19 CL1							
	MEMBER0 27 41 CL1							
	MEMBER0 41 42 CL1							
	MEMBER0 42 16 CL1							
	MEMBER0 23 35 CR1							
	MEMBER0 35 36 CR1							
	MEMBER0 36 6 CR1							
	MEMBER0 8 20 CR1							
	MEMBER0 17 21 CR1							
	MEMBER0 20 21 CR1							

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
	PSTIF							
B	PSTIF RBR MESH1	0.375	2.0	60.	29.0	490.0		
C	PSTIF RBR MESH2	0.500	2.0	60.	29.0	490.0		
	PGRUP							
	PGRUP Z01	0.2500	29.000	0.25036.000				
D	PGRUPCSA1	4.0 A	0.25	4.0	MESH1	5.0 XT		145.0
	PGRUPCSB1	4.0 B	0.25	4.0	MESH1	5.0 YT		
E	PGRUPCSC1	4.0 C	0.25	4.0	MESH2	6.0 XT MESH2	6.0 YT	145.0
	PGRUPCSE1	4.0 E	0.25	4.0	MESH1	5.0 XT MESH2	6.0 YT	145.0
	PGRUP2 INTR	15.	20.					
	PGRUPCSE3	4.0 E	0.25	4.0	MESH2	6.0 XT MESH2	6.0 YT	145.0
	PGRUP2 EDY	40.	1.					
F	PGRUPCSE4	4.0 E	0.25	4.0	MESH1	5.0 XT MESH2	6.0 YT	145.0
G	PGRUP2 CORR	15.	20.					
	PLATE							
	PLATE X01	23	29	35	37SA1			0
	PLATE X02	29	34	37	39SA1			0
	PLATE X03	34	27	39	41SA1			0
	PLATE X04	35	37	36	38SA1			0
	PLATE X05	37	39	38	40SA1			0
	PLATE X06	39	41	40	42SA1			0
	PLATE Y01	18	19	72	76SB1			0
	PLATE Y02	19	16	76	80SB1			0
	PLATE Y03	72	76	73	77SB1			0
	PLATE Y04	76	80	77	81SB1			0
	PLATE Y05	73	77	20	21SB1			0
	PLATE Y06	77	81	21	17SB1			0
	PLATE C101	103	9	56	60SC1			0
	PLATE C102	9	10	60	64SC1			0
	PLATE C103	52	56	53	57SC1			0
	PLATE C104	56	60	57	61SC1			0
	PLATE C105	60	64	61	65SC1			0
	PLATE C106	64	68	65	69SC1			0
	PLATE C105	54	58	55	59SC1			0
	PLATE C106	58	62	59	63SC1			0
	PLATE C107	62	66	63	67SC1			0
	PLATE C108	66	70	67	71SC1			0
	PLATE C109	59	63	26	32SC1			0
	PLATE C110	63	67	32	33SC1			0
	PLATE E101	57	61	107	11SE1			0
	PLATE E102	61	65	11	12SE1			0
	PLATE E103	107	11	58	62SE1			0
	PLATE E104	11	12	62	66SE1			0
	PLATE E301	53	57	114	107SE3			0
	PLATE E302	65	69	12	8SE3			0
	PLATE E303	114	107	54	58SE3			0
	PLATE E304	12	8	66	70SE3			0
	PLATE E401	112	103	52	56SE4			0
	PLATE E402	10	6	64	68SE4			0
	PLATE E403	55	59	28	26SE4			0
	PLATE E404	67	71	33	31SE4			0
	JOINT							
	JOINT 2	10.	-10.	0.				111
	JOINT 3	-3.	-10.	0.	-3.996			111
	JOINT 4	10.	0.	0.				111
	JOINT 6	10.	-10.	10.				
	JOINT 7	-3.	0.	0.	-3.996			111
	JOINT 8	10.	0.	10.				
	JOINT 9	3.	-10.	10.	3.996			
	JOINT 10	6.	-10.	10.	8.004			
	JOINT 11	3.	0.	10.	3.996			
	JOINT 12	6.	0.	10.	8.004			
	JOINT 13	3.	0.	0.	3.996			111
	JOINT 14	20.	-10.	0.				111
	JOINT 15	20.	0.	0.				111

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
JOINT	16	20.	-10.	10.				
JOINT	17	20.	0.	10.				
JOINT	18	13.	-10.	10.	3.996			
JOINT	19	16.	-10.	10.	8.004			
JOINT	20	13.	0.	10.	3.996			
JOINT	21	16.	0.	10.	8.004			
JOINT	22	10.	-20.	0.		111		
JOINT	23	10.	-20.	10.				
JOINT	24	-3.	10.	0.	-3.996		111	
JOINT	25	20.	-20.	0.			111	
JOINT	26	0.	10.	10.				
JOINT	27	20.	-20.	10.				
JOINT	28	-3.	10.	10.	-3.996			
JOINT	29	13.	-20.	10.	3.996			
JOINT	30	10.	10.	0.			111	
JOINT	31	10.	10.	10.				
JOINT	32	3.	10.	10.	3.996			
JOINT	33	6.	10.	10.	8.004			
***** Additional Joint Input Lines *****								
JOINT	103	0.	-10.	10.				
JOINT	107	0.	0.	10.				
JOINT	112	-3.	-10.	10.	-3.996			
JOINT	114	-3.	0.	10.	-3.996			
LOAD								
LOADCN	1				DEAD			
LOAD	6			-0.1319		GLOB JOIN	DEAD WT	
LOAD	8			-0.2638		GLOB JOIN	DEAD WT	
LOAD	9			-0.2638		GLOB JOIN	DEAD WT	
LOAD	10			-0.2638		GLOB JOIN	DEAD WT	
LOAD	11			-0.5277		GLOB JOIN	DEAD WT	
***** Additional Dead Loading *****								
LOADCN	2							
LOAD	6			-0.2777		GLOB JOIN	LIVE	
LOAD	8			-0.5554		GLOB JOIN	LIVE	
LOAD	9			-0.5555		GLOB JOIN	LIVE	
LOAD	10			-0.5555		GLOB JOIN	LIVE	
LOAD	11			-1.1111		GLOB JOIN	LIVE	
LOAD	12			-1.1111		GLOB JOIN	LIVE	
***** Additional Live loading *****								
LOADCN	3							
LOAD	112	2.00000				GLOB JOIN	WIND	
LOAD	114	2.00000				GLOB JOIN	WIND	
LOAD	28	2.00000				GLOB JOIN	WIND	
LOAD	6	2.00000				GLOB JOIN	WIND	
LOAD	8	2.00000				GLOB JOIN	WIND	
LOAD	31	2.00000				GLOB JOIN	WIND	
LOAD	23	2.00000				GLOB JOIN	WIND	
LOAD	27	2.00000				GLOB JOIN	WIND	
LOAD	16	2.00000				GLOB JOIN	WIND	
LOAD	17	2.00000				GLOB JOIN	WIND	
LDCOMB								
LDCOMB	4	140.00	1	170.00	2			
LDCOMB	5	105.00	1	127.50	2	127.50	3	
LDCOMB	6	90.00	1	130.00	3			
END								

The following is a description of selected input lines used the SACS model file for Sample Problem 3. The lines are referenced by the letter in the left margin of the input listing.

A. The concrete analysis options are designated on the CNCOPT input line as follows:

- a. 'BR' in columns 13-14 designates that a first-order linear analysis is to be performed and the structure is to be considered as braced against sidesway.
- b. Member detailed reports for concrete beams along with slab detailed reports are requested.

B. The slab reinforcement pattern named MESH1 is defined using the PSTIF input line with 'MESH1' in columns 11-15.

- a. The pattern is designated as a rebar stiffener pattern type by 'RBP' in columns 7-9.
- b. The rebar diameter, 0.375, and distance from the slab edge, 2.0 is specified.
- c. The yield stress, elastic modulus and density of the rebars are specified as 60.0, 29.0 and 490.0 in columns 35-41, 42-48 and 49-55, respectively.

C. The slab reinforcement pattern named MESH2 is defined using the PSTIF input line with 'MESH2' in columns 11-15.

- a. The pattern is designated as a rebar stiffener pattern type by 'RBP' in columns 7-9.
- b. The rebar diameter, 0.50, and distance from the slab edge, 2.0 is specified.
- c. The yield stress, elastic modulus and density of the rebars are specified as 60.0, 29.0 and 490.0 in columns 35-41, 42-48 and 49-55, respectively.

D. Concrete properties are defined on PGRUP input lines with 'C' specified in column 6. The properties for all slab elements assigned to concrete group named SA1 are defined on the PGRUP line with 'SA1' specified in columns 7-9.

The slab thickness is 4.0 as defined in columns 11-16.

- a. Slabs assigned to group SA1 are one-way slabs spanning in the local X direction as designated by slab type 'A' in column 17.
- b. The concrete compressive strength and density are specified as 4.0 and 145.0, respectively.

-
- c. The slab reinforcement is defined by plate reinforcement section MESH1. The average spacing of the rebars is 5.0 and the rebars span in the local X direction.
 - d. The distance specified on the PSTIF input line is to be measured from the top edge of the plate as designated by 'T' in column 56.

E. The properties for all slab elements assigned to concrete group named SC1 are defined on the PGRUP line with 'SC1' specified in columns 7-9.

The slab thickness is 4.0 as defined in columns 11-16.

- a. Slabs assigned to group SC1 are two-way slabs as designated by slab type 'C' in column 17.
- b. The concrete compressive strength and density are specified as 4.0 and 145.0, respectively.
- c. The first slab reinforcement is defined by plate reinforcement section MESH2. The average spacing of the rebars is 6.0 and the rebars span in the local X direction.
- d. The second slab reinforcement is also defined by plate reinforcement section MESH2. The average spacing of the rebars is 6.0 and the rebars span in the local Y direction.
- e. The distance specified on the PSTIF input line is to be measured from the top edge of the plate for both reinforcement patterns as designated by 'T' in columns 56 and 72.

F. The properties for all slab elements assigned to concrete group named SE4 are defined on the PGRUP line with 'SE4' specified in columns 7-9.

The slab thickness is 4.0 as defined in columns 11-16.

- a. Slabs assigned to group SE4 are two-way slabs that are to be checked for punching shear around a column or concentrated load as indicated by slab type 'E' in column 17.
- b. Concrete compressive strength and density are specified as 4.0 and 145.0, respectively.
- c. The first slab reinforcement is defined by plate reinforcement section MESH1. The average spacing of the rebars is 5.0 and the rebars span in the local X direction.
- d. The second slab reinforcement is also defined by plate reinforcement section MESH2. The average spacing of the rebars is 6.0 and the rebars span in the local Y direction.

- e. The distance specified on the PSTIF input line is to be measured from the top edge of the plate for both reinforcement patterns as designated by 'T' in columns 56 and 72.

G. The critical section to be checked for punching shear of slab group SE4 is defined on the PGRUP2 line immediately following the PGRUP line defining the group.

- a. The critical section for punching is due to a corner column as designated by 'COR' in columns 8-10.
- b. The critical section shape is rectangular as specified by 'R' in column 11.
- c. The critical section dimensions along the plate local X and Y axes are specified in columns 12-17 and 18-23, respectively.

A portion of the output file for the analysis is listed on the following pages.

CONCRETE SAMPLE PROBLEM 3

DATE 12-JUN-1996 TIME 16:24:44 PRE PAGE 1

** PROGRAM OPTIONS **

UNITS

....ENGLISH

EXECUTION

....UNITY CHECK API/AISC LOAD & RESISTANCE FACTOR DESIGN

LOAD

....NO. BASIC LOAD COND. 3

....NO. COMB. LOAD COND. 3

CONCRETE OPTIONS

.... ANALYSIS TYPEBRACED

.... SIDESWAY MOMENT MAGNIFICATION . 1.00

.... COLUMN DETAILED PRINTNO

.... BEAM DETAILED PRINTYES

.... SLAB DETAILED PRINTYES

.... ELEMENT UNITY CHECK OPTIONNO

.... UNITY CHECK RANGE REPORTNO

.... CONCRETE ONLY PRINTNO

***** CONCRETE PLATE REINFORCEMENT BARS *****

SECTION	DIAMETER	DISTANCE	YIELD	ELASTIC	DENSITY
		FROM	STRESS	MODULUS	
		EDGE		1000	
	IN	IN	KSI	KSI	LB/FT3
MESH1	0.375	2.00	60.00	29.00	490.000
MESH2	0.500	2.00	60.00	29.00	490.000

CONCRETE SLAB GROUP PROPERTIES

***** CONCRETE ***** REINFORCEMENT BARS *****

GRUP	PLATE	THICK.	ELASTIC	COMP.	POISSON'S	DIAMETER	DIRECTION	LOCATION	EDGE	SPACING	YIELD	ELASTIC
ID	TYPE		MODULUS	STRENGTH	RATIO				DISTANCE		STRESS	MODULUS
		IN	1000 KSI	KSI		IN			IN	IN	KSI	1000 KSI
SA1	ONE-X	4.00	3.60	4.000	0.250	0.375	X	TOP	2.000	5.000	60.00	29.00
SB1	Y-WAY	4.00	3.60	4.000	0.250	0.375	Y	TOP	2.000	5.000	60.00	29.00
SC1	2-WAY	4.00	3.60	4.000	0.250	0.500	X	TOP	2.000	6.000	60.00	29.00
						0.500	Y	TOP	2.000	6.000	60.00	29.00
SE1	CSUP	4.00	3.60	4.000	0.250	0.375	X	TOP	2.000	5.000	60.00	29.00
						0.500	Y	TOP	2.000	6.000	60.00	29.00
SE3	CSUP	4.00	3.60	4.000	0.250	0.500	X	TOP	2.000	6.000	60.00	29.00

SACS SYSTEM - PLATE INTERNAL LOADS REPORT

PLATE	CONNECTING JOINTS				LOAD CASE	***** MEMBRANE *****			* OUT-OF-PLANE SHEAR *		***** BENDING MOMENTS *****		
	A	B	C	D		NXX K/IN	NYX K/IN	NYZ K/IN	NXZ K/IN	MXX KIPS	MYX KIPS	MYZ KIPS	
X01	23	29	35	37	4	-0.01	0.00	0.00	0.07	0.01	0.00	-0.41	0.50
					5	-0.02	0.00	0.00	0.04	0.01	0.00	0.19	0.38
					6	-0.01	0.00	-0.01	0.00	0.00	0.00	0.43	0.10
X02	29	34	37	39	4	-0.01	0.00	0.00	0.00	0.02	0.00	0.92	0.00
					5	-0.01	0.00	0.00	-0.01	0.02	0.00	0.69	0.00
					6	0.00	0.00	-0.01	-0.01	0.00	0.00	0.16	0.00
X03	34	27	39	41	4	-0.02	0.00	0.00	-0.07	0.01	0.00	-0.38	-0.50
					5	0.00	0.00	0.00	-0.06	0.01	0.00	-0.78	-0.36
					6	0.01	0.00	0.00	-0.02	0.00	0.00	-0.57	-0.08
X04	35	37	36	38	4	0.00	0.00	0.01	0.07	-0.01	0.00	-0.41	-0.50
					5	0.00	0.00	-0.01	0.04	-0.01	0.00	0.18	-0.38
					6	0.00	0.00	-0.01	0.00	0.00	0.00	0.42	-0.10
X05	37	39	38	40	4	0.00	0.00	0.00	0.00	-0.02	0.00	0.92	0.00
					5	-0.01	0.00	-0.01	-0.01	-0.01	0.00	0.69	-0.01
					6	-0.01	0.00	-0.01	-0.01	0.00	0.00	0.16	-0.01
X06	39	41	40	42	4	0.01	0.00	0.00	-0.07	-0.01	0.00	-0.38	0.50
					5	-0.02	0.00	-0.01	-0.06	-0.01	0.00	-0.77	0.38
					6	-0.02	0.00	-0.01	-0.02	0.00	0.00	-0.57	0.09
Y01	18	19	72	76	4	0.00	0.00	0.00	0.01	0.07	-0.45	0.00	0.50
					5	0.00	0.05	0.04	0.00	0.05	-0.35	0.00	0.40
					6	0.00	0.05	0.04	0.00	0.01	-0.10	0.00	0.12
Y02	19	16	76	80	4	0.00	-0.01	0.00	-0.01	0.07	-0.45	0.00	-0.51
					5	0.00	-0.08	0.03	-0.01	0.05	-0.37	0.00	-0.35
					6	0.00	-0.07	0.03	0.00	0.01	-0.11	0.00	-0.06
Y03	72	76	73	77	4	0.00	0.00	0.00	0.02	0.00	0.91	0.00	0.00
					5	0.00	0.01	0.04	0.01	0.00	0.69	0.00	-0.01
					6	0.00	0.01	0.04	0.00	0.00	0.17	0.00	-0.01
Y04	76	80	77	81	4	0.00	-0.01	0.00	-0.02	0.00	0.91	0.00	-0.01
					5	0.00	-0.03	0.03	-0.02	0.00	0.70	0.00	-0.01
					6	0.00	-0.03	0.03	0.00	0.00	0.18	0.00	0.00

SACS SYSTEM - PLATE INTERNAL LOADS REPORT

PLATE	CONNECTING JOINTS				LOAD CASE	***** MEMBRANE *****			* OUT-OF-PLANE	SHEAR *	***** BENDING MOMENTS *****		
	A	B	C	D		NXX K/IN	NYX K/IN	NYZ K/IN			MXX KIPS	MYX KIPS	MYZ KIPS
C101	103	9	56	60	4	-0.01	0.00	0.00	0.02	0.02	0.24	0.93	0.49
					5	-0.03	0.00	-0.03	0.01	0.01	0.24	0.74	0.37
					6	-0.03	0.00	-0.03	0.00	0.00	0.11	0.21	0.09
C102	9	10	60	64	4	-0.01	0.00	0.00	-0.03	0.02	0.22	0.90	-0.50
					5	0.01	0.00	-0.02	-0.02	0.02	0.15	0.66	-0.40
					6	0.01	0.00	-0.03	-0.01	0.00	0.02	0.15	-0.12
C103	52	56	53	57	4	0.00	-0.01	0.00	0.02	0.01	0.77	0.24	0.61
					5	0.00	0.04	-0.01	0.02	0.01	0.63	0.22	0.53
					6	0.00	0.05	-0.01	0.01	0.00	0.19	0.09	0.18
C104	56	60	57	61	4	-0.01	-0.01	0.00	0.00	-0.03	0.95	0.87	0.43
					5	-0.02	0.00	-0.04	-0.01	-0.04	0.66	0.52	0.23
					6	-0.02	0.00	-0.03	-0.01	-0.02	0.12	0.02	-0.01
C105	60	64	61	65	4	-0.01	-0.01	0.00	0.00	-0.03	0.94	0.86	-0.46
					5	-0.03	-0.02	-0.04	0.00	-0.01	0.73	0.73	-0.49
					6	-0.02	-0.02	-0.04	0.00	0.00	0.19	0.25	-0.22
C106	64	68	65	69	4	0.00	-0.02	0.00	-0.02	0.02	0.75	0.22	-0.63
					5	-0.01	-0.05	-0.01	-0.01	0.01	0.53	0.17	-0.45
					6	0.00	-0.04	-0.01	0.00	0.00	0.10	0.05	-0.09
C105	54	58	55	59	4	0.00	-0.01	0.00	0.02	-0.01	0.75	0.24	-0.63
					5	0.00	0.04	0.02	0.03	-0.01	0.68	0.20	-0.55
					6	0.00	0.05	0.01	0.01	-0.01	0.25	0.06	-0.19
C106	58	62	59	63	4	-0.01	-0.01	0.00	0.00	0.03	0.92	0.87	-0.45
					5	-0.02	0.00	0.04	-0.01	0.03	0.73	0.56	-0.23
					6	-0.02	0.01	0.04	-0.01	0.02	0.20	0.06	0.03
C107	62	66	63	67	4	-0.01	-0.01	0.00	0.00	0.03	0.92	0.87	0.44
					5	-0.03	-0.02	0.04	0.00	0.01	0.75	0.74	0.48
					6	-0.03	-0.02	0.05	0.00	-0.01	0.23	0.25	0.23
C108	66	70	67	71	4	0.00	-0.01	0.00	-0.02	-0.01	0.73	0.24	0.61
					5	-0.01	-0.06	0.01	-0.01	-0.01	0.52	0.15	0.42
					6	-0.01	-0.05	0.01	0.00	-0.01	0.10	0.01	0.08

SACS SYSTEM - PLATE INTERNAL LOADS REPORT

PLATE	CONNECTING JOINTS				LOAD CASE	***** MEMBRANE *****			* OUT-OF-PLANE SHEAR *		***** BENDING MOMENTS *****		
	A	B	C	D		NXX K/IN	NYX K/IN	NYZ K/IN	NYZ K/IN	NXZ K/IN	MXX KIPS	MYX KIPS	MYZ KIPS
E101	57	61	107	11	4	-0.01	-0.01	0.00	-0.06	-0.09	-0.85	-0.30	0.20
					5	-0.05	0.02	-0.02	-0.09	-0.13	-1.18	-1.00	-0.14
					6	-0.04	0.02	-0.02	-0.05	-0.07	-0.70	-0.85	-0.26
E102	61	65	11	12	4	-0.01	-0.01	0.00	0.06	-0.09	-0.82	-0.25	-0.23
					5	-0.05	-0.02	-0.03	0.03	-0.03	-0.40	0.38	-0.43
					6	-0.05	-0.01	-0.03	-0.01	0.03	0.07	0.53	-0.31
E103	107	11	58	62	4	-0.01	-0.01	0.00	-0.06	0.09	-0.92	-0.32	-0.20
					5	-0.05	0.02	0.02	-0.08	0.12	-0.93	-0.88	0.18
					6	-0.04	0.03	0.02	-0.05	0.07	-0.41	-0.71	0.30
E104	11	12	62	66	4	-0.01	-0.01	0.00	0.06	0.09	-0.91	-0.28	0.24
					5	-0.06	-0.01	0.03	0.01	0.02	-0.24	0.46	0.58
					6	-0.05	0.00	0.03	-0.02	-0.03	0.29	0.63	0.46
E301	53	57	114	107	4	-0.01	-0.01	0.00	0.03	-0.05	-0.63	-0.30	0.15
					5	-0.03	0.06	0.01	-0.01	-0.03	-0.47	0.28	-0.21
					6	-0.03	0.06	0.01	-0.02	0.00	-0.10	0.47	-0.30
E302	65	69	12	8	4	-0.01	-0.02	-0.01	-0.03	-0.05	-0.60	-0.36	-0.19
					5	-0.09	-0.09	-0.05	-0.04	-0.04	-0.50	-0.50	-0.30
					6	-0.09	-0.08	-0.05	-0.02	-0.02	-0.16	-0.29	-0.20
E303	114	107	54	58	4	-0.01	-0.01	0.00	0.03	0.05	-0.69	-0.32	-0.18
					5	-0.03	0.06	0.00	-0.02	0.02	-0.11	0.42	0.40
					6	-0.03	0.07	0.00	-0.04	-0.01	0.29	0.62	0.51
E304	12	8	66	70	4	-0.01	-0.01	0.00	-0.04	0.06	-0.78	-0.43	0.29
					5	-0.10	-0.10	0.06	-0.05	0.05	-0.59	-0.55	0.42
					6	-0.10	-0.10	0.06	-0.03	0.02	-0.15	-0.31	0.26
E401	112	103	52	56	4	-0.02	-0.02	-0.02	0.10	0.10	-0.55	-0.75	0.30
					5	-0.07	0.01	-0.03	0.06	0.06	0.01	0.03	0.74
					6	-0.06	0.02	-0.03	0.00	0.00	0.33	0.47	0.58

CONCRETE SLAB ALLOWABLE SUMMARY REPORT

PLATE ID	GRUP ID	TYPE	CRIT. LOAD CASE	CRIT. COND.	MAXIMUM UNITY CHECK	***** X-DIRECTION *****			***** Y-DIRECTION *****			IN-PLANE SHEAR XY K/IN
						FORCE X K/IN	MOMENT Y-Y IN-K/IN	SHEAR XZ K/IN	FORCE Y K/IN	MOMENT X-X IN-K/IN	SHEAR YZ K/IN	
X01	SA1	ONE-X	4	SHR-XZ	0.307	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X02	SA1	ONE-X	4	BEND Y-Y	0.427	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X03	SA1	ONE-X	5	BEND Y-Y	0.363	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X04	SA1	ONE-X	4	SHR-XZ	0.307	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X05	SA1	ONE-X	4	BEND Y-Y	0.426	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X06	SA1	ONE-X	5	BEND Y-Y	0.359	0.00	2.2	0.22	0.00	0.0	0.00	0.22
Y01	SB1	ONE-Y	4	SHR-YZ	0.310	0.00	0.0	0.00	0.00	2.2	0.22	0.22
Y02	SB1	ONE-Y	4	SHR-YZ	0.313	0.00	0.0	0.00	0.00	2.2	0.22	0.22
Y03	SB1	ONE-Y	4	BEND X-X	0.424	0.00	0.0	0.00	0.00	2.2	0.22	0.22
Y04	SB1	ONE-Y	4	BEND X-X	0.424	0.00	0.0	0.00	0.00	2.2	0.22	0.22
Y05	SB1	ONE-Y	4	SHR-YZ	0.301	0.00	0.0	0.00	0.00	2.2	0.22	0.22
Y06	SB1	ONE-Y	4	SHR-YZ	0.298	0.00	0.0	0.00	0.00	2.2	0.22	0.22
C101	SC1	TWO WAY	4	BEND Y-Y	0.307	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C102	SC1	TWO WAY	4	BEND Y-Y	0.296	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C103	SC1	TWO WAY	4	BEND X-X	0.254	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C104	SC1	TWO WAY	4	BEND X-X	0.313	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C105	SC1	TWO WAY	4	BEND X-X	0.312	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C106	SC1	TWO WAY	4	BEND X-X	0.249	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C105	SC1	TWO WAY	4	BEND X-X	0.248	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C106	SC1	TWO WAY	4	BEND X-X	0.305	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C107	SC1	TWO WAY	4	BEND X-X	0.304	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C108	SC1	TWO WAY	4	BEND X-X	0.243	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C109	SC1	TWO WAY	4	BEND Y-Y	0.309	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C110	SC1	TWO WAY	4	BEND Y-Y	0.306	0.00	3.0	0.22	0.00	3.0	0.22	0.22
E101	SE1	CSUP	5	BEND Y-Y	0.464	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E102	SE1	CSUP	4	PUNCH-YZ	0.280	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E103	SE1	CSUP	5	BEND Y-Y	0.410	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E104	SE1	CSUP	4	BEND X-X	0.300	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E301	SE3	CSUP	4	PUNCH-YZ	0.234	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E302	SE3	CSUP	5	SHR-XY	0.250	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E303	SE3	CSUP	4	PUNCH-YZ	0.235	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E304	SE3	CSUP	6	SHR-XY	0.266	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E401	SE4	CSUP	4	BEND Y-Y	0.347	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E402	SE4	CSUP	5	BEND Y-Y	0.448	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E403	SE4	CSUP	4	BEND Y-Y	0.340	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E404	SE4	CSUP	5	BEND Y-Y	0.615	0.00	2.2	0.33	0.00	3.0	0.33	0.22

CONCRETE SLAB UNITY CHECK SUMMARY REPORT

						***** X-DIRECTION *****							***** Y-DIRECTION *****			***** IN-PLANE *****
PLATE	GRUP	TYPE	CRIT.	CRIT.	MAXIMUM	*****	X-DIRECTION	*****	*****	*****	*****	*****	*****	*****	*****	
ID	ID		LOAD	COND.	UNITY	FORCE	MOMENT	SHEAR	FORCE	MOMENT	SHEAR	FORCE	MOMENT	SHEAR	SHEAR	
			CASE		CHECK	X	Y-Y	XZ	Y	X-X	YZ	Y	X-X	YZ	XY	
X01	SA1	ONE-X	4	SHR-XZ	0.307	0.307	0.188	0.307	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
X02	SA1	ONE-X	4	BEND Y-Y	0.427	0.427	0.427	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
X03	SA1	ONE-X	5	BEND Y-Y	0.363	0.363	0.363	0.287	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
X04	SA1	ONE-X	4	SHR-XZ	0.307	0.307	0.188	0.307	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
X05	SA1	ONE-X	4	BEND Y-Y	0.426	0.426	0.426	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
X06	SA1	ONE-X	5	BEND Y-Y	0.359	0.359	0.359	0.283	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
Y01	SB1	ONE-Y	4	SHR-YZ	0.310	0.310	0.000	0.000	0.209	0.310	0.000	0.000	0.000	0.000	0.000	
Y02	SB1	ONE-Y	4	SHR-YZ	0.313	0.313	0.000	0.000	0.211	0.313	0.000	0.000	0.000	0.000	0.000	
Y03	SB1	ONE-Y	4	BEND X-X	0.424	0.424	0.000	0.000	0.424	0.005	0.000	0.000	0.000	0.000	0.000	
Y04	SB1	ONE-Y	4	BEND X-X	0.424	0.424	0.000	0.000	0.424	0.007	0.000	0.000	0.000	0.000	0.000	
Y05	SB1	ONE-Y	4	SHR-YZ	0.301	0.301	0.000	0.000	0.163	0.301	0.000	0.000	0.000	0.000	0.000	
Y06	SB1	ONE-Y	4	SHR-YZ	0.298	0.298	0.000	0.000	0.161	0.298	0.000	0.000	0.000	0.000	0.000	
C101	SC1	TWO WAY	4	BEND Y-Y	0.307	0.307	0.307	0.113	0.078	0.085	0.000	0.000	0.000	0.000	0.000	
C102	SC1	TWO WAY	4	BEND Y-Y	0.296	0.296	0.296	0.126	0.072	0.088	0.000	0.000	0.000	0.000	0.000	
C103	SC1	TWO WAY	4	BEND X-X	0.254	0.254	0.079	0.097	0.254	0.066	0.000	0.000	0.000	0.000	0.000	
C104	SC1	TWO WAY	4	BEND X-X	0.313	0.313	0.288	0.014	0.313	0.135	0.000	0.000	0.000	0.000	0.000	
C105	SC1	TWO WAY	4	BEND X-X	0.312	0.312	0.284	0.009	0.312	0.129	0.000	0.000	0.000	0.000	0.000	
C106	SC1	TWO WAY	4	BEND X-X	0.249	0.249	0.073	0.088	0.249	0.074	0.000	0.000	0.000	0.000	0.000	
C105	SC1	TWO WAY	4	BEND X-X	0.248	0.248	0.081	0.096	0.248	0.063	0.000	0.000	0.000	0.000	0.000	
C106	SC1	TWO WAY	4	BEND X-X	0.305	0.305	0.286	0.016	0.305	0.140	0.000	0.000	0.000	0.000	0.000	
C107	SC1	TWO WAY	4	BEND X-X	0.304	0.304	0.287	0.011	0.304	0.138	0.000	0.000	0.000	0.000	0.000	
C108	SC1	TWO WAY	4	BEND X-X	0.243	0.243	0.079	0.092	0.243	0.057	0.000	0.000	0.000	0.000	0.000	
C109	SC1	TWO WAY	4	BEND Y-Y	0.309	0.309	0.309	0.116	0.079	0.087	0.000	0.000	0.000	0.000	0.000	
C110	SC1	TWO WAY	4	BEND Y-Y	0.306	0.306	0.306	0.118	0.076	0.092	0.000	0.000	0.000	0.000	0.000	
E101	SE1	CSUP	5	BEND Y-Y	0.464	0.464	0.464	0.266	0.389	0.386	0.000	0.000	0.000	0.000	0.000	
E102	SE1	CSUP	4	PUNCH-YZ	0.280	0.280	0.117	0.183	0.271	0.280	0.000	0.000	0.000	0.000	0.000	
E103	SE1	CSUP	5	BEND Y-Y	0.410	0.410	0.410	0.249	0.307	0.373	0.000	0.000	0.000	0.000	0.000	
E104	SE1	CSUP	4	BEND X-X	0.300	0.300	0.132	0.184	0.300	0.284	0.000	0.000	0.000	0.000	0.000	
E301	SE3	CSUP	4	PUNCH-YZ	0.234	0.234	0.101	0.142	0.210	0.234	0.000	0.000	0.000	0.000	0.000	
E302	SE3	CSUP	5	SHR-XY	0.250	0.250	0.164	0.187	0.165	0.199	0.000	0.000	0.000	0.000	0.000	
E303	SE3	CSUP	4	PUNCH-YZ	0.235	0.235	0.106	0.146	0.227	0.235	0.000	0.000	0.000	0.000	0.000	
E304	SE3	CSUP	6	SHR-XY	0.266	0.266	0.104	0.136	0.050	0.078	0.000	0.000	0.000	0.000	0.000	
E401	SE4	CSUP	4	BEND Y-Y	0.347	0.347	0.347	0.311	0.183	0.292	0.000	0.000	0.000	0.000	0.000	
E402	SE4	CSUP	5	BEND Y-Y	0.448	0.448	0.448	0.282	0.218	0.253	0.000	0.000	0.000	0.000	0.000	
E403	SE4	CSUP	4	BEND Y-Y	0.340	0.340	0.340	0.310	0.169	0.288	0.000	0.000	0.000	0.000	0.000	
E404	SE4	CSUP	5	BEND Y-Y	0.615	0.615	0.615	0.315	0.267	0.271	0.000	0.000	0.000	0.000	0.000	

4.4 SAMPLE OUTPUT REPORTS

This appendix contains samples of the output reports created for concrete elements. The following table lists the reports illustrated:

Report Description	Page
Concrete Section Properties	A-2
Concrete Reinforcement Properties	A-2
Post Processing Comments	A-3
Concrete Column Element Detail Report	A-4
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***** CONCRETE CIRCULAR SECTION PROPERTIES *****

GRP	SECTION	OD	*** STEEL ***			***** CONCRETE *****			MEMB	K-FACTOR		SECT	***** EFFECTIVE CROSS SECTION PROPERTIES *****			
			ES	DENS	EC	DENS	FC	CLASS		Y	Z		LEN	AXIAL	***** MOMENTS OF INERTIA *****	
		IN	1000	LB/FT3	1000	LB/FT3	KSI					FT	IN**2	IN**4	IN**4	IN**4
CC1	CYLIND	15.0	29.0	490.00	3.6	145.00	4.00	CL	1.0	1.0	0.0	176.71	4970.1	971.6	971.6	

***** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE CIRCULAR *****

GRP	SECTION	* SHEAR REINFORCEMENT *				***** REINFORCEMENT BARS *****				***** MOMENTS OF INERTIA *****						
		FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR	AXIAL	X-X	Y-Y	Z-Z		
		KSI	IN		IN		IN	KSI		DIA	AREA	IN**2	IN**4	IN**4	IN**4	
CC1	CYLIND	60.00	24.0	S	0.375	CIRCLE	7.50	60.00	S	0.75	3.53	0.2	56.0	56.0		

***** CONCRETE RECTANGULAR SECTION PROPERTIES *****

GRP	SECTION	HT	WT	*** STEEL ***			***** CONCRETE *****			MEMB	K-FACTOR		SECT	***** EFFECTIVE CROSS SECTION PROPERTIES *****			
				ES	DENS	EC	DENS	FC	CLASS		Y	Z		LEN	AXIAL	***** MOMENTS OF INERTIA *****	
		IN	IN	1000	LB/FT3	1000	LB/FT3	KSI				FT	IN**2	IN**4	IN**4	IN**4	
CR1	RECT1	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	3.0	360.00	16503.8	8640.0	3375.0	
CR1	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0	
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0	
CR2	RECT3	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	0.0	360.00	16503.8	8640.0	3375.0	
CR2	RECT2	24.0	15.0	29.0	490.00	3.6	145.00	4.00	BM	1.0	1.0	1.7	360.00	16503.8	8640.0	3375.0	

***** REINFORCEMENT PROPERTIES FOR CONCRETE SECTION TYPE RECTANGULAR *****

GRP	SECTION	* SHEAR REINFORCEMENT *				***** REINFORCEMENT BARS *****				***** MOMENTS OF INERTIA *****						
		FY	SPACE	TYPE	DIA.	PATTERN	LOC.	FY	END	BAR	AXIAL	X-X	Y-Y	Z-Z		
		KSI	IN		IN		IN	KSI		DIA	AREA	IN**2	IN**4	IN**4	IN**4	
CR1	RECT1	60.00	12.0	T	0.375	ROW1	2.50	60.00	S	0.50	0.79	0.0	0.0	10.9		
CR1	RECT1	60.00	12.0	T	0.375	ROW2	22.50	60.00	S	0.38	0.23	0.0	0.0	5.7		

*** SACS POST PROCESSOR COMMENTS ***

** THE USER SHOULD TAKE NOTE OF THE FOLLOWING COMMENTS REGARDING THE SACS POST PROCESSOR OUTPUT **

BEAMS

- (1) INTERNAL LOADS FOR MEMBERS ARE PRESENTED IN THE CLASSICAL ENGINEERING SIGN CONVENTION AS DESCRIBED BY TIMOSHENKO
- (2) IF THE AXIAL LOAD ON A MEMBER EXCEEDS THE AISC ALLOWABLE BUCKLING LOAD, THEN THE AXIAL UNITY CHECK VALUE FOR THE MEMBER IS SET EQUAL TO 100 TO INDICATE THAT THE MEMBER HAS BUCKLED
- (3) THE MAXIMUM COMBINED UNITY CHECK CAN BE THE MAXIMUM SHEAR UNITY CHECK IF IT IS GREATER THAN THE MAXIMUM UNITY CHECK DUE TO BENDING AND AXIAL LOAD
- (4) THE FOLLOWING ABBREVIATIONS ARE USED TO DESCRIBE THE CRITICAL UNITY CHECK CONDITIONS:
 - SHEAR - SHEAR
 - CM+BN - COMPRESSION WITH BENDING
 - TN+BN - TENSION WITH BENDING
 - WEB-SH - WEB SHEAR
 - FLG-SH - FLANGE SHEAR
 - EULER - EULER BUCKLING
- (5) THE FOLLOWING ABBREVIATIONS ARE USED TO DESCRIBE THE CRITICAL UNITY CHECK CONDITIONS FOR CONCRETE:
 - COLBUC - COLUMN BUCKLING
 - CM+BN - COMPRESSION WITH BENDING IN COLUMN ELEMENT
 - TN+BN - TENSION WITH BENDING IN COLUMN ELEMENT
 - SHEAR - SHEAR
 - TORS - TORSION
 - BEND-Y - PURE BENDING IN BEAM ELEMENT ABOUT LOCAL Y AXIS
 - REINF - REINFORCEMENT RATIO

PLATES

- (1) MEMBRANE STRESSES ARE GIVEN AT THE NEUTRAL AXIS OF THE PLATE IN THE LOCAL COORDINATE SYSTEM OF THE PLATE . ALSO THE PRINCIPAL MEMBRANE STRESS AND MAXIMUM SHEAR STRESS ARE GIVEN
- (2) THE DIRECT STRESSES RESULTING FROM OUT OF PLANE BENDING ARE GIVEN AT THE UPPER SURFACE OF THE PLATE (POSITIVE LOCAL Z DIRECTION) IN THE LOCAL COORDINATE SYSTEM OF THE PLATE . ALSO THE PRINCIPAL BENDING STRESS AND MAXIMUM SHEAR STRESS ARE GIVEN
- (3) THE MAXIMUM PRINCIPAL STRESS AND MAXIMUM SHEAR STRESS FOR THE COMBINED MEMBRANE AND BENDING STRESS ARE GIVEN . THE UNITY CHECK VALUE IS BASED ON THESE STRESSES

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE COLUMN ELEMENTS

MEMBER	GRP	LOAD CASE	DIST FROM END	AXIAL LOAD	RESULT MOMENT	MOMENT ANGLE	MOM MAGN-IFIER	SHEAR FY	SHEAR FZ	TORSION MX	***** AXIAL	ALLOWABLES BEND	***** SHEAR-Y	***** SHEAR-Z	UNITY TORS. CHECK
			FT	KIPS	IN-KIP	DEG		KIPS	KIPS	IN-KIP	KIPS	IN-KIP	KIPS	IN-KIP	RATIO
1- 101	CC1	4	0.0	-27.1	0.0	0.000	1.000	2.3	2.9	0.0	510.55	0.00	22.54	22.54	0.13
		5	0.0	-19.6	0.0	0.000	1.000	-0.6	2.2	0.0	510.55	0.00	22.19	22.19	0.10
		6	0.0	-8.3	0.0	0.000	1.000	-1.7	0.6	0.0	510.55	0.00	21.66	21.66	0.08
		4	4.5	-25.8	199.2	37.778	1.000	2.3	2.9	0.0	147.32	1136.62	22.48	22.48	0.18
		5	4.5	-18.7	118.6	5.497	1.000	1.0	2.2	0.0	179.85	1144.26	22.14	22.14	0.10
		6	4.5	-7.4	58.8	-54.389	1.000	-0.1	0.6	0.0	142.93	1130.57	21.62	21.62	0.05
		4	9.0	-24.5	398.4	37.778	1.000	2.3	2.9	0.0	58.67	953.27	22.41	22.41	0.42
		5	9.0	-17.7	260.0	24.746	1.000	2.6	2.2	0.0	64.83	953.18	22.09	22.09	0.27
		6	9.0	-6.6	68.9	-6.475	1.000	1.6	0.6	0.0	103.00	1076.07	21.58	21.58	0.07
3- 103	CC1	4	0.0	-23.4	0.0	0.000	1.000	-2.3	2.9	0.0	510.55	0.00	22.36	22.36	0.13
		5	0.0	-18.2	0.0	0.000	1.000	-2.6	2.2	0.0	510.55	0.00	22.12	22.12	0.12
		6	0.0	-10.2	0.0	0.000	1.000	-1.6	0.6	0.0	510.55	0.00	21.75	21.75	0.07
		4	4.5	-21.9	199.4	-37.808	1.000	-2.3	2.9	0.0	122.19	1113.30	22.29	22.29	0.18
		5	4.5	-17.1	183.7	-49.981	1.000	-2.6	2.2	0.0	99.51	1068.00	22.07	22.07	0.17
		6	4.5	-9.2	90.6	-67.812	1.000	-1.6	0.6	0.0	110.53	1087.35	21.70	21.70	0.08
		4	9.0	-20.4	398.8	-37.808	1.000	-2.3	2.9	0.0	49.38	967.79	22.22	22.22	0.41
		5	9.0	-16.0	367.5	-49.981	1.000	-2.6	2.2	0.0	42.38	973.76	22.02	22.02	0.38
		6	9.0	-8.3	181.3	-67.812	1.000	-1.6	0.6	0.0	44.34	970.53	21.66	21.66	0.19
7- 107	CC1	4	0.0	-26.9	0.0	0.000	1.000	-2.2	-2.9	0.0	510.55	0.00	22.53	22.53	0.13
		5	0.0	-20.9	0.0	0.000	1.000	-2.6	-2.2	0.0	510.55	0.00	22.25	22.25	0.12
		6	0.0	-10.3	0.0	0.000	1.000	-1.6	-0.6	0.0	510.55	0.00	21.75	21.75	0.07
		4	4.5	-25.2	198.8	217.671	1.000	-2.2	-2.9	0.0	143.65	1133.50	22.45	22.45	0.18
		5	4.5	-19.6	183.2	229.903	1.000	-2.6	-2.2	0.0	118.52	1105.52	22.19	22.19	0.17
		6	4.5	-9.2	90.7	247.816	1.000	-1.6	-0.6	0.0	110.01	1086.66	21.70	21.70	0.08
		4	9.0	-23.5	397.5	217.671	1.000	-2.2	-2.9	0.0	56.75	956.79	22.37	22.37	0.41
		5	9.0	-18.4	366.4	229.903	1.000	-2.6	-2.2	0.0	48.57	966.78	22.13	22.13	0.38
		6	9.0	-8.1	181.4	247.816	1.000	-1.6	-0.6	0.0	43.37	970.63	21.65	21.65	0.19
5- 105	CC1	4	0.0	-28.0	0.0	0.000	1.000	2.3	-2.9	0.0	510.55	0.00	22.58	22.58	0.13
		5	0.0	-20.3	0.0	0.000	1.000	-0.6	-2.2	0.0	510.55	0.00	22.22	22.22	0.10
		6	0.0	-9.0	0.0	0.000	1.000	-1.7	-0.6	0.0	510.55	0.00	21.69	21.69	0.08
		4	4.5	-26.2	199.1	142.339	1.000	2.3	-2.9	0.0	149.62	1138.30	22.49	22.49	0.17
		5	4.5	-18.9	118.8	174.639	1.000	1.0	-2.2	0.0	182.37	1144.41	22.15	22.15	0.10
		6	4.5	-7.8	58.8	234.387	1.000	-0.1	-0.6	0.0	151.26	1137.71	21.63	21.63	0.05
		4	9.0	-24.3	398.3	142.339	1.000	2.3	-2.9	0.0	58.28	954.09	22.40	22.40	0.42
		5	9.0	-17.5	260.1	155.389	1.000	2.6	-2.2	0.0	64.17	954.25	22.09	22.09	0.27
		6	9.0	-6.6	68.9	186.439	1.000	1.6	-0.6	0.0	103.00	1076.06	21.58	21.58	0.07

SACS-IV SYSTEM MEMBER DETAIL REPORT FOR CONCRETE BEAM ELEMENTS

MEMBER	GRP	LOAD CASE	DIST FROM END FT	*****		APPLIED		*****		***** ALLOWABLES *****				CRIT. COND	UNITY CHECK RATIO
				AXIAL LOAD KIPS	MOMENT MY IN-KIP	SHEAR FZ KIPS	TORSION MX IN-KIP	REINF. RATIO	MOMENT MY IN-KIP	SHEAR FZ KIPS	TORSION MX IN-KIP	REINF. RATIO			
101- 105	CRL	4	0.0	-2.9	-112.7	7.4	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56	
		5	0.0	-2.2	-84.5	5.5	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42	
		6	0.0	-0.6	-38.5	2.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15	
		4	10.0	-2.9	332.7	0.0	-2.5	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.53	
		5	10.0	-2.2	249.5	0.0	-1.8	0.005	217.5	8.2	78.0	0.033	BEND-Y	1.15	
		6	10.0	-0.6	80.0	0.0	0.1	0.005	217.5	8.2	74.6	0.033	BEND-Y	0.37	
		4	20.0	-2.9	-101.6	-7.3	-2.5	0.009	331.0	13.1	22.0	0.017	SHEAR	0.56	
		5	20.0	-2.2	-76.2	-5.5	-1.8	0.009	331.0	13.1	22.0	0.017	SHEAR	0.42	
6	20.0	-0.6	-39.0	-2.0	0.1	0.009	331.0	13.1	19.4	0.017	SHEAR	0.15			
103- 117	CLL	4	0.0	-2.4	-47.0	5.7	-3.6	0.009	331.0	13.1	24.4	0.017	SHEAR	0.44	
		5	0.0	-1.8	-35.2	4.3	-2.7	0.009	331.0	13.1	24.4	0.017	SHEAR	0.33	
		6	0.0	-0.5	-41.8	2.1	0.2	0.009	331.0	13.1	19.9	0.017	SHEAR	0.16	
		4	2.5	-2.4	97.1	3.9	-3.6	0.005	217.5	8.2	24.0	0.033	SHEAR	0.47	
		5	2.5	-1.8	72.9	2.9	-2.7	0.005	217.5	8.2	24.0	0.033	SHEAR	0.36	
		6	2.5	-0.5	12.6	1.6	0.2	0.005	217.5	8.2	19.8	0.033	SHEAR	0.19	
		4	5.0	-2.4	186.3	2.1	-3.6	0.005	217.5	8.2	28.2	0.033	BEND-Y	0.86	
		5	5.0	-1.8	139.7	1.5	-2.7	0.005	217.5	8.2	28.2	0.033	BEND-Y	0.64	
6	5.0	-0.5	52.2	1.1	0.2	0.005	217.5	8.2	20.1	0.033	BEND-Y	0.24			
101- 111	CRL	4	0.0	-2.3	-273.6	17.1	-248.8	0.004	894.9	29.8	340.3	0.022	TORS	0.73	
		5	0.0	-2.6	-141.9	12.1	-186.6	0.004	894.9	29.2	343.2	0.022	TORS	0.54	
		6	0.0	-1.6	-10.8	4.6	-40.1	0.004	894.9	43.4	306.2	0.022	TORS	0.13	
		4	3.0	-2.3	315.0	15.6	-248.8	0.010	348.8	15.5	282.0	0.024	SHEAR	1.01	
		5	3.0	-2.6	273.7	11.0	-186.6	0.010	348.8	15.2	287.6	0.024	BEND-Y	0.78	
		6	3.0	-1.6	136.7	3.6	-40.1	0.010	348.8	17.8	250.2	0.024	BEND-Y	0.39	
		4	3.0	-2.3	315.0	15.6	-248.8	0.009	925.4	19.1	271.2	0.024	TORS	0.92	
		5	3.0	-2.6	273.7	11.0	-186.6	0.009	925.4	18.7	275.3	0.024	TORS	0.68	
		6	3.0	-1.6	136.7	3.6	-40.1	0.009	925.4	26.9	243.5	0.024	TORS	0.16	
		4	6.7	-2.3	957.4	13.6	-248.8	0.009	925.4	18.1	279.5	0.024	BEND-Y	1.03	
5	6.7	-2.6	723.9	9.5	-186.6	0.009	925.4	17.6	283.5	0.024	BEND-Y	0.78			
6	6.7	-1.6	267.5	2.4	-40.1	0.009	925.4	26.9	275.2	0.024	BEND-Y	0.29			
103- 112	CRL	4	0.0	-2.3	-210.6	16.6	314.7	0.004	894.9	27.2	351.7	0.022	TORS	0.89	
		5	0.0	-2.6	-266.9	13.2	236.0	0.004	894.9	27.7	349.7	0.022	TORS	0.67	
		6	0.0	-1.6	-190.7	6.1	36.7	0.004	894.9	43.4	274.3	0.022	BEND-Y	0.21	
		4	3.0	-2.3	359.0	15.0	314.7	0.010	348.8	14.5	305.1	0.024	SHEAR	1.04	
		5	3.0	-2.6	186.1	12.0	236.0	0.010	348.8	14.7	300.0	0.024	SHEAR	0.82	
		6	3.0	-1.6	10.7	5.1	36.7	0.010	348.8	17.8	216.4	0.024	SHEAR	0.29	

SACS-IV SYSTEM CONCRETE ELEMENT UNITY CHECK REPORT
ACI 318R-89 (REVISED 92)

MEMBER	GRP	MAX. CRIT. CHECK	COND	LOAD NO.	DIST FROM END FT	* * * * *				U N I T Y				C H E C K				C O M P O N E N T S				* * * * *			
						SHEAR	*TORSION*	*REINF*	*BEAMS*	*****	*****	*****	*****	*****	*****	*****	*****	COLUMNS	*****	*****	*****	*****	*****	*****	*****
						FZ	MX	RATIO	MY	AXIAL	EULER	BUCKLING	SHEAR	BEND	MOM	CMY	CMZ								
										FX	Y-Y	Z-Z	FY												
1- 101	CC1	0.42	CM+BN	4	9.00	0.130	0.000	0.559		0.418	0.010	0.010	0.101	0.418	1.00	0.60	0.60								
3- 103	CC1	0.41	CM+BN	4	9.00	0.131	0.000	0.559		0.412	0.008	0.008	0.102	0.412	1.00	0.60	0.60								
7- 107	CC1	0.41	CM+BN	4	9.00	0.130	0.000	0.559		0.415	0.010	0.010	0.101	0.415	1.00	0.60	0.60								
5- 105	CC1	0.42	CM+BN	4	9.00	0.130	0.000	0.559		0.417	0.010	0.010	0.101	0.417	1.00	0.60	0.60								
101- 105	CRL	1.53	BEND-Y	4	10.00	0.006	0.031	0.161	1.529																
103- 117	CLL	0.86	BEND-Y	4	5.00	0.251	0.126	0.161	0.856																
101- 111	CR1	1.03	BEND-Y	4	6.67	0.752	0.890	0.368	1.035																
103- 112	CR1	1.10	TORS	4	3.00	0.877	1.097	0.368	0.388																
105- 113	CR1	1.03	BEND-Y	4	6.67	0.761	0.921	0.368	1.029																
107- 114	CR1	1.05	BEND-Y	4	6.67	0.773	0.978	0.368	1.048																
111- 112	CR2	1.08	BEND-Y	4	5.00	0.023	0.105	0.368	1.077																
113- 114	CR2	1.07	BEND-Y	4	5.00	0.024	0.040	0.368	1.074																
111- 113	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136																
112- 114	CT1	1.14	BEND-Y	4	15.00	1.083	1.084	0.161	1.136																
117- 107	CLL	1.38	BEND-Y	4	7.50	0.330	0.704	0.161	1.384																

SACS-IV SYSTEM CONCRETE MEMBER GROUP SUMMARY REPORT
ACI 318R-89 (REVISED 92)

GRP ID	CRITICAL MEMBER	LOAD COND	MAX. UNITY CHECK	DIST FROM END FT	***** APPLIED LOADS *****				***** ALLOWABLE LOADS *****				CRIT COND	EFFECTIVE KL/R		
					AXIAL KIPS	BEND IN-KIP	LOADS TORSION IN-KIP	***** REINF RATIO	AXIAL KIPS	EULER KIPS	BEND IN-KIP	LOADS TORSION IN-KIP		***** REINF RATIO	Y	Z
CC1	1- 101	4	0.42	9.0	-24.51	398.4	0.0	0.017	58.7	2412.7	953.3	38.6	0.031	CM+BN	28.8	28.8
CRL	101- 105	4	1.53	10.0	-2.89	332.7	-2.5	0.005			217.5	78.0	0.033	BEND-Y	52.3	35.6
CLL	117- 107	4	1.38	7.5	-2.89	301.1	53.9	0.005			217.5	76.6	0.033	BEND-Y	39.2	26.7
CR1	103- 112	4	1.10	3.0	-2.26	359.0	314.7	0.009			925.4	286.9	0.024	TORS	11.5	18.5
CR2	111- 112	4	1.08	5.0	-2.26	996.6	32.9	0.009			925.4	312.1	0.024	BEND-Y	11.5	18.5
CT1	111- 113	4	1.14	15.0	-0.03	247.2	-2.9	0.005			217.5	2.7	0.033	BEND-Y	52.3	43.9

SACS-IV CONCRETE MEMBER UNITY CHECK RANGE SUMMARY

GROUP I - UNITY CHECKS GREATER THAN 1.33

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****					SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST		
		UNITY CHECK	LOAD CASE		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS	KLY/RX	KLZ/RZ	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE	
101-	105	CRL	1.53	4	10.0	-2.9	332.7	0.8	0.0	0.0	52.26	35.55	1.15	5	0.56	4
117-	107	CLL	1.38	4	7.5	-2.9	301.1	-0.8	0.0	-1.5	39.20	26.66	1.04	5	0.86	4

GROUP II - UNITY CHECKS GREATER THAN 1.00 AND LESS THAN 1.33

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****					SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST		
		UNITY CHECK	LOAD CASE		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS	KLY/RV	KLZ/RZ	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE	
101-	111	CR1	1.03	4	6.7	-2.3	957.4	1.5	0.0	13.6	11.55	18.48	1.01	4	0.92	4
103-	112	CR1	1.10	4	3.0	-2.3	359.0	-0.1	0.0	15.0	11.55	18.48	1.07	4	1.04	4
105-	113	CR1	1.03	4	6.7	-2.3	952.2	-1.6	0.0	13.5	11.55	18.48	1.01	4	0.95	4
107-	114	CR1	1.05	4	6.7	-2.3	969.7	1.5	0.0	13.1	11.55	18.48	1.01	4	1.00	4
111-	112	CR2	1.08	4	5.0	-2.3	996.6	1.1	0.0	-0.6	11.55	18.47	1.07	4	1.05	4
113-	114	CR2	1.07	4	5.0	-2.3	994.3	-1.2	0.0	-0.7	11.55	18.47	1.07	4	1.05	4
111-	113	CT1	1.14	4	15.0	0.0	247.2	0.4	0.0	-5.7	52.26	43.86	1.14	4	1.11	4
112-	114	CT1	1.14	4	15.0	0.0	247.1	-0.3	0.0	-5.7	52.26	43.86	1.14	4	1.11	4

GROUP III - UNITY CHECKS GREATER THAN 0.00 AND LESS THAN 0.50

MEMBER	GRP	MAX.		DIST FROM END FT	***** CRITICAL LOADING *****						SLENDERNESS RATIO		SECOND-HIGHEST		THIRD-HIGHEST	
		UNITY CHECK	LOAD CASE		AXIAL LOAD KIPS	MOMENT MY IN-KIP	MOMENT MZ IN-KIP	SHEAR FY KIPS	SHEAR FZ KIPS	KLY/RY	KLZ/RZ	UNITY CHECK	LOAD CASE	UNITY CHECK	LOAD CASE	
1-	101	CC1	0.42	4	9.0	-24.5	314.9	244.1	2.3	2.9	28.80	28.80	0.27	5	0.18	4
3-	103	CC1	0.41	4	9.0	-20.4	315.1	-244.5	-2.3	2.9	28.80	28.80	0.38	5	0.19	6
7-	107	CC1	0.41	4	9.0	-23.5	-314.7	-242.9	-2.2	-2.9	28.80	28.80	0.38	5	0.19	6
5-	105	CC1	0.42	4	9.0	-24.3	-315.3	243.3	2.3	-2.9	28.80	28.80	0.27	5	0.17	4

CONCRETE SLAB ALLOWABLE SUMMARY REPORT

PLATE ID	GRP ID	TYPE	CRIT. LOAD CASE	CRIT. COND.	MAXIMUM UNITY CHECK	***** X-DIRECTION *****			***** Y-DIRECTION *****			IN-PLANE SHEAR XY K/IN
						FORCE X K/IN	MOMENT Y-Y IN-K/IN	SHEAR XZ K/IN	FORCE Y K/IN	MOMENT X-X IN-K/IN	SHEAR YZ K/IN	
X01	SA1	ONE-X	4	SHR-XZ	0.307	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X02	SA1	ONE-X	4	BEND Y-Y	0.427	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X03	SA1	ONE-X	5	BEND Y-Y	0.363	0.00	2.2	0.22	0.00	0.0	0.00	0.22
X04	SA1	ONE-X	4	SHR-XZ	0.307	0.00	2.2	0.22	0.00	0.0	0.00	0.22
C107	SC1	TWO WAY	4	BEND X-X	0.304	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C108	SC1	TWO WAY	4	BEND X-X	0.243	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C109	SC1	TWO WAY	4	BEND Y-Y	0.309	0.00	3.0	0.22	0.00	3.0	0.22	0.22
C110	SC1	TWO WAY	4	BEND Y-Y	0.306	0.00	3.0	0.22	0.00	3.0	0.22	0.22
E101	SE1	CSUP	5	BEND Y-Y	0.464	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E102	SE1	CSUP	4	PUNCH-YZ	0.280	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E103	SE1	CSUP	5	BEND Y-Y	0.410	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E104	SE1	CSUP	4	BEND X-X	0.300	0.00	2.2	0.33	0.00	3.0	0.33	0.22
E301	SE3	CSUP	4	PUNCH-YZ	0.234	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E302	SE3	CSUP	5	SHR-XY	0.250	0.00	3.0	0.23	0.00	3.0	0.23	0.22
E303	SE3	CSUP	4	PUNCH-YZ	0.235	0.00	3.0	0.23	0.00	3.0	0.23	0.22

CONCRETE SLAB UNITY CHECK SUMMARY REPORT

PLATE ID	GRP ID	TYPE	CRIT. LOAD CASE	CRIT. COND.	MAXIMUM UNITY CHECK	***** X-DIRECTION *****			***** Y-DIRECTION *****			IN-PLANE SHEAR XY
						FORCE X	MOMENT Y-Y	SHEAR XZ	FORCE Y	MOMENT X-X	SHEAR YZ	
X01	SA1	ONE-X	4	SHR-XZ	0.307	0.307	0.188	0.307	0.000	0.000	0.000	0.000
X02	SA1	ONE-X	4	BEND Y-Y	0.427	0.427	0.427	0.001	0.000	0.000	0.000	0.000
X03	SA1	ONE-X	5	BEND Y-Y	0.363	0.363	0.363	0.287	0.000	0.000	0.000	0.000
X04	SA1	ONE-X	4	SHR-XZ	0.307	0.307	0.188	0.307	0.000	0.000	0.000	0.000
C107	SC1	TWO WAY	4	BEND X-X	0.304	0.304	0.287	0.011	0.304	0.138	0.000	0.000
C108	SC1	TWO WAY	4	BEND X-X	0.243	0.243	0.079	0.092	0.243	0.057	0.000	0.000
C109	SC1	TWO WAY	4	BEND Y-Y	0.309	0.309	0.309	0.116	0.079	0.087	0.000	0.000
C110	SC1	TWO WAY	4	BEND Y-Y	0.306	0.306	0.306	0.118	0.076	0.092	0.000	0.000
E101	SE1	CSUP	5	BEND Y-Y	0.464	0.464	0.464	0.266	0.389	0.386	0.000	0.000
E102	SE1	CSUP	4	PUNCH-YZ	0.280	0.280	0.117	0.183	0.271	0.280	0.000	0.000
E103	SE1	CSUP	5	BEND Y-Y	0.410	0.410	0.410	0.249	0.307	0.373	0.000	0.000
E104	SE1	CSUP	4	BEND X-X	0.300	0.300	0.132	0.184	0.300	0.284	0.000	0.000
E301	SE3	CSUP	4	PUNCH-YZ	0.234	0.234	0.101	0.142	0.210	0.234	0.000	0.000
E302	SE3	CSUP	5	SHR-XY	0.250	0.250	0.164	0.187	0.165	0.199	0.000	0.000
E303	SE3	CSUP	4	PUNCH-YZ	0.235	0.235	0.106	0.146	0.227	0.235	0.000	0.000

5 INPUT LINES

CONCRETE MEMBER GROUP

COLUMNS	COMMENTARY
GENERAL	THE GROUP LINES DESCRIBE GROUPS OF MEMBERS HAVING IDENTICAL STRUCTURAL, MATERIAL AND CODE CHECK PARAMETERS. MEMBERS WHOSE CROSS SECTIONS VARY BETWEEN THE END JOINTS (NON-PRISMATIC MEMBERS) CAN BE DESCRIBED WITH UP TO TWENTY (20) DIFFERENT CROSS SECTION TYPES. FOR THIS TYPE OF DESCRIPTION, THE MEMBER GROUP LABEL IS SUCCESSIVELY REPEATED SUCH THAT EACH 'GRUP' ENTRY WILL DEFINE THE CROSS SECTION AND SEGMENT LENGTH FOR SEGMENTS MOVING FROM JOINT A TO JOINT B.
(1- 4)	ENTER 'GRUP'. THE FIRST OF THIS DATA BLOCK MUST BE A BLANK 'GRUP' HEADER LINE.
(5)	ENTER 'C' TO DESIGNATE GROUP AS A CONCRETE GROUP.
(6- 8)	ENTER A UNIQUE GROUP LABEL FOR THIS MEMBER GROUP. IF THIS LABEL IS REPEATED ON SUCCESSIVE LINES THE PROGRAM WILL ASSUME IT IS A NON-PRISMATIC GROUP WITH MULTIPLE SEGMENTS.
(10-16)	ENTER A SECTION LABEL FROM A 'SECT' DATA OR FROM THE DEFAULT SECTION LIBRARY. THE LABEL MUST BE LOCATED IN THE FIELD AS IT APPEARS ON THE SECTION LINE.
(17)	THE FOLLOWING DESIGNATIONS CAN BE APPLIED TO A GROUP FOR REDESIGN. LEAVE BLANK FOR DEFAULT TO 'REDESIGN' LINE. 'D' - DECREASE SIZE ONLY 'E' - CONSTANT DEPTH (O.D.), DECREASE SIZE ALLOWED 'G' - MINIMUM WEIGHT, DECREASE SIZE ALLOWED 'I' - INCREASE SIZE ONLY 'J' - CONSTANT DEPTH (O.D.), INCREASE SIZE ONLY 'L' - MINIMUM WEIGHT, INCREASE SIZE ONLY 'U' - USER SPECIFIED DESIGN, DECREASE SIZE ALLOWED 'X' - SKIP THIS GROUP DURING REDESIGN.
(18-29)	ENTER STEEL PROPERTIES. NOTE: 'M' PREFIX IS FOR MEGA.
(31-45)	ENTER CONCRETE PROPERTIES. DEFAULT FOR EC IS CALCULATED FROM DENSITY AND Fc BY $E_c = 33(\text{DENSITY} * 1.5) * \text{SQRT}(F_c)$.

COLUMNS	COMMENTARY
(47)	MEMBER CLASSIFICATION. ENTER '1' COLUMN (AXIAL + FLEXURE) BRACED AGAINST SIDESWAY; '2' COLUMN (AXIAL + FLEXURE) UNBRACED AGAINST SIDESWAY; '3' BEAM (FLEXURE ONLY); '4' SKIP OUTPUT FOR THIS BRACED COLUMN GROUP; '5' SKIP OUTPUT FOR THIS UNBRACED COLUMN GROUP; '6' SKIP OUTPUT FOR THIS BEAM GROUP; '8' INCLUDE THIS PREVIOUSLY SKIPPED GROUP IN POST.
(48-51)	ENTER SUPPORT THICKNESS. ONE HALF OF THIS VALUE IS SUBTRACTED FROM BOTH ENDS OF THE MEMBER FOR EULER BUCKLING ALLOWABLES AND TO DETERMINE THE LOCATION OF THE FACE OF THE MEMBER.
(52-59)	EFFECTIVE LENGTH FACTORS - SEE APPENDIX.
(60-64)	YIELD STRESS OF SHEAR REINFORCEMENT.
(65-69)	SHEAR REINFORCEMENT SPACING.
(70)	SHEAR REINFORCEMENT TYPE. ENTER 'T' TIES OR STIRRUPS 'S' SPIRALS
(71-76)	ENTER THE REINFORCEMENT BAR DIAMETER.
(77-80)	NON-PRISMATIC MEMBER SEGMENT LENGTH. TWO FORMS FOR SPECIFYING THE SEGMENT LENGTHS ON SUCCESSIVE 'GRUP' LINES ARE AVAILABLE. (A) LENGTH SPECIFICATION. EACH SEGMENT LENGTH FROM 1 TO N ARE ENTERED IN FEET. ANY ONE SEGMENT LENGTH CAN BE LEFT BLANK ALLOWING THE PROGRAM TO CALCULATE ITS LENGTH. (B) FRACTIONAL SPECIFICATION. EACH SEGMENT LENGTH FROM 1 TO N IS ENTERED AS A FRACTION OF THE TOTAL MEMBER LENGTH. ALL SEGMENT LENGTHS MUST BE ENTERED AND SUM TO EXACTLY 1.0.

LINE LABEL	CON- CRETE GROUP	GROUP LABEL	SECTION LABEL	RE- DESIGN CODE SEE ABOVE	STEEL PROPERTIES		CONCRETE PROPERTIES			MEMBER CLASS SEE ABOVE	SUPPORT THICKNESS	EFFECTIVE LENGTH FACTORS		SHEAR REINFORCEMENT				MEMBER SEGMENT LENGTH (20 MAX)
					Es	DENSITY	Ec	DENSITY	Fc			KY	KZ	SY	SPACING	REINF TYPE	BAR DIAM	
GRUP	C																	
1-- 4	5	6-- 8	10--16	17	18--23	24--29	31--35	36--40	41--45	47	48--51	52--55	56--59	60--64	65--69	70	71--76	77--80
DEFAULT					29.0 ENGL	490.0 ENGL		145.0 ENGL	4.0 ENGL	'1'		1	1	50.0 ENGL		'T'		
ENGLISH					MPSI	LB/CU.FT	MPSI	LB/CU.FT	KSI		FT			KSI	IN		IN	FT
METRIC (KN)					MN/SQ.CM	TONNE/CU.M	MN/SQ.CM	TONNE/CU.M	KN/SQ.CM		M			KN/SQ.CM	CM		CM	M
METRIC (KG)					MKG/SQ.CM	TONNE/CU.M	MKG/SQ.CM	TONNE/CU.M	KG/SQ.CM		M			KG/SQ.CM	CM		CM	M

LOAD CONDITION HEADER LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	THIS HEADER LINE IS THE FIRST LINE OF EACH BASIC LOAD CONDITION IN THE DATA SET.
GENERAL	THE PROGRAMS PROVIDE FOR THE ANALYSIS OF A VIRTUALLY UNLIMITED NUMBER OF BASIC LOAD CONDITIONS.
(1- 6)	ENTER THE LOAD CONDITION HEADER LINE LABEL 'LOADCN'.
(7-10)	ENTER THE BASIC LOAD CONDITION NAME.
(47-50)	ENTER THE LOAD TYPE: 'SWAY' - DESIGNATES A LOAD CASE THAT IS ASSUMED TO CAUSE APPRECIABLE SIDE SWAY. THIS IS THE DEFAULT. 'DEAD' - DESIGNATES A LOAD CASE THAT CAUSES NO APPRECIABLE SIDE SWAY.

LINE LABEL	BASIC LOAD CONDITION NAME	LEAVE BLANK	LOAD TYPE	LEAVE BLANK
LOADCN				
1-- 6	7-->10	14-----46	47--50	51-----80
DEFAULT			'SWAY'	

CONCRETE SLAB TO COLUMN CONNECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA TYPE IS USED TO DESCRIBE THE COLUMN LOCATION AND SIZE FOR CONCRETE SLAB TO COLUMN CONNECTION. THIS DATA IS USED TO CALCULATE THE PUNCHING SHEAR ALLOWABLE FOR CONCRETE SLABS IN THE VICINITY OF A COLUMN OR CONCENTRATED LOAD.
(8-10)	ENTER THE CRITICAL SECTION LOCATION. NOTE THAT THIS IS THE LOCATION OF THE COLUMN OR CONCENTRATED LOAD WITH RESPECT TO THE OVERALL STRUCTURE, NOT WITH RESPECT TO THE CONCRETE PLATE ELEMENT TO WHICH IT IS ATTACHED. 'INT' FOR AN INTERIOR COLUMN. 'EDX' FOR A COLUMN ON AN EDGE PARALLEL TO LOCAL X AXIS. 'EDY' FOR A COLUMN ON AN EDGE PARALLEL TO LOCAL Y AXIS. 'COR' FOR A CORNER COLUMN.
(11)	ENTER THE SHAPE OF THE COLUMN OR CONCENTRATED LOAD FOOTPRINT. 'R' FOR RECTANGULAR 'C' FOR CIRCULAR
(12-23)	ENTER THE DIMENSIONS OF THE COLUMN IN THE LOCAL PLATE COORDINATES.

LINE LABEL	COLUMN LOCATION	COLUMN SHAPE	COLUMN DIMENSIONS		LEAVE BLANK
			X	Y	
PGRUP2					
1-- 6	8--10	11	12<--17	18<--23	24-----80
DEFAULT	'INT'	'R'			
ENGLISH			IN	IN	
METRIC			CM	CM	

CONCRETE SLAB GROUP DESCRIPTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO DESCRIBE CONCRETE SLAB CROSS SECTION PROPERTIES. THIS DATA FOLLOWS THE 'PSTIF' RECORDS, IF ANY.
(7- 9)	ENTER SLAB GROUP LABEL. THIS LABEL IS USED FOR SUBSEQUENT REFERENCING ON PLATE RECORDS.
(17)	ENTER 'A' FOR ONE WAY SLAB IN THE LOCAL X DIRECTION. ENTER 'B' FOR ONE WAY SLAB IN THE LOCAL Y DIRECTION. ENTER 'C' FOR TWO WAY SLAB. ENTER 'D' FOR GENERAL CONCRETE ELEMENT. ENTER 'E' FOR TWO WAY SLAB AT A COLUMN CONNECTION. NOTE: IF 'E' IS SELECTED, THEN THIS DATA RECORD MUST BE IMMEDIATELY FOLLOWED BY A 'PGRUP2' RECORD GIVING THE SLAB TO COLUMN CONNECTION DETAILS.
(36-41)	ENTER THE LOCAL Z OFFSETS TO BE USED WITH ALL PLATES OF THIS GROUP.
(42-48)	FOR STEEL REINFORCEMENTS, ENTER THE SECTION LABEL CORRESPONDING TO A REBAR DESCRIBED ON A 'PSTIF' RECORD. THE 'PSTIF' RECORD MUST BE AN 'RBR' TYPE.

COLUMNS	COMMENTARY
(49-54)	ENTER SPACING BETWEEN REBAR CENTERLINES.
(55)	ENTER 'X' OR 'Y' FOR REBARS IN LOCAL X OR Y DIRECTIONS.
(56)	ENTER 'B' FOR BOTTOM REBARS (REBAR LOCATED FROM SLAB BOTTOM) 'T' FOR TOP REBARS (REBAR LOCATED FROM SLAB TOP) 'S' FOR SYMMETRIC REBARS (BOTTOM AND TOP)
(58-72)	ENTER DATA FOR SECOND SET OF REBARS. IF MORE THAN TWO SETS OF REBARS ARE DESIRED,THEY CAN BE DESCRIBED ON ADDITIONAL 'PGRUP' RECORDS HAVING THE SAME GROUP LABEL FOR A MAXIMUM OF FIVE SETS.
(73-80)	ENTER THE CONCRETE WEIGHT DENSITY TO BE USED FOR WEIGHT CALCULATION.

LINE LABEL	PLATE GROUP LABEL	AUTO. NEUTRAL AXIS OFFSET 'Z'	CONCRETE SLAB THICKNESS	SLAB TYPE	CONCRETE PROPERTY DATA			PLATE LOCAL Z OFFSET	FIRST STIFFENER				SECOND STIFFENER				SEASTATE WEIGHT DENSITY
					E ---- 1000	POISSON'S RATIO U	COMP. STRENGTH FPC		SECTION LABEL	AVERAGE SPACING	'X' OR 'Y'	'B' 'T' OR 'S'	SECTION LABEL	AVERAGE SPACING	'X' OR 'Y'	'B' 'T' OR 'S'	
PGRUPC																	
1-- 6	7<-- 9	10	11<--16	17	18<--23	24<--29	30<--35	36<--41	42<--48	49<--54	55	56	58<--64	65<--70	71	72	73<--80
DEFAULT					CALC.	0.3	4.0 ENGL				'X'	'T'			'X'	'T'	145.0 ENGL
ENGLISH			IN		KSI		KSI	IN		IN				IN			LB/CU.FT
METRIC (KN)			CM		KN/SQ.CM		KN/SQ.CM	CM		CM				CM			TONNE/CU.M
METRIC (KG)			CM		KG/SQ.CM		KG/SQ.CM	CM		CM				CM			TONNE/CU.M

PLATE REINFORCEMENT BAR DESCRIPTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA TYPE IS USED TO DESCRIBE PLATE REBAR PROPERTIES FOR CONCRETE SLABS AND WALLS.
(7- 9)	ENTER 'RBR' TO IDENTIFY THE PLATE STIFFENER TYPE.
(11-17)	THE SECTION LABEL IDENTIFIES THIS STIFFENER FOR SUBSEQUENT REFERENCE ON A 'PGRUP' DATA RECORD. ANY ALPHANUMERIC NAME OF UP TO SEVEN CHARACTERS CAN BE USED. IT SHOULD BE RECOGNIZED THAT THIS LABEL MUST MATCH EXACTLY WITH THE CORRESPONDING ENTRY ON A 'PGRUP' RECORD WITH REGARDS TO CONTENT AND LOCATION WITHIN THE SEVEN COLUMN FIELD. FOR EXAMPLE, FOUR BLANKS FOLLOWED BY 'ABC' WOULD NOT MATCH 'ABC' FOLLOWED BY FOUR BLANKS.
(21-27)	ENTER THE REBAR DIAMETER.
(28-34)	ENTER THE LOCATION OF THE REBAR AS MEASURED FROM THE TOP OF THE SLAB FOR TOP LOCATION AND FROM THE BOTTOM OF THE SLAB FOR BOTTOM LOCATION.
(35-41)	ENTER THE REBAR YIELD STRESS.
(42-48)	ENTER THE REBAR ELASTIC MODULUS.
(49-55)	ENTER THE REBAR DENSITY.

LINE LABEL	STIFFENER TYPE	SECTION LABEL	REINFORCEMENT DETAILS					LEAVE BLANK
			DIAMETER	DISTANCE FROM EDGE	YIELD STRESS	ELASTIC MODULUS	DENSITY	
PSTIF	RBR							
1-- 5	7-- 9	11<--17	21<--27	28<--34	35<--41	42<--48	49<--55	56--80
DEFAULT					50.0 ENGL	29.0 ENGL	490.0 ENGL	
ENGLISH			IN	IN	KSI	MPSI	LB/CU.FT	
METRIC (KN)			CM	CM	KN/SQ.CM	MN/SQ.CM	TONNE/CU.M	
METRIC (KG)			CM	CM	KG/SQ.CM	MKG/SQ.CM	TONNE/CU.M	

CONCRETE REINFORCEMENT BAR PATTERN SECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO SPECIFY THE CROSS-SECTION PROPERTIES OF CONCRETE REINFORCEMENT BAR PATTERNS.
(1- 4)	ENTER 'SECT' ON EACH LINE OF THIS SET. THE FIRST LINE IN THIS SET SHOULD CONTAIN ONLY THE WORD 'SECT' AS A HEADER.
(6-12)	ENTER THE UNIQUE CROSS-SECTION LABEL FOR THIS PATTERN SECTION TYPE.
(16-18)	ENTER THE REINFORCEMENT BAR PATTERN TYPE FROM THE LIST BELOW. 'RRP' - ROW REINFORCEMENT BAR PATTERN 'CRP' - CIRCULAR REINFORCEMENT BAR PATTERN 'BRP' - BOX REINFORCEMENT BAR PATTERN 'PRP' - PRISMATIC REINFORCEMENT SECTION
(50-55)	ENTER THE REINFORCEMENT BAR DIAMETER.
(56-76)	ENTER CROSS-SECTION DETAILS AS ILLUSTRATED IN THE ACCOMPANYING FIGURES.

LINE LABEL	REBAR PATTERN SECTION LABEL	PATTERN TYPE	.	CROSS SECTION DETAILS				
			TYPE	A	B	C	D	E
			RRP	BAR DIAMETER	NUMBER OF BARS	N.A.	WIDTH	N.A.
			CRP	BAR DIAMETER	NUMBER OF BARS	N.A.	DIAMETER	GAMMA ANGLE (DEG)
			BRP	BAR DIAMETER	NUMBER HORIZ BARS	NUMBER VERT BARS	WIDTH	HEIGHT
			PRP	AREA OF REINF	N.A.	N.A.	N.A.	N.A.
SECT								
1-- 4	6--12	16<--18	49	50<--55	56<--60	61<--66	67<--71	72<--76
DEFAULT								
ENGLISH				IN			IN	IN
METRIC				CM			CM	CM

CONCRETE CROSS SECTION DATA LINE

COLUMNS	COMMENTARY
GENERAL	THIS DATA IS USED TO SPECIFY THE CROSS-SECTION PROPERTIES OF CONCRETE MEMBERS. REINFORCEMENT BAR PATTERNS FOR THIS SECTION ARE SPECIFIED ON THE 'SECT2' LINE IMMEDIATELY FOLLOWING THIS SECTION LINE.
(1- 4)	ENTER 'SECT' ON EACH LINE OF THIS SET. THE FIRST LINE IN THIS SET SHOULD CONTAIN ONLY THE WORD 'SECT' AS A HEADER.
(6-12)	ENTER THE UNIQUE CROSS SECTION LABEL FOR THIS CROSS SECTION TYPE.
(15)	ENTER 'R' IF REINFORCEMENT IS DEFINED ON THE 'SECT2' LINE IMMEDIATELY FOLLOWING.

COLUMNS	COMMENTARY
(16-18)	ENTER THE CROSS SECTION TYPE FROM THE LIST BELOW. THE CROSS SECTION TYPE IS USED TO SPECIFY THE EQUATIONS FOR STRESS CALCULATIONS AND STIFFNESS PROPERTIES (SEE APPENDIX) . 'CCS' - CONCRETE CIRCULAR SECTION 'CRS' - CONCRETE RECTANGULAR SECTION 'CTS' - CONCRETE TEE SECTION 'CRL' - CONCRETE RIGHT L SECTION 'CLL' - CONCRETE LEFT L SECTION 'CIS' - CONCRETE I SECTION
(19-48)	ENTER THE STIFFNESS PROPERTIES - AXIAL AREA, POLAR MOMENT OF INERTIA, MOMENT OF INERTIA ABOUT LOCAL Y AND Z AXES. NOTE: IF THE STIFFNESS PROPERTIES ARE LEFT BLANK, THEN THE PROGRAM WILL CALCULATE THE PROPERTIES FROM THE CROSS SECTION DETAIL DIMENSIONS.
(50-76)	ENTER CROSS SECTION DETAILS AS ILLUSTRATED IN THE ACCOMPANYING FIGURES.

LINE LABEL	CROSS SECTION LABEL	REIN FORCED	CROSS SECTION TYPE	STIFFNESS PROPERTIES				.	CROSS SECTION DETAILS				
				AX	J	IY	IZ	TYPE	A	B	C	D	E
								CCS	O.D.	N.A.	N.A.	N.A.	N.A.
								CRS	HEIGHT	WIDTH	N.A.	N.A.	N.A.
								CTS	HEIGHT	WEB WIDTH	EFF FLANGE WIDTH	FLANGE THICK	N.A.
								CLL	HEIGHT	WEB WIDTH	EFF FLANGE WIDTH	FLANGE THICK	N.A.
								CRL	HEIGHT	WEB WIDTH	EFF FLANGE WIDTH	FLANGE THICK	N.A.
								CIS	HEIGHT	WEB WIDTH	EFF FLANGE WIDTH	FLANGE THICK	N.A.
SECT													
1-- 4	6--12	15	16<--18	19<--24	25<--32	33<--40	41<--48	49	50<--55	56<--60	61<--66	67<--71	72<--76
DEFAULT													
ENGLISH				SQ.IN	IN**4	IN**4	IN**4		IN	IN	IN	IN	
METRIC				SQ.CM	CM**4	CM**4	CM**4		CM	CM	CM	CM	

CONCRETE CROSS SECTION DATA LINE 2

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS A CONTINUATION LINE FOR THE CONCRETE CROSS SECTION DATA LINE. IT IS REQUIRED FOR ANY CROSS SECTION CONTAINING STEEL REINFORCEMENT BARS. IT PROVIDES THE REINFORCEMENT DATA FOR THE CONCRETE SECTION ON THE 'SECT' LINE SPECIFIED IMMEDIATELY ABOVE THIS LINE. UP TO THREE REINFORCEMENT PATTERN SECTIONS ARE ALLOWED. ONLY SECTIONS DESIGNATED AS CONCRETE REINFORCEMENT PATTERN TYPES IN THE 'SECT' RECORDS CAN BE SPECIFIED.
(1- 5)	ENTER LINE LABEL 'SECT2'.
(9-15)	ENTER THE CONCRETE REINFORCEMENT PATTERN SECTION IDENTIFIER FOR THE FIRST REINFORCEMENT PATTERN.
(16-20)	ENTER THE DISTANCE FROM THE TOP OF THE CONCRETE SECTION TO THE CENTROID OF THE REINFORCEMENT PATTERN.
(21-25)	ENTER THE YIELD STRENGTH OF THE REINFORCEMENT BARS IN THE FIRST REINFORCEMENT PATTERN.

COLUMNS	COMMENTARY
(26)	ENTER THE REBAR END CONDITION TO BE USED FOR DEVELOPMENT LENGTH CALCULATIONS IN THE FIRST REINFORCEMENT PATTERN. ENTER 'S' - STRAIGHT END 'C' - CONTINUOUS BAR 'H' - HOOKED END
(28-32)	ENTER THE REBAR DIAMETER OVERRIDE FOR THE FIRST REINFORCEMENT PATTERN IF A SIZE OTHER THAN THAT SPECIFIED ON THE REINFORCEMENT BAR PATTERN SECTION LINE IS TO BE USED.
(33-80)	ENTER THE DETAILS FOR THE SECOND AND THIRD REINFORCEMENT PATTERNS IF APPLICABLE.

LINE LABEL	FIRST REINFORCEMENT BAR PATTERN					SECOND REINFORCEMENT BAR PATTERN					THIRD REINFORCEMENT BAR PATTERN				
	PATTERN SECTION ID	D DIMENSION	SY	BAR END COND	BAR DIAM OVERRIDE	PATTERN SECTION ID	D DIMENSION	SY	BAR END COND	BAR DIAM OVERRIDE	PATTERN SECTION ID	D DIMENSION	SY	BAR END COND	BAR DIAM OVERRIDE
SECT2															
1-- 5	9--15	16<--20	21<--25	26	28<--32	33--39	40<--44	45<--49	50	52<--56	57--63	64<--68	69<--73	74	76<--80
DEFAULT			50.0 ENGL	'S'				50.0 ENGL	'S'				50.0 ENGL	'S'	
ENGLISH		IN	KSI		IN		IN	KSI		IN		IN	KSI		IN
METRIC (KN)		CM	KN/SQ.CM		CM		CM	KN/SQ.CM		CM		CM	KN/SQ.CM		CM
METRIC (KG)		CM	KG/SQ.CM		CM		CM	KG/SQ.CM		CM		CM	KG/SQ.CM		CM